

OPTICAL SENSING IN MULTIMODAL MARINE MONITORING

VIABILITY TESTS IN TOULON, FRANCE
CONCEPCIÓN AND ISLA DE CHAÑARAL, CHILE

End-of-study internship - SYSMER
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Matthieu JUNG

Julie Patris, Hervé Glotin, Iván Hinojosa, Valentin Gies, Valentin Barchasz,
Franck Malige, Lyudmyla Yushchenko

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Abstract

This study presents the design and implementation of an autonomous, open-source, low-cost (<450€), and low-power optical sensor developed to complement acoustic studies, particularly in the context of monitoring NBHF (Narrow-Band High-Frequency) species using instruments deployed off the Chilean coast. Beyond cetaceans, the system also provides valuable applications for fish monitoring, enabling visual assessment of species presence, abundance, and behavior. The system enables continuous image capture over periods ranging from 4 to 8 days.

The prototype, called UWOS (UnderWater Optical Sensor), is built from readily available components and can be synchronized with the Qualilife HighBlue (QHB, developed by the SMIoT laboratory at the University of Toulon) passive acoustic recorder. It integrates a wide-angle camera within a watertight housing, offering a 120° field of view per unit. Laboratory tests characterized optical distortions, power consumption, and waterproofing reliability. Field deployments were then carried out in Toulon (France) and off Concepción (Chile), demonstrating autonomous operation under real conditions and the ability to interact reliably with the QHB system.

The combination of UWOS and QHB, named COPEPOP (Coordinando Opto-acústica para el Estudio de Peces en el Océano Pacífico), was successfully deployed at Isla Chañaral, providing complementary visual data to acoustic monitoring. This simple, reproducible, and cost-effective system offers a promising complementary tool for extending the temporal and spatial scope of underwater observations, with potential applications in scientific research, education, and citizen science.

Keywords

Underwater video – Medium-term monitoring – Open source hardware – Species monitoring – Bio-acoustic

Glossary

AI Artificial Intelligence

CIAN International Center for Artificial Intelligence in Natural Acoustics

COPEPOP The combination of UWOS and QHB : "Coordinando Opto-acústica para el Estudio de Peces en el Océano Pacífico"

FOV Field of View: refers to the extent of the observable world that a camera can capture at any given moment

FFMPEG FFmpeg is a collection of free software tools for processing audio or video streams

FS Functional Specifications

IHM Human-Machine Interface

IA Artificial Intelligence

IRP International Research Project

LCA Life Cycle Assessment

LIS Laboratory of Computer Science and Systems

NBHF Narrow-Band High-Frequency

PAM Passive Acoustic Monitoring

PLA *Polylactic acid* ($(C_3H_4O_2)_n$), is a biodegradable polymer under industrial composting conditions

QHB Qualilife HighBlue

RPI Raspberry Pi

RTC Real-Time Clock

SBC Single-Board Computer

SSI Scuba Schools International is a for-profit organization that teaches the skills required for scuba diving and freediving

TDOA Time Difference of Arrival

UART Type of serial communication link

UWOS UnderWater Optic Sensor

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PART I

Introduction and Context

1

Introduction

The observation of marine ecosystems is a major scientific and environmental challenge in the context of climate change and increasing material and acoustic pollution. Among the most widely used non-invasive tools today, passive acoustic monitoring (*PAM*) enables the detection and recording of species' sounds and vocalizations over varying durations, without disturbing their environment. In recent years, passive monitoring of marine ecosystems has greatly advanced thanks to technological progress in sensors and automated analysis [Boom et al. \[1\]](#). Combined with artificial intelligence algorithms, PAM now allows for the detection, classification, and tracking of certain species with growing efficiency, and even for spatial localization [Giraudet and Glotin \[2\]](#).

However, acoustic analysis alone has limitations in several contexts. It does not always enable reliable species identification, particularly when multiple species produce similar acoustic signatures, as is the case with NBHF (Narrow-Band High-Frequency) species found in Chilean waters [Patris et al. \[3\]](#). In addition to marine mammals, it is now well established that many fish species are soniferous, while numerous others are considered potential sound producers [Looby et al. \[4\]](#). Moreover, acoustic analysis does not provide visual information about size, morphology, abundance, or non-vocal behaviors, and is not well suited for direct observation of interspecies interactions or anthropogenic disturbances.

In this regard, the addition of a visual sensor represents a significant advancement, as suggested by [Reyes et al. \[5\]](#). By combining imagery and sound, it becomes possible to enrich datasets to validate acoustic classifications, estimate population sizes more precisely, and better understand observed behaviors. This kind of multimodal approach—still rare in field deployments due to technical, energy, and cost constraints—is now more accessible thanks to recent innovations in open-source, low-cost embedded systems [Mouy et al. \[6\]](#), [Purser et al. \[7\]](#).

The present work is part of this dynamic and aims to design, test, and integrate an autonomous, low-cost, low-power image capture module to be coupled with an existing acoustic recorder (QHB, developed by the SMIoT laboratory at the University of Toulon), as part of observation missions targeting *NBHF* species off the coast of Chile.

2

Company Presentation

This internship was carried out within a highly interdisciplinary and international framework, involving several laboratories and partner institutions. To provide a clearer picture of the environment in which the project took place, this chapter introduces the different structures that hosted and supported me throughout the internship. These units, both local and international, brought together complementary expertise in computer science, bioacoustics, marine ecology, mechanics, and instrumentation.

2.1 Company History

The Laboratoire d'Informatique et des Systèmes (LIS) was established in January 2018, following the merger of the Laboratoire des Sciences de l'Information et des Systèmes (LSIS) and the Laboratoire d'Informatique Fondamentale (LIF). This merger brought together and strengthened expertise in computer science, control theory, signal processing, and data analysis. Since its creation, LIS has conducted both fundamental and applied research addressing scientific, technological, and societal challenges, while fostering academic and industrial collaborations at national and international levels.

The center is particularly active in research valorization through partnerships, European projects, and the development of technological platforms. Its location within Aix-Marseille University (AMU) and the SeaTech engineering school in Toulon supports the development of specific focus areas such as underwater acoustics, bioacoustics, and artificial intelligence applied to complex systems.

2.2 Description of the Corporate Structure

The LIS is a Joint Research Unit (UMR 7020) under the supervision of CNRS, Aix-Marseille University (AMU), and the University of Toulon (UTLN). Created in 2018 by merging two laboratories, the LIS has around 375 members, including 190 permanent researchers and faculty members. The Toulon-based teams of LIS are hosted at the SeaTech engineering school.

LIS is organized into four research divisions: Systems Analysis and Control (ACS), Computing, Signal and Image (SI), and Data Science (SD). The Toulon teams are distributed across these divisions and conduct research in diverse areas such as computer science, automation, and signal/image processing.

Among LIS's key initiatives is the International Center for Artificial Intelligence in Natural Acoustics (CIAN). Originating from the ADSIL Chair in marine bioacoustics (2020–2024), CIAN brings together researchers specializing in the study of sounds produced by living organisms. I had the opportunity to carry out my internship within this unit. Their objective is to understand the subtleties of natural soundscapes in order to derive valuable insights into animal behavior, ecological dynamics, and to contribute to conservation efforts. The center employs cutting-edge technologies and innovative methodologies to capture, analyze, and interpret bioacoustic data.

Note

At present, CIAN does not have a formal legal status, but it brings together researchers, engineers, and PhD students. For more information: <https://cian.lis-lab.fr>

My supervisors at LIS were Julie Patris (Figure 2.2), my tutor, and Hervé Glotin, co-supervisor (Figure 2.1).



Figure 2.1: Hervé Glotin



Figure 2.2: Julie Patris

2.3 Description of the Department Where the Internship Took Place

My internship took place within the LIS DYNI's (DYNamiques de l'Information¹) Toulon branch, specifically within the team of the International Center for Artificial Intelligence in Natural Acoustics (CIAN). This department is specialized in the analysis of natural soundscapes, with a particular focus on marine bioacoustics. Researchers study sounds emitted by wildlife (especially cetaceans) in their natural environments using advanced artificial intelligence techniques (deep learning, signal processing, automatic classification, etc.).

CIAN also leads interdisciplinary projects at the crossroads of ecology, AI, and engineering. It benefits from international collaborations and access to unique datasets gathered during sea expeditions, via acoustic buoys, or from sensor networks deployed in protected areas. Within this context, I had the opportunity to contribute to an applied research project, under the guidance of experts in bioacoustics and environmental data processing.

2.4 Partners and Scientific Collaboration

As previously mentioned, this internship took place within a highly interdisciplinary and international research framework, bringing together multiple academic and technical partners. Each of these collaborators played a key role in the successful design, testing, and deployment of the instrument.

2.4.1 SMIoT — Scientific Microsystems for the Internet of Things

The SMIoT platform, hosted at the University of Toulon, focuses on the development of low-power embedded electronics for sensor networks and scientific applications. It is responsible for designing and manufacturing the Qualilife HighBlue (QHB) recorder Barchasz et al. [8], which was used as the base platform in this project Patris et al. [3]. This ultra low-power system can record up to six audio channels

¹<https://www.lis-lab.fr/dyni/>

at 256 kHz , with a total energy consumption of just 2 W . During this internship, my primary contacts at SMIoT were Valentin Barchasz (Figure 2.3) and Valentin Gies (Figure 2.4).



Figure 2.3: Valentin Barchasz



Figure 2.4: Valentin Gies

2.4.2 COSMER

The COSMER Laboratory at the University of Toulon specializes in the design, simulation, and optimization of eco-engineered mechanical systems, with particular expertise in autonomous robotic platforms for marine, underwater, amphibious, and terrestrial environments. I previously worked with COSMER during my final year engineering project on the development of an autonomous sailing drone, MARIUS [Jung, Caudron, and Autet \[9\]](#). Their expertise in mechatronics, prototyping, and additive manufacturing provided strong technical support throughout this internship, especially during the design and mechanical validation phases.

2.4.3 University of Concepción

The Catholic University of Concepción (UCSC) is a key scientific partner through the CNRS International Research Project (IRP) "DCDB", for *Détection et Classification IA des Biosonars des dauphins et marsouins du Cône Sud*, which focuses on the study of marine biodiversity and in particular NBHF cetacean species. The university collaborates closely with the LIS laboratory, particularly through the work of Dr. Iván Hinojosa (Fig.2.5), a marine ecologist and key local coordinator. UCSC provided both scientific support and logistical infrastructure during the field deployment of the instrument off the Chilean coast.

2.4.4 Austral University of Chile

The connection with the Universidad Austral de Chile (UACH) is notably established through Dr. Mauricio Soto Gamboa (Fig.2.6), who also works on NBHF dolphins in the Valdivia Bay. He has previously collaborated with Julie Patris and Franck Malige on projects within this framework.



Figure 2.5: Dr. Iván Hinojosa Toledo - UCSC



Figure 2.6: Dr. Mauricio Soto Gamboa - UACH

2.4.5 ExploraSub

ExploraSub is one of the scuba diving schools located in Caleta Chañaral, in the Atacama region of Chile. It is managed by César Villarroel, a photographer, explorer, diving instructor, and filmmaker. He provides valuable support as a key field actor for the implementation of bioacoustic projects in the region.



Figure 2.7: César Villarroel

3

Context and Objectives

This chapter sets the stage for the project by presenting its scientific and technical context. It first introduces the Qualilife HighBlue (QHB) recorder, the main acoustic instrument with which the optical sensor was designed to be integrated. It then outlines the specific interest of conducting such research in Chile. Finally, the overall objectives of the internship and the methodology adopted to manage the project are detailed, providing the framework that guided the subsequent design, implementation, and field deployment phases.

3.1 Qualilife HighBlue (QHB) recorder

The primary instrument used in the experiment is the Qualilife HighBlue (QHB) recorder, developed by SMIoT, University of Toulon, and shown in Figure 3.1. The recorder features the following characteristics (see also [Barchasz et al. \[8\]](#)):



Figure 3.1: QHB recorder

- Sampling rates up to 512 Ksps (kilo-samples per second), corresponding to a frequency range up to 256 kHz. Recording can be scheduled according to user requirements.
- Up to six synchronous recording channels, with precise synchronization.
- Adjustable sampling depth of 8, 16, or 24 bits. In 24-bit mode, the recorder's self-noise is limited to the two least significant bits, meaning 22 bits are truly significant for recording. This enhances signal quality and detection range compared to standard recorders, especially in quiet environments.
- Each channel has an adjustable differential gain: X1, X10, X20, or X100.
- Anti-aliasing filtering
- Sensor hub capability: The QHB includes a 9-axis IMU sensor (MEMS accelerometer, magnetometer, and gyroscope) and supports additional sensors via UART, SPI, and I2C extension buses, depending on user requirements.

The QHB recorder was housed in a custom-made enclosure designed to withstand pressures up to 100 m depth, ensure stable ground placement, and power the system with a set of 24 D alkaline batteries. A reference for the QHB recorder can be found at: <http://bioacoustics.lis-lab.fr/smiot>.

3.2 What About Chile?

The Chilean coast is of particular interest for the study of marine species that use Narrow-Band High-Frequency (*NBHF*) acoustic signals [Patris et al. \[3\]](#), [Reyes et al. \[5\]](#), [Reyes et al. \[10\]](#). This area lies along one of the most productive upwelling systems in the world (the Humboldt Current), and hosts exceptional marine biodiversity, including many odontocete cetacean species, some of which are endemic or poorly studied. Human activity in this highly sensitive region may add stress to species already at risk of extinction [Jaramillo-Legorreta et al. \[11\]](#), [Silva et al. \[12\]](#). Among these, several species have evolved highly specialized acoustic communication modes, characterized by the emission of *NBHF* signals (>100 kHz), enabling efficient echolocation in complex or shallow environments.

This phenomenon is one of the most striking examples of evolutionary convergence in the animal kingdom: different odontocete lineages, living in similar environments, independently developed *NBHF* signals. These signals are thought to serve multiple adaptive functions: enabling fine prey detection at short ranges and potentially reducing the risk of detection by predators such as orcas, whose hearing does not cover such high frequencies.



Figure 3.2: Location of the mission stopover in Chile

In this context, the support of Ivan Hinojosa has been particularly valuable. Since 2015, he has been conducting research in underwater bioacoustics, notably illustrated in [Carrasco et al. \[13\]](#), where the authors assess the importance of underwater soundscapes, especially for larvae and macrobenthos. More recently, Ivan has also been deeply involved in studying the sounds produced by fish along the Chilean coast.

3.3 Objectives of the internship

The primary goal of this project is to design, develop, and validate an embedded optical sensing system to complement passive acoustic monitoring (PAM) for the identification and observation of marine species. This visual sensor is intended to be integrated into an existing bioacoustic monitoring platform in order to enable robust image/sound correlation within a non-intrusive ecological study framework.

More specifically, the objectives of this work are as follows:

- To overcome the limitations of acoustic analysis alone, particularly in ecosystems where multiple species produce vocalizations in overlapping frequency bands, making acoustic discrimination challenging.
- To provide visual evidence of species presence, enabling fine-grained identification and supporting the validation of biological hypotheses based on acoustic data.
- To develop a compact, waterproof, and energy-efficient optical system capable of autonomous operation over several days in offshore environments.
- To ensure precise temporal synchronization with existing acoustic systems, enabling reliable multi-modal data correlation and post-processing.
- To experimentally validate the complete system both in laboratory conditions (power consumption, optical distortion, sensor comparison) and in real-world deployments, including a scientific field mission in Chile.
- To anticipate future integration of AI-based post-processing, by structuring the visual data in a way that facilitates the development of supervised learning algorithms for automatic species detection.

Note

The project was carried out under the constraint that all mechanical development and fabrication had to be completed in Toulon, leveraging locally available components, tools, and materials. The activities in Chile were primarily dedicated to completing the software implementations in collaboration with local partners, including meetings, scientific and technical exchanges with end-users, refining the understanding of constraints and requirements, as well as conducting scaled tests and a first real-world deployment, which successfully produced data in the Isla Chañaral.

3.4 Project Management and Working Methodology

Given the multidisciplinary nature of the project—ranging from electronics and embedded software to mechanical design and field deployment—a structured and iterative project management approach was adopted.

The project was carried out following an agile-inspired methodology, broken down into multiple iterative phases:

- **Specification phase:** Initial meetings with researchers from the DYNi section and the CIAN team allowed the definition of the functional and technical requirements. A detailed requirement document and component selection report were produced at this stage.
- **Design phase:** The hardware architecture, sensor interfaces, and power management system were designed based on the requirements. Concurrently, the mechanical structure was modeled in 3D CAD, with a focus on waterproofing, ease of deployment, and integration with existing systems.

- **Implementation phase:** The system was built, coded, and assembled. Low-level embedded software was developed for sensor acquisition, power optimization, and data logging. A particular effort was made to ensure robustness, reliability, and synchronization with acoustic data streams.
- **Testing and validation:** A series of laboratory tests were performed, including power consumption profiling, sensor comparison, and image quality evaluation. The prototype was then first tested in France, before being deployed in real conditions during a scientific expedition in Chile.

3.4.1 Collaboration and Resource Management

Throughout the internship, collaboration was maintained with multiple stakeholders:

- The SMIoT and COSMER laboratories (Université de Toulon) provided guidance on embedded systems and power efficiency.
- In Chile, close cooperation was established with the Universidad Católica de la Santísima Concepción for field testing and data acquisition.

A lab notebook was maintained to document the daily progress, design decisions, encountered issues, and solutions implemented. Version control (Git) was used throughout the project to manage software development and hardware schematics.

In order to estimate the time allocated to each task, we created a GANTT chart, being as exhaustive as possible. We therefore have a project start date of 03/05/2025, with a one-week interruption from 16/05 to 22/05. The prototype must be validated before this week.

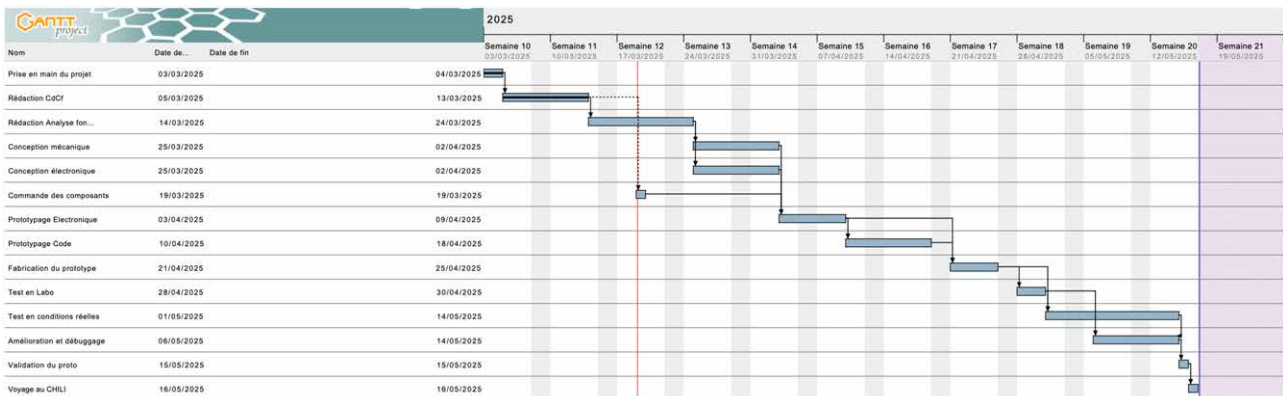


Figure 3.3: GANTT chart

PART II

System Design and Engineering

4

Technical Specifications and Operational Constraints

This chapter introduces the technical specifications and operational constraints that framed the development of the system. It establishes the functional requirements, highlights the environmental and logistical challenges inherent to underwater deployments, and details the design envelope used to balance robustness, energy autonomy, and data acquisition needs.

Marine wildlife monitoring imposes a number of strict technical constraints. Furthermore, the instrument had to be compact, easy to integrate with the QHB¹ system, and capable of recording high-resolution video while consuming minimal power.

The specifications were also influenced by logistical and environmental factors, including transportation to remote sites, exposure to seawater and variable lighting conditions, and the need for autonomous operation over several days. The sensor's design thus required a careful balance between robustness, modularity, simplicity of use, and minimal environmental impact.

This chapter outlines the core functional requirements, details the mechanical, electrical, and software-level constraints, and introduces the design envelope that framed all further development steps.

4.1 Functional Specifications (FS)

4.1.1 Life Cycle Analysis

The life cycle of a low-power underwater UWOS² module follows several stages, from its design to its end-of-life recycling:

- Design and development phase
- Manufacturing phase
- Transport phase
- Operation and usage phase
- End-of-life and recycling

In our case, due to the limited timeline (less than 3 months of R&D), the design, manufacturing, and operational phases are likely to overlap. Indeed, the developed solution may remain at the prototype stage during this project.

¹Qualilife HighBlue

²UnderWater Optic Sensor

4.1.2 External Environmental Elements (EEE)

The underwater environment imposes a set of physical and chemical constraints that directly influence system design (pressure, corrosion, salinity, visibility, etc.).

External Environmental Elements are defined as components of the environment that interact with the optical system. In our context, we identify the following EEEs:

- Existing recording system
- End user
- Regulations and standards
- Underwater environment

In the diagrams shown in the following section, the EEE *Existing system* is broken down into several sub-elements for improved clarity.

4.1.3 Use Case and Operational Context

4.1.3.1 APTE Diagram

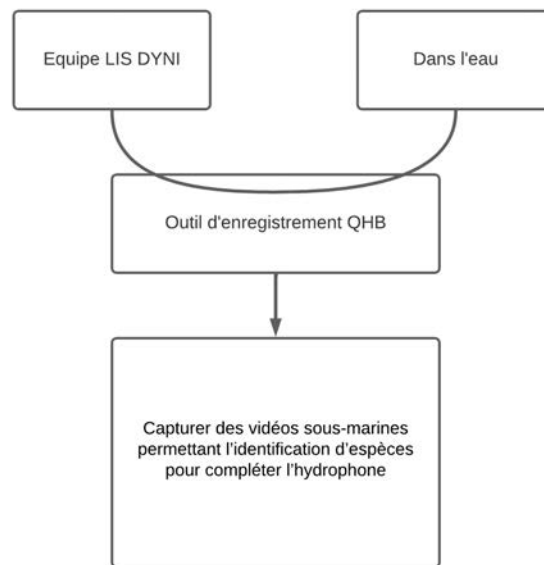


Figure 4.1: APTE Diagram

The “horned beast” diagram provides a clear and concise visualization of the various elements of the system and their relationships. It helps understand who will use the system, how, and for what purpose.

However, it must be complemented by a second, more detailed diagram that outlines the system’s specific functions.

4.1.3.2 Actor Interaction Diagram

This second diagram defines the key functions the system must fulfill. It helps identify the main project objectives from the start. The primary goal of this project is to enhance the existing acoustic recording tool by adding a visual sensor.

From this, we can derive the following functional objectives:

- Capture visual information
- Enable species identification up to a certain range
- Ensure autonomous recording synchronized with audio
- Guarantee a battery life of 2 to 4 days

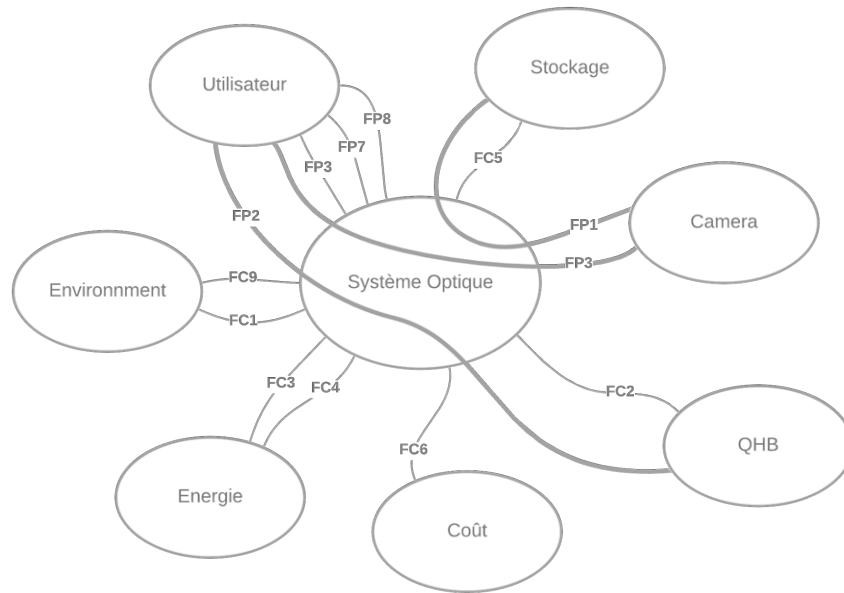


Figure 4.2: PIEUVRE Diagram

The complete description of all functions is provided in Appendix B. However, the main functions can be summarized as follows:

Function Name	Short Description
FP1	Capturing Underwater Videos
FP2	Video triggering
FP3	Informing the user of the recording status
FC1	Mechanical resistance
FC2	Interfacing with existing systems
FC3	Battery life
FC5	Storage
FC6	Cost
...	...

Table 4.1: Principal Functions and Constraints

4.2 Technical Functional Analysis (TFA)

In this project, we will not formalize the system using a traditional SADT or FAST diagram due to time constraints. However, since we aim to adopt a systems-thinking approach, we describe the architecture of

each subsystem through structured tables, which closely resemble the logic of an SADT diagram. These tables allow us to represent the architecture of each component and sub-component as well as their interactions.

This section marks the beginning of the design phase of our solution. A complete set of supporting files can be found in Appendix.B.

4.3 Failure Modes

It is crucial from the beginning of the project to anticipate possible failure modes in order to define how the system should respond in the event of mechanical or electronic malfunction.

The following table summarizes the failure modes associated with each key function of the system, extract of the table that can be found in Appendix.B :

ID	Function Name	Event	Undesired Outcome	Severity (0–1)	...
1.1	Optical acquisition	Camera failure	No image captured	1	...
2.1	Audio/Video synchronization	Temporal offset between sound and image	Inconsistent analysis	0.8	...
3.2	Data storage	No recording saved	Analysis impossible	1	...
...					

Table 4.2: Failure Modes (excerpt; full version in Appendix B)

4.4 Project Planning and Management

Task Name	Start Date	End Date
Project Familiarization	03/03/2025	04/03/2025
Drafting of Functional Specification	05/03/2025	13/03/2025
Drafting of Technical Functional Analysis	14/03/2025	24/03/2025
Mechanical Design	25/03/2025	02/04/2025
Electronic Design	25/03/2025	02/04/2025
Component Ordering	03/04/2025	03/04/2025
Electronic Prototyping	04/04/2025	10/04/2025
Software Prototyping	11/04/2025	21/04/2025
Prototype Fabrication	22/04/2025	28/04/2025
Laboratory Testing	29/04/2025	01/05/2025
Field Testing (Real Conditions)	02/05/2025	15/05/2025
Debugging and Improvements	07/05/2025	15/05/2025
Prototype Validation	16/05/2025	16/05/2025

Table 4.3: Tasks scheduled over the first 3 months

5

State of the Art and Component Selection

The selection of components must meet a set of precise technical criteria defined in the specifications (see Chapter 4), including: low power consumption, compactness, mechanical robustness, software compatibility, ease of integration, and an optimal cost-to-performance ratio. These requirements are further constrained by the specific challenges of the marine environment: prolonged immersion, hydrostatic pressure, salinity, low ambient light, and the impossibility of human intervention during acquisition phases.

This chapter presents a comparative analysis of potential hardware configurations for the system architecture. Particular attention is given to the selection of the optical sensor, processing module, power supply system, and acquisition interfaces. The goal is to define a balanced configuration that maximizes data quality while minimizing power consumption and implementation complexity.

Before detailing the final choice, it is important to review the main solutions proposed in both the scientific literature and the commercial sector.

5.1 State of the Art

5.1.1 Autonomous Stations

Cameras with Enclosure

In [Cappo, Speare, and De'Ath \[14\]](#), the authors describe the design and use of a *Baited Remote Underwater Video Station* (BRUVS) equipped with a Sony Hi-8 Handicam (model TR516E), protected within a dedicated housing. This type of solution is simple and robust, but offers little flexibility for integration with other sensors or for advanced control over acquisition parameters.

Action Cameras

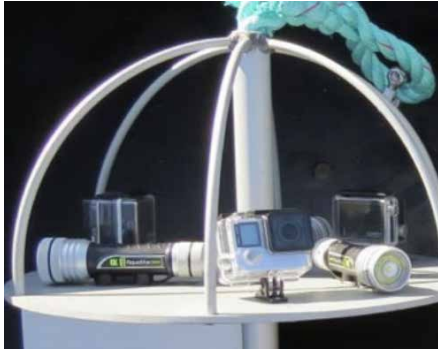
Several recent studies have employed commercially available action cameras as the core of autonomous underwater imaging systems. For example, [Lelièvre et al. \[15\]](#) reports the use of *GoPro® HERO7* cameras. These solutions benefit from the compactness, ruggedness, and image quality of consumer-grade cameras, but they offer limited flexibility in terms of hardware customization, triggering, and integration with other scientific instruments.

Similarly, in [Ferrari et al. \[16\]](#), dedicated to sperm whale studies off Mauritius, action cameras were combined with hydrophones to enable synchronized audio-video acquisition, with the old version of the QHB, named JASON. While action cameras offer plug-and-play deployment and proven waterproofing, their closed architectures limit the level of customization and interoperability required for integration into complex, multi-sensor platforms.

Moreover, these solutions cannot be used for medium- or long-term missions due to their high power consumption.

Single-Board Computers (SBCs)

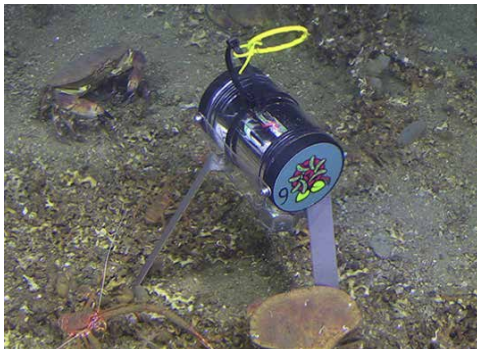
Another approach involves designing custom-built imaging systems based on single-board computers (SBCs), which provide greater control over acquisition parameters, power management, and sensor integration. Among the most popular SBCs for rapid prototyping is the Raspberry Pi, which combines low cost, a large developer community, and compatibility with a wide range of camera modules. Several works, such as [Purser et al. \[7\]](#), [Pelletier et al. \[17\]](#), [Sheehan et al. \[18\]](#), and [Mouy et al. \[6\]](#), describe underwater camera systems built around Raspberry Pi boards, enabling programmable triggers, tailored acquisition workflows, and direct integration with additional sensors. [Pham et al. \[19\]](#) even developed a 360° camera using a Raspberry Pi combined with a video synchronization HAT, but relying on an NVIDIA Jetson Nano, which generates significantly more heat and consumes much more power than a Raspberry Pi.



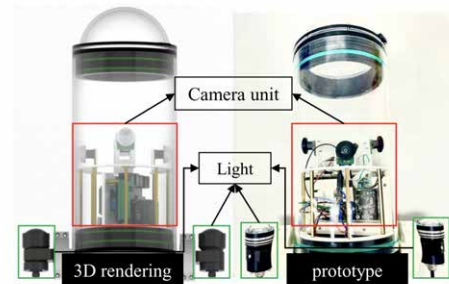
(a) PelagiCam [Sheehan et al. \[18\]](#)



(b) KOSMOS camera [Pelletier et al. \[17\]](#)



(c) PlasPI [Purser et al. \[7\]](#)



(d) UWA360CAM [Pham et al. \[19\]](#)

Figure 5.1: Examples of autonomous camera systems described in the literature.

5.1.2 All-in-One Solutions

An alternative approach consists in using fully integrated “all-in-one” scientific camera systems, which combine imaging, power management, data storage, and sometimes environmental sensors within a single housing. These devices are designed for ease of deployment and minimal maintenance, but tend to be expensive and less adaptable to specific research needs.

For instance, [Del-Rio et al. \[20\]](#) describes a self-contained underwater observation platform deployed off the coast of Spain and connected by a submarine cable, capable of long-term autonomous operation and delivering high-quality data, but requiring substantial investment and specialized maintenance.

5.1.3 Comparative Assessment

Solution Type	Advantages	Limitations	QHB Integration
Cameras with Enclosure	Proven waterproofing, robustness, ease of deployment	Limited sensor integration, minimal control over acquisition parameters	Low
Action Cameras	Compact, robust, high-quality imaging, accessible cost	Closed architecture, limited triggering and customisation options	Medium
SBC-Based Systems	Fully customisable, open-source, programmable acquisition and power control	Requires engineering effort, waterproofing to be designed	High
All-in-One Scientific Systems	Ready-to-use, optimized for long-term deployment, integrated sensors	High cost, limited adaptability, proprietary components	Medium

Table 5.1: Summary comparison of autonomous imaging approaches.

5.2 Choice of System Components

5.2.1 Main Microcontroller

For sensor management and data processing, a **Raspberry Pi Zero 2W** was selected. This single-board computer offers several advantages: it is physically compact and cost-effective, its moderate power consumption meets the autonomy requirements discussed in Chapter 8, and it supports an extensive software ecosystem, including video processing libraries, ArduCam drivers, and GPIO¹ management tools. The platform also benefits from an active open-source community, which facilitates technical support and the integration of new functionalities. From a hardware perspective, it provides a complete set of interfaces, including CSI ports for cameras, GPIO pins for serial communication (UART) with the QHB, and support for high-capacity microSD cards.

Power management is handled by a **Witty Pi 4 Mini**, an add-on module that combines a real-time clock (RTC) for time synchronisation with a programmable ON/OFF control system. This allows the instrument to operate only during required recording periods, optimising battery life without compromising data collection.

5.2.2 Optical Sensors

The Raspberry Pi platform is compatible with a wide range of optical sensors, from native modules to third-party models supported through dedicated drivers. The choice was based on a balance between price, low-light sensitivity, resolution, and physical footprint. A comparative table of the shortlisted options is presented in Appendix C. Two models were selected for field testing: the **Sony IMX708**, a high-resolution sensor suitable for detailed imaging in standard light conditions, and the **Sony IMX462**, optimised for low-light performance and therefore well adapted to underwater environments with limited illumination. Both sensors are evaluated in Chapter 8 to determine the optimal compromise between image quality and energy efficiency for field deployment.

¹General-Purpose Input/Output port



(a) Sensor IMX462



(b) Sensor IMX708

Figure 5.2: Selected sensors

5.2.3 Waterproofing Solution

Ensuring the camera's waterproofing posed several challenges. Beyond simple resistance to water ingress, the enclosure needed to withstand the hydrostatic pressure at the intended deployment depth, while also allowing optical recording without significant distortion.

Several approaches have been reported in the literature. [Moore et al. \[21\]](#) proposed encapsulating the sensor in resin to achieve a watertight seal. [Purser et al. \[7\]](#) suggested the fabrication of custom-machined housings, whereas [Mouy et al. \[6\]](#) described the use of commercially available PVC components.

In our case, the requirement for a smooth, optically clear dome made these solutions less suitable. We therefore opted for a proven, commercially available option: the *Watertight Enclosure* from BlueRobotics (Fig. 5.3).

**Figure 5.3:** BlueRobotics watertight enclosure

Note

BlueRobotics enclosures were already used in the first versions of the QHB recorder and are also standard components in the company's BlueROV platforms.

For modularity, we selected the four-hole flange variant. Two of these ports are reserved for the pressure-relief valve and the power supply connection to the external module. If additional modules are added in the future, at least one more port will be required for synchronisation between systems, meaning that a minimum of three functional feedthroughs should be anticipated.

5.3 Choice of Power Source

Selecting an appropriate power source is a critical aspect of designing an autonomous instrument. Two main categories were considered: rechargeable batteries and primary (non-rechargeable) cells.

5.3.1 Rechargeable Options

Li-ion 18650

Lithium-ion 18650 cells offer high energy density and a long cycle life, making them attractive from a waste-reduction perspective. However, they present several drawbacks for this application, including transportation restrictions, safety risks (thermal runaway), and relatively high cost.

LiFePO₄

Lithium iron phosphate (LiFePO₄) cells provide improved safety and thermal stability compared to conventional Li-ion cells, with a slightly lower energy density. They remain a viable option for long-term sustainability, though availability in the required form factor is more limited in certain regions. However, their adoption can be hindered by the relatively high cost of this technology, which may be prohibitive for certain large-scale or budget-constrained applications.

5.3.2 Primary Cells

Ultimately, we selected alkaline D cells as the primary power source. This decision was motivated by their widespread availability, ease of replacement in the field, absence of explosion risk, and predictable performance over the required operating period. While less environmentally sustainable than rechargeable solutions, their logistical reliability and safety profile made them the most practical choice for the present deployment.



(a) D-Cells - Duracell



(b) D-Cells - Varta

Figure 5.4: Exemple of D-Cells

5.4 Bill of materials

In this section, most of the screws and nuts are not listed because they are already purchased by the various labs included in the project. We can broadly estimate the maximum cost of the hardware at around 70 – 80€.

Designator	Component	Number	Cost per unit	Total cost	Source of materials	Material type
	Raspberry Pi Zero 2WH	1	24,90€	24,90€	–	Semiconductor
	Witty Pi 4 Mini	1	30,50€	30,50€	–	Semiconductor
	Pololu DC-DC 5 Vcc S13V15F5	1	11,15€	11,15€	–	Semiconductor
	Camera connectic SC1128	1	1,60€	1,60€	–	Semiconductor
	JST PH	1	1,30 €	1,30 €	–	Semiconductor
	ArduCAM 12 MP IMX708 - B0310	1	34,90€	34,90€	–	Semiconductor
	ArduCAM 2 MP IMX462 - B0333	1	59,00€	59,00€	–	Semiconductor
	Lens M12 1.56mm M25156H14	1	14,00€	14,00€	–	Lens
	ON / OFF switch	1	1,75€	1,75€	–	Semiconductor
	Kingston Canvas Select Plus SDCS2/512GB	1	62,09€	62,09€	–	Semiconductor
	BlueRobotic Watertight Enclosure	1	166,67€	166,67€	–	Plastic

Table 5.2: Bill of materials

6

3D Design, Modeling and Simulation

Design file name	File type	Open source license	Location of the file
bride_tube_2p.STL	STL file	BSD-3	GitHub
bride_tube_2p.pdf	PDF file	–	–
rail_50mm_69mm_cam.STL	STL file	BSD-3	GitHub
rail_50mm_69mm_cam.pdf	PDF file	–	–
rack_IMX708.STL	STL file	BSD-3	GitHub
rack_IMX708.pdf	PDF file	–	–
support_double_UWOS.STL	STL file	BSD-3	GitHub
support_double_UWOS.pdf	PDF file	–	–

Table 6.1: Design Files

6.1 Electronic Architecture

6.1.1 Design and Electronic Schematic

After defining the required components, it is essential to design an architecture that enables their coherent and reliable integration. The core of the system is based on a *Raspberry Pi Zero 2W* board, combined with a *Witty Pi 4 Mini* power management module, directly connected via the 40-pin GPIO interface (see Figure 6.1).

Power supply is provided by a *Pololu S13V15F5* regulator, capable of delivering a stable 5 V output voltage for an input voltage range between 2.8 V and 22 V. This flexibility allows powering the system from various sources while ensuring the stability required for sensitive components.

The module is designed to accommodate two possible versions of sensors: the *IMX708* or the *IMX462*, both compatible with M12-mount optics. This standard makes it possible to consider a wide range of focal lengths, with fields of view from 90° to 180°, depending on the observation requirements. The final choice of optics will be validated during upcoming experimental tests.

Data is stored on a 512 GB microSDXC card (Sandisk Extreme), providing sufficient capacity for prolonged high-resolution video recordings. The main battery is housed in the central tube of the system, which also accommodates the QHB.

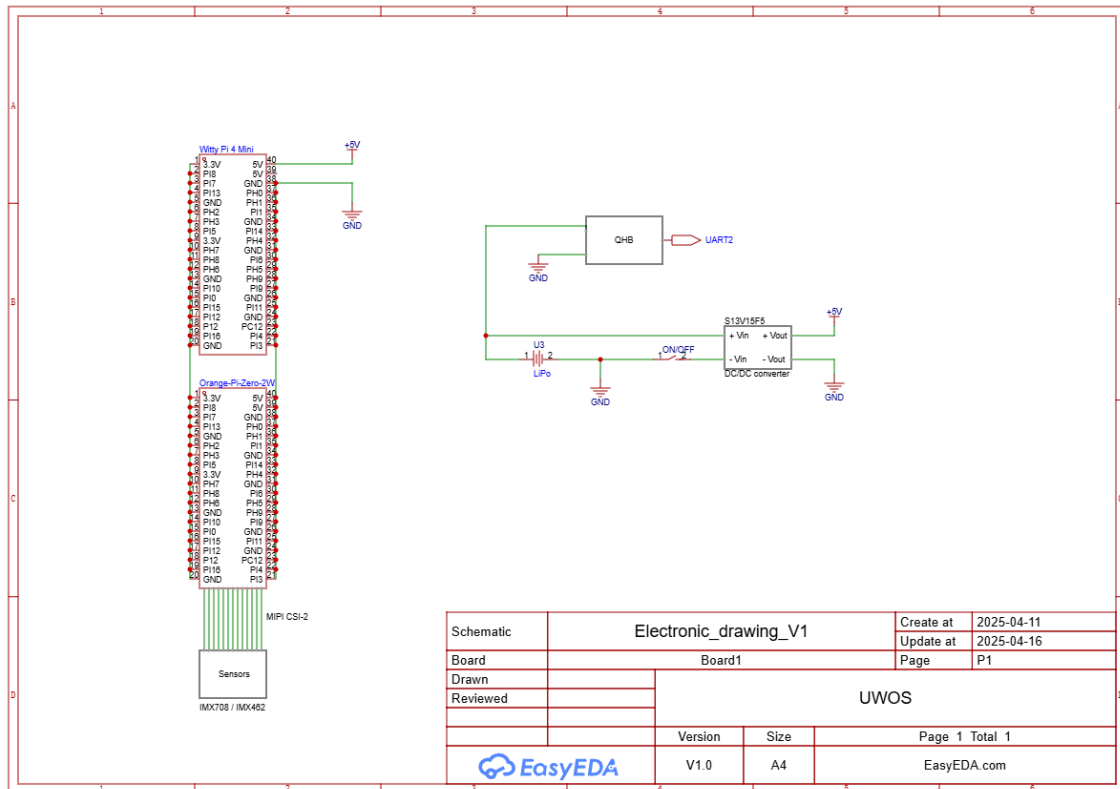
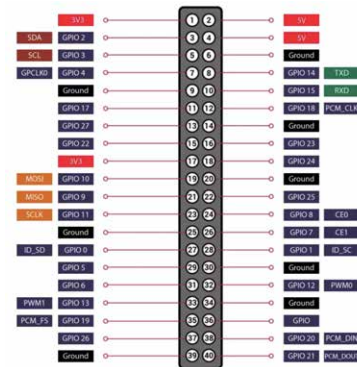


Figure 6.1: Electronic Drawing



(a) Photo: Raspberry Pi + Witty Pi 4 + IXM708



(b) Raspberry Pi 02W Pinout

Figure 6.2: PCB and Sensor Photo & Pinout

6.1.2 Wiring and Soldering

In a constrained environment subject to vibrations, wiring plays a crucial role in ensuring the robustness and maintainability of the system. Connectors were soldered to the main critical components, particularly for the UART connection and the input/output of the voltage regulator.

Note

Due to the extremely limited space inside the module, it was not possible to add a dedicated connector to the Raspberry Pi's UART pins. An alternative solution was therefore implemented: the serial link was soldered directly to the corresponding pins of the Witty Pi. To secure this connection and facilitate access during possible maintenance operations, a strain relief point was added using a zip tie, allowing the cable to be neatly fixed without any risk of pulling on the solder joints.

6.2 3D Design and Modeling

6.2.1 SolidWorks Assembly of a Module

Based on the precise definition of functional requirements and the electronic components to be integrated (see Chap. 5), it becomes possible to estimate the minimum required dimensions for each module. This step is crucial to optimize volume, weight, and associated costs, particularly in a marine deployment context where compactness and robustness are key criteria. The enclosure choice naturally focused on the waterproof tubes offered by Blue Robotics, known for their compatibility with underwater environments. For maximum compactness, the smallest available model was selected: a tube with a 2-inch diameter and a length of 100 mm. The objective is to verify the complete integration of the electronics within this constrained volume. To achieve this, all components were modeled in 3D or downloaded from online libraries such as GrabCAD or supplier platforms. Once gathered, a complete assembly was created in SolidWorks to validate part arrangement, identify potential mechanical interferences, and facilitate the planning of future printing and assembly operations.



Figure 6.3: SolidWorks Assembly of a UWOS Module

6.2.2 Internal Rail and Rack

To assemble all components inside the tube, an internal rack system is required. Some parts are available in 3D directly from the Blue Robotics website; however, racks are only available for tubes of 150 mm length or more. Therefore, the part had to be redesigned. Below is the newly modeled part.

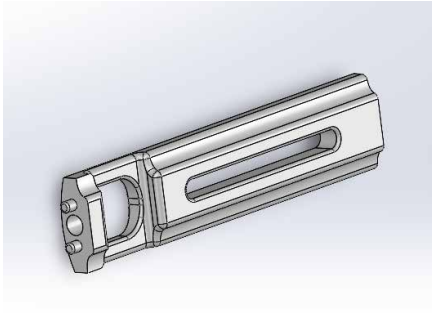


Figure 6.4: Internal Rail — View #1

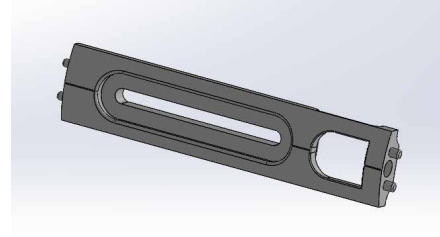


Figure 6.5: Internal Rail — View #2

This rail was designed to fit perfectly into the Blue Robotics tube through reverse engineering of an existing rail from the same brand. It is also designed to accommodate racks accordingly. No stress analysis was performed on this part, as it is normally protected from any external loads. One side is inserted and screwed onto a “Cap” (see Fig. 6.4), while the other simply slides into the holes provided for this purpose.

Note

It should be noted that the CAD files provided on the website did not match the parts received. As a result, this part underwent further modifications to match the actual physical part.

6.2.3 Module Flange

The flange of a module provides the interface between the Blue Robotics tube, which has no built-in features for integration into an existing system, and the tube developed by SMIoT for the QHB. The flanges clamp the tube using polyurethane foam gaskets.

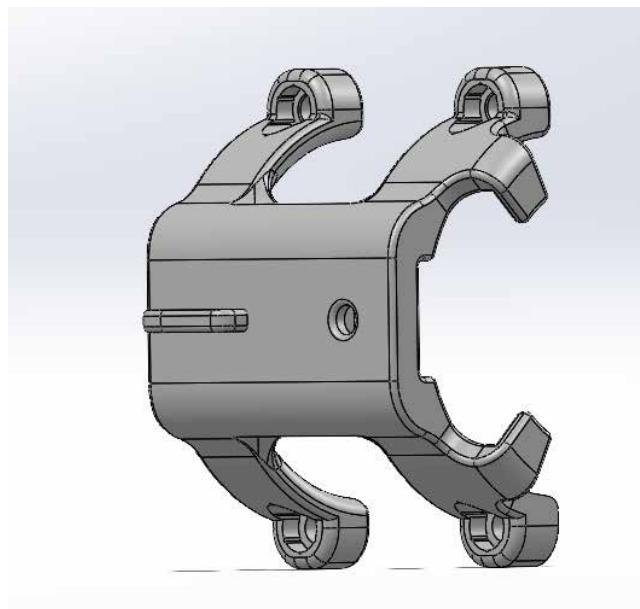


Figure 6.6: Flange 3D Model

6.2.4 Dual-Module Support

The shape of this support was designed to offer four fixed positions (0 , $\pi/6$, $\pi/3$, $\pi/2$) in order to minimize positioning uncertainties. The position is locked using an M5 screw, whose nut is housed in the flange

described in Section 6.2.3. To reduce the buoyancy of the part, material was removed wherever possible without compromising structural integrity (see Section 6.3.1).

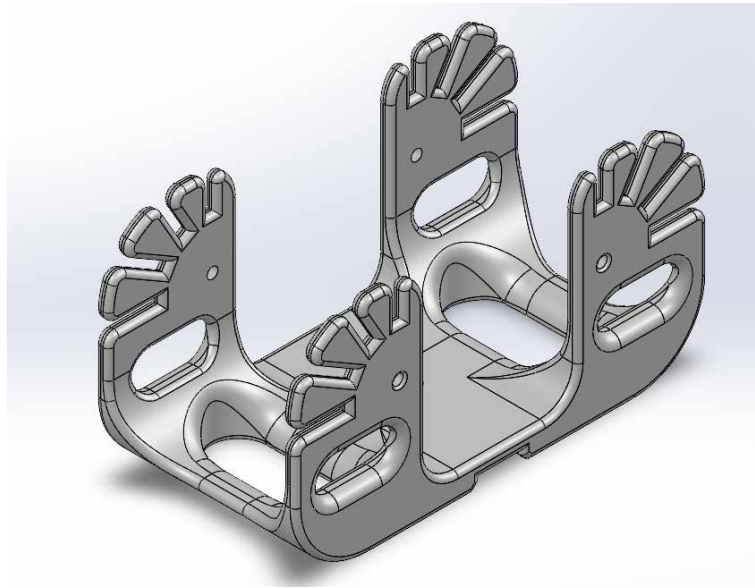


Figure 6.7: Dual-Module Support 3D Model

6.2.5 SolidWorks Assembly of UWOS

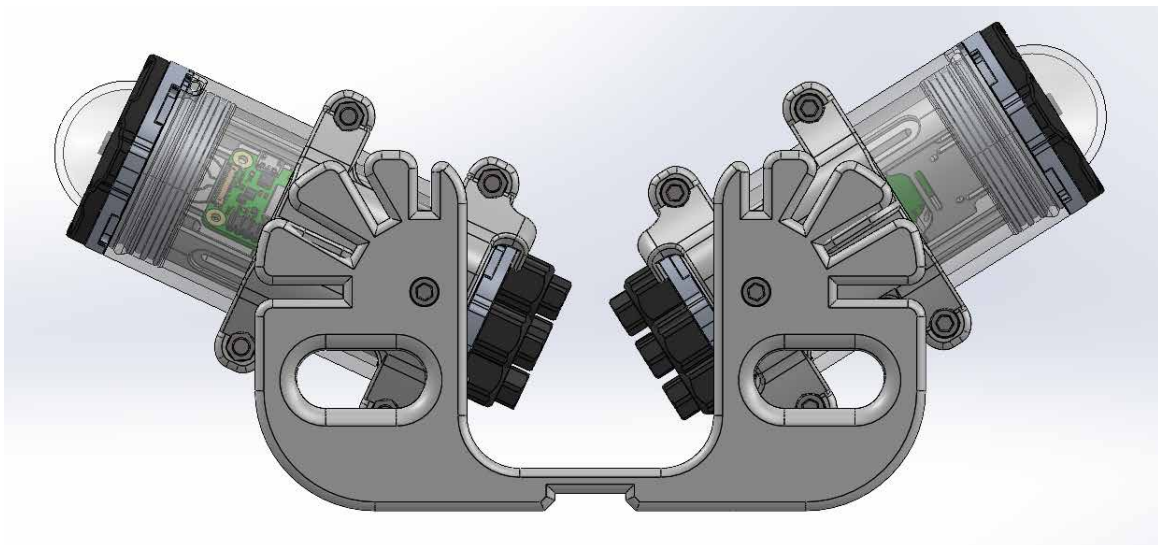


Figure 6.8: SolidWorks Assembly of two UWOS Module

The complete assembly, including two modules and all associated fasteners, weighs a total of 1.6 kg, which is slightly above the specifications but remains within the tolerance range.

6.3 Simulation

6.3.1 Static Pressure Test

Dual-Camera Support

In order to validate the design of the camera support, static pressure tests were performed using the online software *SimScale*. A load of 10 kg was applied along the $-\vec{y}$ direction on the groove at an angle of $\frac{\pi}{6}$ radians (see Figure 6.7), as this is the position most frequently used. Gravity was also modeled along the $-\vec{y}$ axis.



Figure 6.9: Simulation Front View

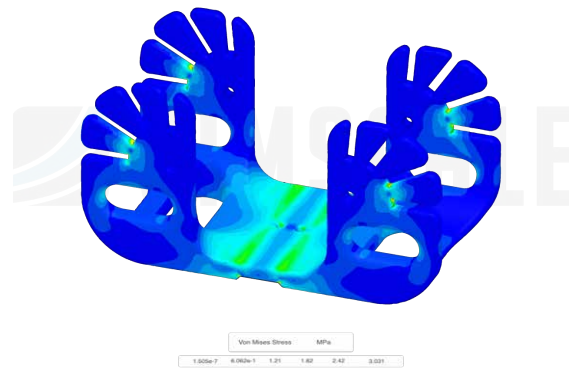


Figure 6.10: Simulation 3/4 Top View

The highest stress concentration is clearly located at the interface with the QHB tube. The maximum Von Mises stress reaches 3.031 MPa, which is well below the typical Young's modulus of a 3D-printed PLA part, approximately 50 MPa. We can therefore calculate the safety factor as follows:

$$\text{Safety Factor} = \frac{\text{Elastic Limit}}{\text{Applied Stress}} = \frac{50 \text{ MPa}}{3.031 \text{ MPa}} = 16.5 \quad (6.1)$$

Note

These simulations were conducted on a single-block PLA mesh, not on a laminated structure. However, they provide a useful order of magnitude estimate.

The integration with the QHB could still be modified to accommodate a wider fitting. The displacement of the part remains minimal. The primary purpose of this simulation was to confirm that the material removal does not compromise the structural integrity of the part.

To-Do

A potential improvement would be to perform a topological optimization of this part to remove even more material and further reduce its buoyancy.

No static simulations were performed on other parts, as they are not expected to be subjected to significant loads. If they were, it would imply that other components have already failed, which would be an undesirable situation.

6.3.2 Preliminary Hydrodynamic and Acoustic Considerations

A preliminary assessment of the hydrodynamic and acoustic effects induced by the device was considered, particularly to estimate the potential buoyancy generated by the module's geometry, as well as the possibility of acoustic disturbances such as "flute effect" resonances occurring in open cavities exposed to flow.

Initially, numerical fluid dynamics (CFD) simulations were planned to characterize these phenomena. However, due to time constraints, it was decided not to pursue these analyses immediately. This decision is based on two main considerations: firstly, the exposed surface area of the tube housing the sensor is significantly smaller than that of the main QHB tube, making its buoyancy negligible at the overall system scale; secondly, any potential acoustic disturbances will be directly observed during real-world testing and can be characterized retrospectively if necessary.

This pragmatic approach allows prioritization of available resources while leaving open the possibility of a more in-depth analysis in a future phase of optimization or multiphysics modeling of the system.

6.4 Assembly

6.4.1 UWOS Assembly

At first, only the components required to assemble a single complete module were printed, despite the support piece being designed to accommodate two modules. The parts were produced at the COSMER Laboratory.



Figure 6.11: Assembled Support and Clamp

Note**Material Choice: Justification for Using PLA for a Structure Submerged in a Marine Environment**

Poly(lactic acid) ($(C_3H_4O_2)_n$ (PLA)), a biodegradable polymer widely used in additive manufacturing, was selected for the construction of our prototype structure. This choice is based on several mechanical, environmental, and practical considerations, supported by recent studies. First, the work by Upadhyay, Mishra, and Kumar [22] demonstrated that PLA printed by fused filament deposition (FDM, or 3D printing) retains stable mechanical properties (elastic modulus of 1101 MPa, tensile strength of 52.96 MPa) after 30 days of immersion in a simulated seawater solution, without significant alteration of its microstructure or critical performance loss. Furthermore, tribological wear remains moderate, provided that surface roughness is limited by post-print polishing.

These results are corroborated by Menezes et al. [23] and Chaudhary et al. [24], who quantified a slow mechanical degradation of PLA immersed in saltwater, with a ductility loss below 1% over one week at temperatures near 15 °C. Moreover, the effects of the marine environment on crystallinity and mechanical strength become significant only after several weeks of exposure at temperatures above 30 °C. In our application, with the structure submerged at an average temperature of 15 °C for less than 7 days, all results point towards sufficient dimensional and mechanical stability.

PLA also offers the advantages of being easily printable, cost-effective, and compatible with complex geometries, making this material particularly relevant for rapid prototyping of short-duration submerged structures.

**(a)** Rails and Racks with PCBs**(b)** Pre-assembly of a Module**Figure 6.12:** UWOS Module Pre-assembly

For this first prototype, UWOS was assembled using the IMX708 sensor. The main reason was mechanical: the dimensions provided by the supplier did not match the received IMX462 component, which could not fit inside the 100 mm housing tube. In addition, preliminary tests indicated that the IMX708 produced significantly smaller video files compared to the IMX462 (see Chap. 8), which directly contributes to extending storage capacity and therefore deployment autonomy. Combined with its satisfactory image quality, these arguments motivated the choice of the IMX708 for the prototype stage.

Like the support and clamp, the internal rails and racks were 3D printed, but this time at the SMIoT laboratory. During assembly, it became apparent that space was tight, particularly due to the camera flex cable. Indeed, this cable must perform a full turn to be correctly oriented for the connector on the sensor.

Note

A new version of the rack was thus designed: by rotating the sensor PCB 180°, the flex cable now exits on the opposite side, properly aligned without twisting. This change significantly improves the mechanical durability of the system in the long term.

This adjustment also highlighted the space constraint within the module: despite careful CAD modeling, component integration remains very tight and allows practically no margin.

6.4.2 Complete Assembly

Before departing for Chile, we conducted a first full assembly of the system in Toulon to verify the waterproofness of all components.

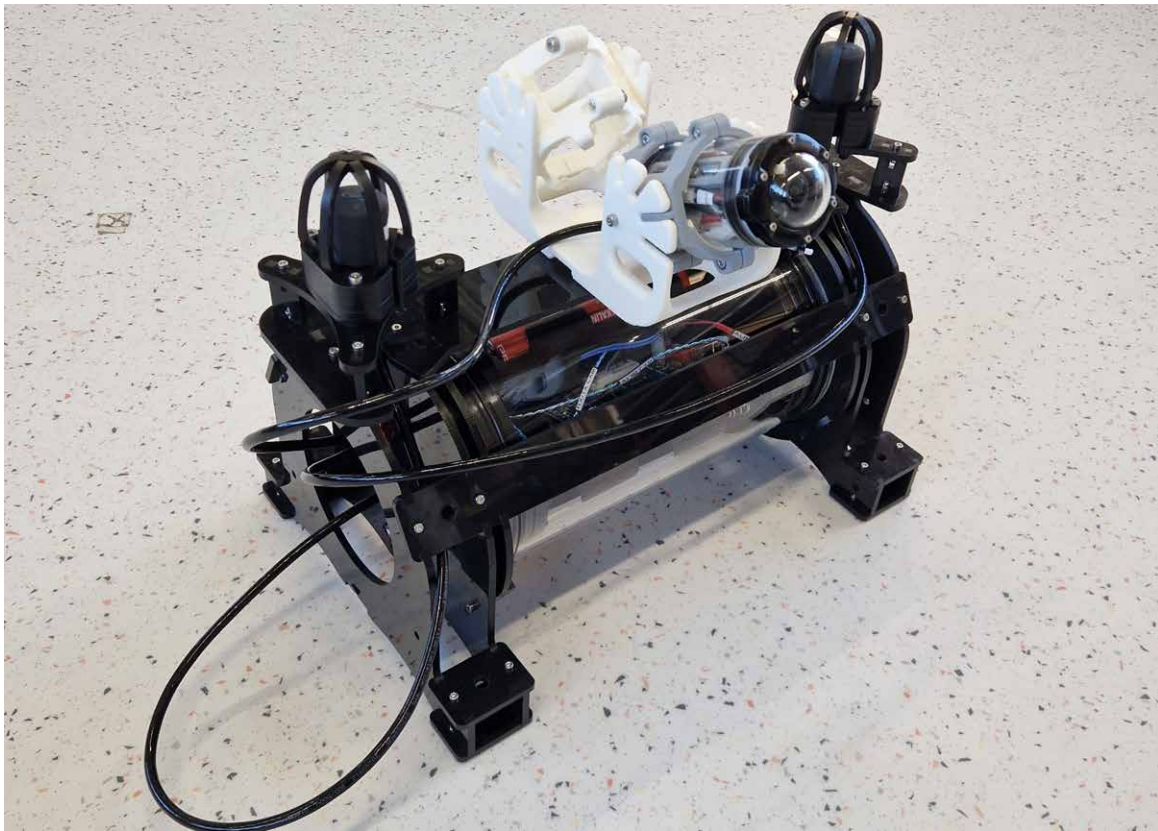


Figure 6.13: Complete Assembly

This test was crucial. Upon the first immersion, a leak was detected at the QHB tube. Fortunately, although all electronic components were already installed, no damage occurred. However, only a few hours before the departure to Chile, this naturally caused some stress.

We quickly undertook a thorough inspection of the housing with the SMIoT team, coated with lemon-scented soap to aid leak detection. The source of the problem was located: it was one of the penetrators, due to a residual chip from the laser cutting. This debris had inserted itself between the O-ring and the housing wall.

Ultimately, after cleaning and reassembling, the system was submerged overnight in the bathtub of our shared apartment for final validation. No further leaks were observed, allowing us to proceed with the field campaign with peace of mind.

Note

The QHB tube was specifically developed for this project and was only finalized the day before departure. This explains the limited testing conducted in Toulon; ideally, we would have performed much more extensive tests.

PART III

Software and Power Management

7

Embedded Software Initialization and Debugging

Given the autonomous nature of the device and the challenges associated with marine deployments—such as limited access, power cycles, and difficult retrieval conditions—the embedded software had to be both robust and self-reliant. The system must be able to boot, initialize hardware components, and start recording video data without user intervention.

To achieve this, a lightweight boot sequence was implemented using shell scripts and Python, designed to manage camera configuration, storage checks, timestamping, and file handling. A particular emphasis was placed on ensuring that the system would recover gracefully from interruptions and that it could log its status for post-deployment analysis.

This chapter outlines the boot process, the core logic of the recording routine, and the debugging strategies implemented through onboard logs and visual indicators.

7.1 Camera Initialization

The system allows the execution of custom programs at specific points during the Raspberry Pi startup and shutdown sequence. Commands placed in `beforeScript.sh` are executed before the schedule script, for instance to generate or modify it. Commands in `afterStartup.sh` run immediately after the schedule script, for example to act based on the scheduled shutdown time. Finally, commands in `beforeShutdown.sh` are executed just before shutting down the Raspberry Pi, for instance to send a notification or perform cleanup tasks.

7.1.1 Camera Driver Installation

The selected camera modules are not natively supported by the Raspberry Pi OS, and therefore require additional driver installation and configuration. The drivers are provided by the manufacturer and can be found at: <https://docs.arducam.com>.

After downloading and installing the appropriate driver package, the system configuration file must be edited to enable the correct overlay:

```
sudo nano /boot/firmware/config.txt
# Locate the section: [all]
# Add the following line according to the camera model:
dtoverlay=arducam-pivariety # or IMX708 if applicable
# Save the file and reboot.
```

Once the driver is installed and the configuration updated, the camera functionality can be tested with:

```
libcamera-hello
```

This command captures a short test video using default parameters, allowing verification of camera detection and basic operation.

7.1.2 Configuration File

To simplify system setup and avoid modifying the source code directly, a dedicated configuration file is used. This file is parsed at runtime using Python's `configparser` library, enabling easy adjustments to recording parameters without recompilation.

In addition to video settings, the file also contains QHB timing parameters for synchronisation between the acoustic and optical modules.

```
[Settings_UWOS]
output_file = islachanaral.h264
width = 1920
height = 1080
framerate = 30
codec = h264
bitrate = 10000000
nopreview = True
disable_display = True
mission_name = islachanaral_090825
[Settings_QHB]
Shutdown_Duration = 60
Preparing_Duration = 20
Recording_Duration = 540
Stopping_Duration = 10
```

This structure allows the recording behaviour to be fully customised, including resolution, frame rate, codec, bitrate, and mission naming, while keeping synchronisation parameters clearly separated from video settings.

7.2 Boot Sequence

To ensure complete autonomy of the system during sea deployments, the device was designed to automatically start video recording as soon as it powers on, without any manual intervention. This feature is crucial in marine environments where physical access to the instrument is limited or even impossible once deployed.

7.2.1 Main.py

The launch of the main program is automated via the `afterStartup.sh` script provided by the *Witty Pi 4 Mini* module. This script executes the `main.py` file located in the `/home/uwos/bin/` directory. The main script reads parameters from a configuration file (see Section 9), then initializes a state machine described in more detail later in the synchronization section.

Once the execution conditions are met, `main.py` launches a second script as a subprocess responsible for capturing video using the `libcamera` library. This decoupling separates the control logic (synchronization, states) from the capture process itself, allowing `main.py` to continue monitoring and logging statuses. Before starting the video recording, available space on the SD card is checked. If the space is insufficient, the script stops and schedules a shutdown in 45 seconds to save battery.

Note

The minimum available space threshold is configurable in the `main.py` source code and not in the `config.txt` file due to time constraints.

To-Do

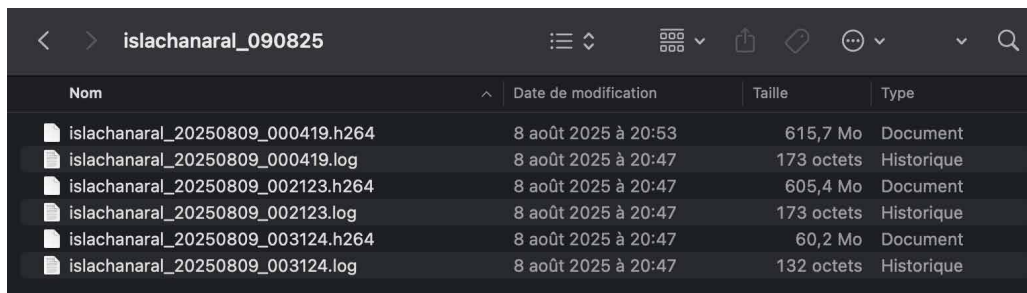
Currently, the absence of the camera is not automatically detected.

A log file is generated for each new mission to enable post-deployment monitoring. This file is saved in the directory:

/home/uwos/DATA/MISSION_NAME/logs.txt

This boot sequence ensures systematic and reproducible behavior at every power-up, making the device suitable for autonomous and extended measurement campaigns in marine environments.

Additionally, an XXXXX.log file is created for each video file. It stores the timestamp information of UART frame reception to ensure synchronization.



Nom	Date de modification	Taille	Type
islachanaral_20250809_000419.h264	8 août 2025 à 20:53	615,7 Mo	Document
islachanaral_20250809_000419.log	8 août 2025 à 20:47	173 octets	Historique
islachanaral_20250809_002123.h264	8 août 2025 à 20:47	605,4 Mo	Document
islachanaral_20250809_002123.log	8 août 2025 à 20:47	173 octets	Historique
islachanaral_20250809_003124.h264	8 août 2025 à 20:47	60,2 Mo	Document
islachanaral_20250809_003124.log	8 août 2025 à 20:47	132 octets	Historique

Figure 7.1: Example of a mission folder

This is what can be found inside a .log file, with synchronization management detailed in Chapter 9.

```
RECEPTION_UART_UTC: 2025-08-09 00:04:19.131346
START_REC_UWOS: 2025-08-09 00:04:19.132117
START_REC_QHB: 2025-08-09 00:04:39.123545
STOP_REC_QHB: 2025-08-09 00:13:39.008383
```

7.2.2 Automated launch via subprocess

As described above, the video capture is triggered from the main control script using Python's subprocess module. This approach allows the execution of system commands in a new process while dynamically passing parameters to the recording script rec.py, without having to modify its source code for each mission.

The following command is used:

```
subprocess.Popen([
    'python3',
    '/home/uwos/bin/rec.py',
    '--output_file', output_file,
    '--duration', str(recording_duration_UWOS),
    '--width', str(width),
    '--height', str(height),
    '--framerate', str(framerate),
    '--codec', codec,
    '--bitrate', str(bitrate),
    '--nopreview', str(nopreview),
    '--disable_display_flag', str(disable_display_flag)
])
```

Each argument is converted to a string if necessary, then passed to rec.py, which uses Python's argparse library to parse them. This mechanism makes it possible to adjust parameters such as resolution, duration, frame rate, or codec directly from the configuration file, without editing the recording script.

Once launched, `rec.py` builds the appropriate `libcamera-vid` command and automatically renames the output video file following the format (see Fig.7.1):

YYYYMMDD_HHMMSS.h264

All videos are saved in the `.h264` format (MPEG-4), the standard used for Blu-ray discs and many online video platforms.

Note

The `h264` format, also known as MPEG-4 AVC, is widely adopted due to its high compression efficiency and broad compatibility.

During testing, raw `.h264` files showed playback issues in VLC, such as resolution mismatches and dropped frames. After reviewing similar reports on user forums, it was found that playback using `FFplay` — a utility from the `FFmpeg`¹ package — resolved these issues.

To streamline post-processing, a dedicated script was developed to automatically convert all `.h264` files in a directory into `.mp4` format. The script's operation is detailed in Annex A, and its source code is available on GitHub: <https://github.com/JungMatthieu/UWOS>.

Algorithm 1 Recording a video with libcamera-vid

Data: Configuration file `config.txt` (optional)

Result: Video recorded with the given parameters

```

1 if configuration file exists then
2   | Read parameters: filename, duration, resolution, codec, etc.
3 end
4 else
5   | Use default values
6 end
7 Generate a filename with timestamp
8 if disable_display option is true then
9   | Disable graphical display
10 end
11 Build the libcamera-vid command with all parameters
12 Execute the command if error during execution then
13   | Display an error message
14 end
15 else
16   | Display a success message
17 end
18 if display was disabled then
19   | Re-enable display
20 end
```

7.3 Embedded Debugging Tools

SSH Connection

One of the advantages of the Raspberry Pi Zero 2W is its built-in Wi-Fi connectivity, which enables wireless remote access via SSH. This feature is particularly useful in field conditions where physical access to the device is limited.

¹An open-source suite for audio/video processing and conversion

For this project, a smartphone was configured as a mobile hotspot, allowing the Raspberry Pi to connect automatically whenever the hotspot is available. Once the Wi-Fi credentials are stored in the Raspberry Pi's network configuration, the connection is established without user intervention.

To perform on-site diagnostics, a terminal session can be opened directly from an Android device using the Termux application. This setup enables real-time verification of running processes, file creation, and overall system health — even in remote or offshore environments.

```

14:54 No More
0:00 gpio -g wfi 4 falling
root 798 0.0 0.0 0 0 ? S 17:51
0:00 [irq/199-wiringpi_gpio_irq]
root 974 0.2 2.2 16956 9928 ? S 17:53
0:00 python3 /home/uwos/bin/rec.py --output_file /home/uwos/
root 977 37.0 17.9 540904 78212 ? SL 17:53
0:28 libcamera-vid -t 82000 --width 1920 --height 1080 --fra
root 1102 0.3 1.9 14472 8508 ? Ss 17:53
0:00 sshd: uwos [priv]
uwos 1112 1.3 2.0 16304 8872 ? Ss 17:53
0:00 /lib/systemd/systemd --user
uwos 1113 0.0 0.8 36396 3772 ? S 17:53
0:00 (sd-pam)
uwos 1132 0.0 1.3 14732 5976 ? S 17:53
0:00 sshd: uwos@pts/0
uwos 1133 0.3 0.8 8452 3788 pts/0 Ss 17:53
0:00 -bash
root 1204 0.0 0.3 6660 1608 ? S 17:54
0:00 sleep 5
uwos 1209 200 0.9 11720 4072 pts/0 R+ 17:54
0:00 ps aux
uwos@raspberrypi:~$
Broadcast message from root@raspberrypi (Thu 2025-07-24 17:5
4:39 UTC):

The system will power off now!

Connection to 192.168.191.254 closed by remote host.
Connection to 192.168.191.254 closed.
~$
ESC / - HOME ↑ END PGUP
TAB CTRL ALT ← ↓ → PGDN

```

Figure 7.2: Remote access to the Raspberry Pi via SSH using Termux on Android

For example, to verify that the recording scripts are running, the following command can be used:

```
ps aux | grep python
```

This command lists all currently running Python processes on the Raspberry Pi. In Fig. 7.2, the `ps aux` output shows several active processes. The seventh line from the top corresponds to the `Libcamera` subprocess, confirming that a video recording is in progress.

8

Power Management & Optimization

Energy management is one of the most critical aspects of the UWOS design. The system must operate autonomously for several days without the possibility of external power supply or battery recharging, while remaining reliable once deployed in the field. This constraint makes rigorous optimization of energy consumption a key requirement from the outset.

To meet these challenges, the instrument was designed to alternate between deep sleep and active recording phases, thereby reducing unnecessary power draw while preserving data quality. This chapter presents the methods implemented to manage and optimize power usage, from hardware architecture to software scheduling.

8.1 Power Management - Cycle ON/OFF

Among the possible strategies for reducing power consumption, one option considered was the use of programmable ON/OFF cycles via the Witty Pi 4 Mini. However, due to time constraints and integration challenges, this functionality could not yet be fully implemented during the internship. The principle and its potential applications are nevertheless presented here, while the synchronization aspects related to the QHB are discussed in more detail in Chapter 9.

Witty Pi 4 Mini

The Witty Pi 4 Mini is an add-on board in the form of a hat that adds a real-time clock and power management capabilities (see Chap. 4). To operate these cycles, it requires a specific file named `schedule.wpi`:

```
# Generated by GUI
BEGIN 2025-04-26 00:00:00
END 2025-05-30 23:59:59
ON M15
OFF M15
```

This file contains all the necessary information to generate ON/OFF cycles for our system. Although this synchronization is no longer required in the current setup (see Chap. 9), it is important to highlight that such an adaptation remains technically possible if future cycles were added on the QHB side.

The cycles are therefore independent and overlap with the cycles of the QHB on which UWOS synchronizes to record, the startup (Startup) and scheduled shutdown (Shutdown) functionalities of the Witty Pi are fully exploited to ensure reliable and energy-efficient system management.

8.2 Power Optimization

To validate the energy and functional performance of our camera, a series of laboratory tests was implemented. The main goal is to evaluate the impact of different hardware and software configurations on energy consumption.

8.2.1 Experimental Design - Taguchi Method

The experimental protocol relies on a rigorous design of experiments allowing exploration of factor combinations: sensor type, GPU output activation, image resolution, frame rate, and simulated lighting conditions. Using a Taguchi L18 table, it is possible to significantly reduce the number of necessary experiments while maintaining a reliable analysis of the main effects.

All standard Taguchi tables are orthogonal tables. For the study of fractional factorial designs, it suffices to select a standard Taguchi table. The plan used is a Taguchi design $L18(2^1 \cdot 3^7)$, allowing us to test 1 factor at 2 levels and 7 factors at 3 levels in only 18 trials:

$$L_{18} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 3 & 3 & 3 & 3 & 3 & 3 & 3 \\ 1 & 2 & 1 & 1 & 2 & 2 & 3 & 3 & 3 \\ 1 & 2 & 2 & 2 & 3 & 3 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 & 1 \\ 1 & 3 & 2 & 3 & 2 & 1 & 3 & 1 & 2 \\ 1 & 3 & 3 & 1 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 & 2 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 & 1 & 3 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 1 & 1 & 3 & 1 \\ 2 & 2 & 1 & 2 & 3 & 1 & 3 & 2 & 1 \\ 2 & 2 & 2 & 3 & 1 & 2 & 1 & 3 & 2 \\ 2 & 2 & 3 & 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 & 1 & 3 & 2 \\ 2 & 3 & 2 & 1 & 3 & 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 2 & 1 & 2 & 3 & 2 & 1 \end{pmatrix} \quad (8.1)$$

We thus find our 18 rows corresponding to the 18 trials we will need to perform, and the 8 columns of the different factors. Columns 6 to 8 are not used at the moment. However, they can be made available for adding secondary factors such as room temperature, noise, etc. We therefore map the factors as follows:

Factor	Levels	Type	Taguchi Slot
Sensor	IMX462/IMX708	2 levels	Col1, A_i
Display	ON/OFF	2 levels	Col2, B_j
Resolution	720p/1080p/4K	3 levels	Col3, C_k
Frame Rate	15fps/20fps/30fps	3 levels	Col4, D_l
Lighting	Low/Medium/High	3 levels	Col5, E_m

Table 8.1: Power Consumption: L18 plan mapping

Factor 2, "Display", only has 2 states; to fit it into the table, we duplicate one of its states, the one most likely, which is "OFF". We can thus write the following equation, which allows estimating the consumption of a given configuration:

$$Y = \mu + A_i + B_j + C_k + D_l + E_m + \epsilon \quad (8.2)$$

With:

- Y : The measured response (here, consumption in W)
- μ : The overall average of the responses
- $A_i, B_j, \dots$: The effects of the factors at their levels $i, j, \dots$
- ϵ : The residual error (noise, imprecision, etc.)

We define μ :

$$\mu = \frac{1}{18} \sum_{r=1}^{18} Y_r \quad (8.3)$$

Setup, Method, and Results

To carry out this experimental design, a room where brightness can be controlled is required, as well as equipment to measure the power consumption of our system. Since no such room or equipment was available at LIS, the experiments below were conducted in the COSMER facilities thanks to *Charly Peraud*. The controllable light source used was a *GoDox VL150* spotlight, and the power supply was a *TENMA 72-10495 Digital-control DC Power Supply 0-30V 5AX2*.

Method

The power supply and the camera with the desired sensor are placed on a table. Opposite, on the wall at $240cm$, a photo of a fish is placed. The light source is then positioned at a 45° angle relative to the camera/fish axis:

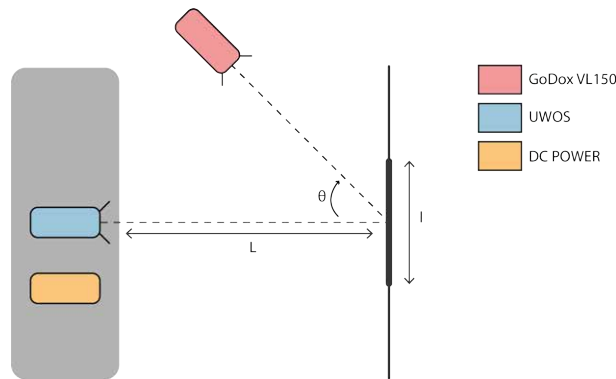


Figure 8.1: Method diagram – electrical consumption

With:

- θ : angle of incidence of the light source
- L : Distance between the camera and the wall (here $240cm$)
- I : width of the image on the wall (here $30cm$)



Figure 8.2: Photo of the setup

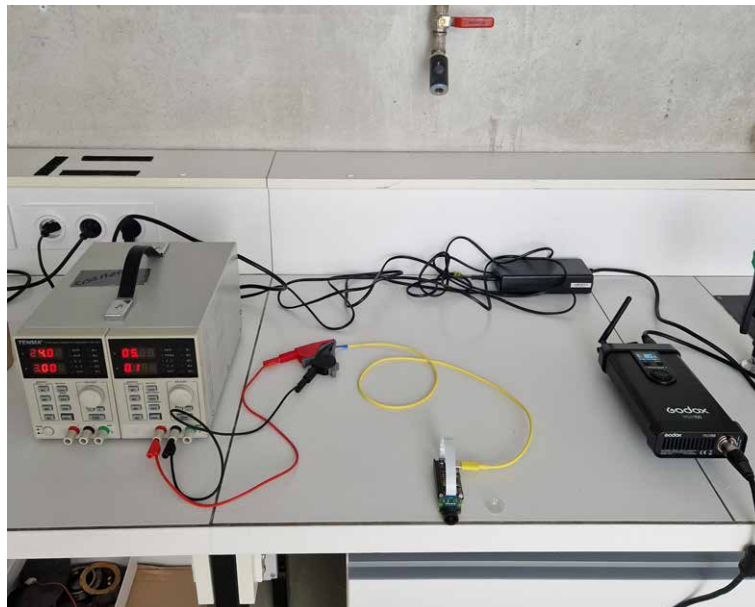


Figure 8.3: Experiment table

The power of the light source is adjusted from a control box (on the right in Fig. 8.3). Unfortunately, we did not have a lux meter, so the adjustment was made solely based on the percentage of maximum power (≈ 61000 Lux) displayed on the control box screen. The parameters chosen were: Low = 0%, Medium = 20%, High = 80%. The capture duration is the same for all videos and was arbitrarily set to 10s, allowing time to observe consumption while avoiding edge effects.

Note

At 0%, the source obviously emits no light; however, since the room is not completely sealed, some light passes under the door, and this residual light allows testing the low-light capabilities of the IMX462 sensor (see Fig.8.4).



Figure 8.4: Screenshot IMX462 with 0% brightness



Figure 8.5: Screenshot IMX708 with 0% brightness

Results

After performing the 18 experiments and compiling everything into a table (see Table.8.2), we calculate the factor averages to determine their impacts on the power consumption of our system (see Figure.8.6):

Run	Sensors	Display	Resolution	FPS	Light	Power (W)
1	IMX708	OFF	720p	15	20%	0.220A
2	IMX708	OFF	1080p	30	80%	0.340A
3	IMX708	OFF	1080p	15	0%	0.250A
...	...	...	...	...	...	...
18	IMX462	OFF	1080p	15	20%	0.270A

Table 8.2: Excerpt from the Experiment Table (see appendix B)

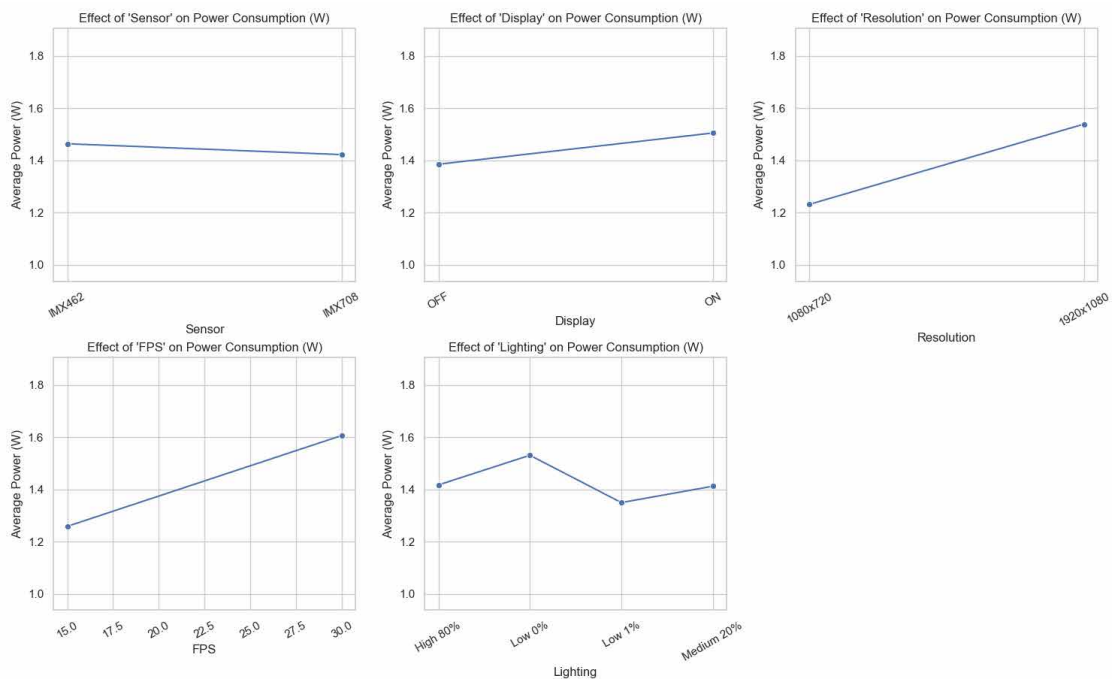


Figure 8.6: Study results of parameters on power consumption (W)

Several conclusions can be drawn from these graphs. The first is that the ambient brightness in which the video is taken seems to have no impact on electrical consumption. Next, no surprising information

appears. Indeed, as expected, consumption increases with a higher FPS count and similarly increases with resolution. However, the framerate seems to have a slightly greater impact than resolution. Thus, we can lean towards the following capture parameters:

- Resolution: Full HD 1920x1080
- Sensor: to be verified with sea tests
- Framerate: 15 FPS
- Display: OFF

It is worth recalling that these values correspond to the consumption of a single module; therefore, for a 360° camera, consumption should roughly double, though this remains to be verified in tests.

One of the key result of these measurements is that the electrical consumption of the UWOS remains stable, in the range of 1.2–1.6 W, regardless of recording conditions. This is a fundamental point, as it guarantees that the system does not “drop out” and that its energy budget can be reliably estimated.

Note

The power consumption in deep sleep mode achieved thanks to the Witty 4 is less than 0.001A, which corresponds to less than 0.005W! By comparison, the QHB consumes 1.8W during recording.

Note

However, be aware that when reducing resolution, the full-format image is "cropped" (see Figure.8.7).



Figure 8.7: Screenshot IMX708 at 1080x720p



Figure 8.8: Screenshot IMX708 at 1920x1080p

We can also note the significant difference in average file size between the two sensors:

$$\begin{aligned} S_{avg,IMX708} &= 1.72 \text{ MB} \quad \text{or} \quad 0.172 \text{ MB/s} \\ S_{avg,IMX462} &= 9.31 \text{ MB} \quad \text{or} \quad 0.931 \text{ MB/s} \end{aligned}$$

This corresponds to an increase of:

$$\frac{S_{avg,IMX462} - S_{avg,IMX708}}{S_{avg,IMX708}} = \frac{9.31 - 1.72}{1.72} \approx 441\%$$

However, these theoretical averages must be interpreted with caution. Field experiments showed that file size depends strongly on ambient light and scene dynamics, with differences of up to a factor 20 between day and night (see Chap. 14). For instance, the IMX708 produced about 600 MB for 9 minutes of daytime video compared to only 20 MB for 9 minutes recorded at night !

9

Synchronization Protocol

Managing synchronization between the two electronic boards is a fundamental issue. If one board is recording while the other is turned off, the instrument loses its purpose. One of the main challenges is to determine the recording duration in advance, since it is not possible to start an unlimited recording and simply shut down the Raspberry Pi afterwards.

This task was mostly handled in Chile, although initially planned to be completed in France. Tight deadlines and accumulated delays were the main reasons for this shift.

9.1 Use of the Witty Pi 4 Cycle

As mentioned in the previous chapter, the first approach relies on estimating the recording duration using the startup and shutdown timestamps stored in the Raspberry Pi system logs. The program extracts the last lines of the log to identify the next scheduled shutdown and startup events, and from this it computes the allocated video recording time.

This method is relatively flexible, as it allows for dynamic adjustment of recording time based on the planned cycles. However, it has a major drawback: there is no real interaction with the QHB (the audio board). UART frames from the QHB are only logged, without any direct exchange.

Even if the two RTC clocks show only a small drift, this setup does not guarantee precise synchronization between audio and video recordings. Furthermore, since it relies solely on initial synchronization, the solution is not robust against potential runtime anomalies.

9.2 Synchronization Based on QHB Status

It was therefore decided to implement synchronization based on the status information transmitted through the UART interface. It is important to note that no modification of the QHB firmware or behavior was possible; hence, the solution had to be fully adapted to the existing setup. We first describe the UART frame format transmitted by the QHB via UART2.

UART Frame

The QHB sends a status frame via UART2 every second. This protocol is described in the official QHB documentation [Barchasz and Gies \[25\]](#), and is configured as follows:

Data	Unit	Hexa	Explanation
SoF	UInt8	0xFE	Start of Frame
MSB_CMD	UInt8	0XX	Most significant byte of command ID
LSB_CMD	UInt8 file	0XX	Least significant byte of command ID
MSB_DataLength	UInt8	0XX	Most significant byte of PayloadLength
LSB_DataLength	UInt8	0XX	Least significant byte of PayloadLength
Payload	DataLength x UInt8	0XX	Payload Datas
Checksum	UInt8	0XX	Checksum =XOR bitwise across the entire frame except the checksum

Table 9.1: UART2 protocol, QHB

The serial link is configured with the following parameters:

```
# === Configuration UART ===
ser = serial.Serial(
    '/dev/ttyAMA0',
    baudrate=115200,
    bytesize=serial.EIGHTBITS,
    parity=serial.PARITY_NONE,
    stopbits=serial.STOPBITS_ONE,
    timeout=1
)
```

The Payload field indicates the current status of the QHB:

0x00: SHUTDOWN_In_Progress	0x05: DATA_SEND_In_Progress
0x01: PREPARING_In_Progress	0x06: Preparing_For_DATA_SEND
0x02: RECORDING_In_Progress	0x07: DATA_SEND_Finished
0x03: STOPPING_In_Progress	0x08: IDLE_In_Progress
0x04: CNN_DETECTION_In_Progress	

The idea is therefore to exploit these status updates to ensure synchronization, with a precision limited by the transmission rate (1 Hz). This provides a temporal granularity of about one second, which is acceptable as a first implementation.

Note

At this stage, it is estimated that the species likely to pass in front of the camera will not move at speeds where a one-second temporal uncertainty would compromise the analysis.

To-Do

It would be interesting to evolve the UART protocol so that, in addition to the periodic status frame, the QHB sends a dedicated message signaling the exact start of recording. This would reduce temporal uncertainty to the transmission and processing delay of a single message, i.e., well below one second.

GRAFCET Modeling

To guarantee precise synchronization between QHB and UWOS, a sequential model of their operation was designed using GRAFCET. This representation makes it possible to clearly visualize internal states, waiting times, and critical transitions relevant to acoustic synchronization.

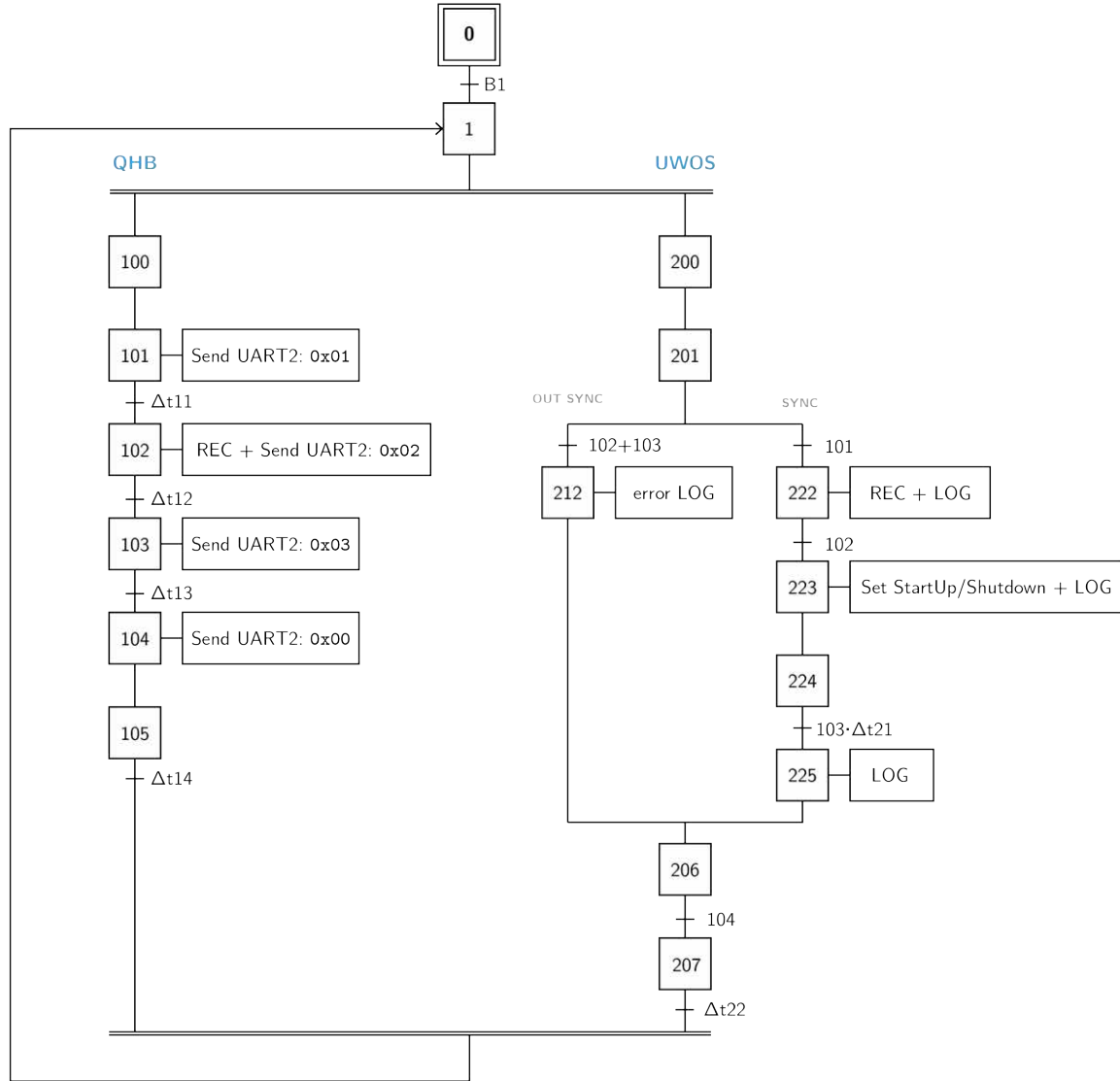


Figure 9.1: GRAFCET diagram for UWOS–QHB synchronization

The following tables summarize the different states and transitions. Table 9.2 presents the functional states of the synchronization process:

Before presenting the transitions, we recall a section of the JConfig.CFG file of the QHB, which defines execution phase durations:

```

Shutdown_Duration=60; # Time period of wait time between each record (in seconds)
Preparing_Duration=5; # Time to boot and prepare Pic32 (SD Card or HDD differs)
Recording_Duration=120; # Recording time period (in seconds)
Stopping_Duration=5; # Time to stop Pic32 (SD Card or HDD differs)
  
```

These durations are used as time intervals between states, and thus define the transitions in the GRAFCET. The correspondences are as follows:

Synchronization States		Unit	Action
0	Initial step	–	–
1	Power on	–	–
100	Initialization	QHB	–
101	PREPARING in progress	QHB	TX UART
102	RECORDING in progress	QHB	REC + TX UART
103	STOPPING in progress	QHB	TX UART
104	SHUTDOWN in progress	QHB	TX UART
105	Turned off	QHB	–
200	Initialization	UWOS	–
201	Waiting for: 0x01	UWOS	–
222	Start RECORDING + LOG	UWOS	REC + LOG
223	Waiting for: 0x02	UWOS	Configure next setup & shutdown + LOG
224	Waiting for: 0x03 & Δt_{21}	UWOS	–
225	End RECORDING, final LOG	UWOS	LOG
212	Sync error (timeout or misalignment)	UWOS	Error LOG
206	Waiting for: 0x00	UWOS	–
207	Turned off	UWOS	–

Table 9.2: Main functional states of the QHB–UWOS synchronization GRAFCET

$$\Delta t_{11} = \text{Preparing_Duration}$$

$$\Delta t_{12} = \text{Recording_Duration}$$

$$\Delta t_{13} = \text{Stopping_Duration}$$

$$\Delta t_{14} = \text{Shutdown_Duration} - \Delta t_{11} - \Delta t_{13}$$

For example, with the current configuration:

$$\Delta t_{14} = 60 - 5 - 5 = 50 \text{ s}$$

Transition	Explanation
B1	Power-on switch
100	Initialization
101	PREPARING in progress, first reception of 0x01
102	RECORDING in progress, first reception of 0x02
103	RECORDING in progress, first reception of 0x03
104	SHUTDOWN in progress, first reception of 0x00
Δt_{11}	Preparing_Duration
Δt_{12}	Recording_Duration
Δt_{13}	Stopping_Duration
Δt_{14}	Shutdown_Duration – Δt_{11} – Δt_{13}
Δt_{21}	Recording_Duration (not in the same file)
Δt_{22}	Duration automatically determined between shutdown and startup upon reception of 0x02
Δt_{ϵ}	Recording margin

Table 9.3: Transition overview of the QHB/UWOS synchronization GRAFCET

All of these transitions define the logical behavior of the two systems. To better understand their real dynamics, it is useful to represent their development on a time scale.

Time Flow

To complement the formal Graftet modeling, we present several scenarios illustrating the time flow of the synchronization phases between the QHB and the UWOS. These representations aim to concretely visualize the sequence of states, the times associated with each phase, as well as any errors that may occur during the process.

Two types of scenarios are presented:

- Standard scenarios showing nominal operation with power-on and synchronized recording (Fig. 9.2a).
- Scenarios with intentionally introduced synchronization errors, used to analyze the system's robustness and limitations (Fig. 9.2b).

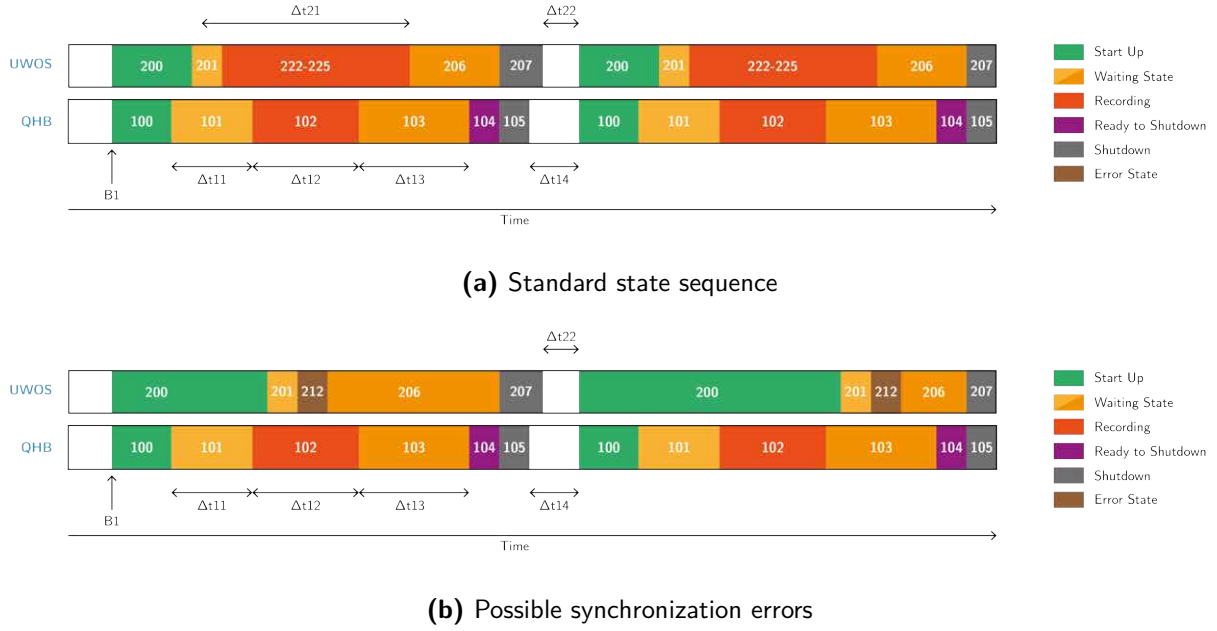


Figure 9.2: Temporal synchronization stages between QHB and UWOS

To guarantee reliable synchronization between the QHB and the UWOS, it is essential that the UWOS does not stop recording before the QHB. Such a mismatch would compromise the temporal overlap required for proper correlation between the two instruments.

To avoid desynchronization, we impose the following timing constraints:

$$\Delta t_{13} > \Delta t_{11} > 10 \text{ s}$$

$$\Delta t_{21} = \Delta t_{11} + \Delta t_{12} + \Delta t_{\epsilon}$$

For instance, if UWOS recording starts at the beginning of phase 101, Δt_{21} must ensure that the recording continues until the complete stop of the QHB recording. The margin Δt_{ϵ} is therefore indispensable.

Conversely, if the UWOS recording is triggered late, at the end of phase 101, a phase 103 that is too short could cause both systems to stop simultaneously before the QHB recording ends, leading to partial data loss or file corruption.

Thus, the recording margin Δt_{ϵ} plays a critical role in system reliability, ensuring sufficient overlap regardless of timing variations.

UWOS Software Management

The embedded software continuously analyzes frames received via UART from the QHB. Based on the received statuses, the system automatically triggers or schedules different actions (recording, shutdown, restart), thereby ensuring robust synchronization between the two modules. The operation is described in the pseudo-code below:

Algorithm 2 UART frame processing and recording cycle management

Input: UART frames containing status codes

Output: A recording started and a restart cycle scheduled

```

1 Initialize serial communication
  Define cycle durations (preparing, recording, stopping, shutdown)
  Initialize indicators recording_in_progress, next_shutdown_done, next_startup_done

2 while program is running do
3   Read a UART frame

4   if frame is invalid then
5     Skip to the next one
6   end
7   Extract status from the frame

8   if status = 1 (PREPARING) then
9     if recording is not in progress then
10      Start the recording script
11      Set recording_in_progress to true
12    end
13  else if status = 2 (RECORDING) then
14    if restart cycle not yet scheduled then
15      Compute time_shutdown = now + recording duration
16      Compute time_startup = time_shutdown + shutdown duration
17      Schedule shutdown via WittyPi
18      Schedule restart via WittyPi
19      Update indicators
20    end
21  end
22 end

```

Special care was taken for error handling. If a received frame is invalid (e.g., wrong format, CRC error, empty frame), it is simply ignored to prevent unintended actions. This mechanism ensures that UWOS never triggers critical processes based on corrupted data.

Note

For further implementation details, complete source codes (QHB, UWOS, synchronization, logging, etc.) are publicly available on GitHub: <https://github.com/JungMatthieu/UWOS>.

10

Graphical User Interface

A clear and intuitive user interface is a crucial aspect of any scientific instrument, as it directly conditions how easily researchers can operate the system, retrieve data, and react to unexpected events during deployments. At the same time, there is a fundamental contradiction: it is nearly impossible to design a truly functional interface without extensive field trials, discussions with users in real conditions, and direct observation of potential problems or system failures.

In this context, the work presented in this chapter should be seen as a first step. Initial investigations were carried out to explore possible interface designs and to identify suitable tools and frameworks. However, a fully operational interface can only be developed in later stages of the project, once sufficient feedback from real deployments and user interactions has been gathered.

10.1 Objectives

To simplify the use of UWOS and avoid repeatedly exploring the different configuration files for the camera, power management, and other modules, we developed a Python script that gathers all this information into a single interface. The main objective of creating a Graphical User Interface (GUI) was to provide a simple, intuitive, and accessible tool for configuring and controlling the onboard optical module. This interface was designed to offer:

- configuration of video parameters (duration, resolution, framerate, codec, etc.);
- configuration of audio parameters (duration, sampling rate, etc.);
- scheduling of ON/OFF cycles for the Witty Pi 4 Mini (not yet fully implemented);
- transfer of configuration files via SSH;
- import of recorded video files via SSH;
- automatic conversion of H264 files into standard MP4 format;
- automatic synchronization of audio and video files;
- automatic generation of the `schedule.wpi` file for the Witty Pi.

10.2 Prototype Version

Video Conversion Software

The first program developed with Tkinter focused on converting video files from H264 to MP4 using FFmpeg. Without this step, it was impossible to properly play back the videos recorded by the camera. A

standalone executable (.exe) was created for Windows users. More details about this software can be found in Appendix A.

Version 1

The first complete graphical interface was then implemented using PyQt5. It integrated nearly all the functions specified in the objectives. However, after discussion with my supervisor, we concluded that this interface was not sufficiently user-friendly and lacked key information, such as the estimated recording duration for a given configuration. This information was displayed in the interface screenshot but not yet functional in the code.

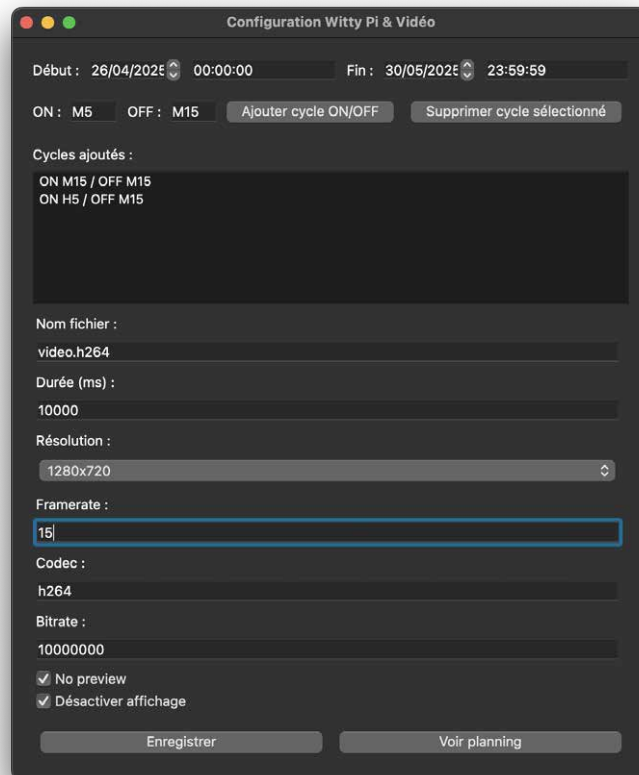


Figure 10.1: Configuration Software (GUI prototype)

Implemented Features

This prototype version integrated the following key functionalities:

- Input fields for video capture parameters (duration, resolution, framerate, codec, etc.);
- Automatic generation of the `schedule.wpi` configuration file;
- Secure transfer (`scp`) script for retrieving videos;
- Automatic conversion of H264 → MP4 using `ffmpeg`;
- Automated writing of the `afterStartup.sh` script to launch recordings;

Note

A user guide is available here: <https://github.com/JungMatthieu/UWOS>. It explains how to use the QHB recorder in combination with UWOS, from assembly and wiring to configuration files.

The interface also allows scheduling ON/OFF cycles, defining alternating sequences as follows:
In the ON or OFF section, the user specifies the duration of the state using time units:

- D = Days (e.g., D2 means 2 days)
- H = Hours (e.g., H3 means 3 hours)
- M = Minutes (e.g., M15 means 15 minutes)
- S = Seconds (e.g., S30 means 30 seconds)

Tip

For example, to define an ON state lasting one hour and 30 minutes, you can write:

```
ON H1 M30
```

Note

This software must be run from an external computer where the Raspberry Pi SD card is mounted, since the Raspberry Pi does not have a graphical desktop environment capable of executing this script directly.

Although codec and bitrate parameters can be modified, the default values are optimized for best performance. Similarly, preview and display deactivation options (see Chapter 8) are present but still relatively simple. The figure below illustrates the basic functionality of our GUI:

Algorithm 3 UWOS Configuration Interface

Data: Optional configuration files `config.txt` and `schedule.wpi`**Result:** Updated configuration files and ON/OFF scheduling saved

```
1 Create input fields: start/end dates, ON/OFF cycles, video options Initialize the cycle list
2 if config.txt AND schedule.wpi exist then
3   | Read parameters: filename, duration, resolution, framerate, codec, bitrate, options Read start/end
    | dates and ON/OFF cycles Update interface fields accordingly
4 end
5 if user clicks "Add cycle" then
6   | Add selected (ON, OFF) cycle to the list
7 end
8 if user clicks "Delete cycle" then
9   | Remove selected cycle from the list if no cycle is selected then
10  | | Display an error message
11  | end
12 end
13 if user clicks "Save" then
14   | Save schedule.wpi with BEGIN, END, ON/OFF values Save config.txt with all parameters if
    | success then
15   | | Display confirmation message
16   | end
17 end
```

10.3 Graphical Redesign

To overcome the limitations identified in the Tkinter prototype, a complete redesign of the graphical interface was undertaken. This new version is based on the PyQt5 framework, which enables the creation of modern, modular, and more robust applications. It also provides enhanced visual customization, improved ergonomics, and better compatibility with high-resolution screens and touchscreen devices.

Figma Design Phase

Before starting the technical migration, a complete mockup of the application was designed using the collaborative tool Figma. This step allowed us to structure the different interface views efficiently and to ensure an optimized user experience for field operations. The main objectives of this phase were:

- Define a clear hierarchy of information and interactions;
- Optimize ergonomics for field use (large clickable areas, readable fonts, high contrast);
- Plan navigation across views (configuration, scheduling, transfer, logs, etc.);
- Facilitate translation of design elements into PyQt5 widgets.

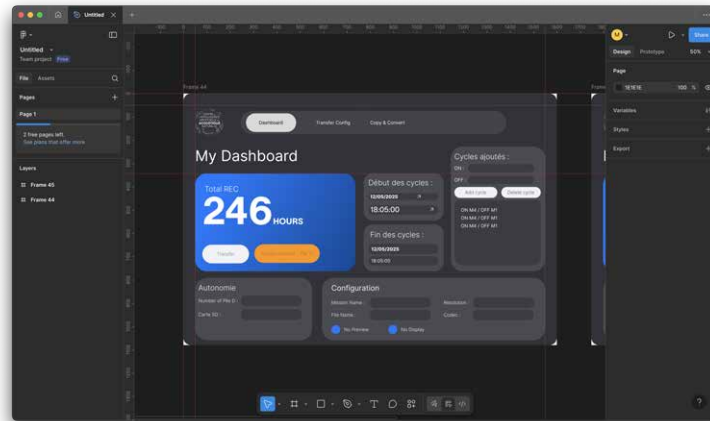
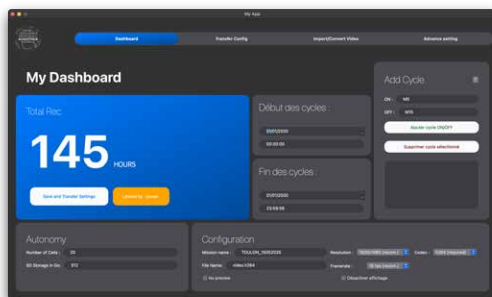


Figure 10.2: Figma prototype of the GUI

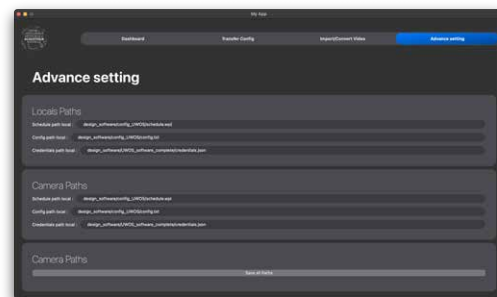
The Figma mockup thus served as a direct reference for the development of the final interface.

Implementation with PyQt5

The PyQt5 version of the application retained all the functionalities of the initial prototype while adding the missing improvements, namely: **autonomy estimation** and a truly **user-friendly interface**.



(a) Main Configuration Tab



(b) Path Management Tab

Figure 10.3: Screenshots of the final application

The final interface is fully autonomous, does not require an internet connection, and is designed also for field use, with optional touchscreen control.

Software Architecture

The development with PyQt5 required a clear structuring of the code and its modules. As the application grew in complexity, it became essential to define a rigorous project tree, illustrated in Figure 10.4.

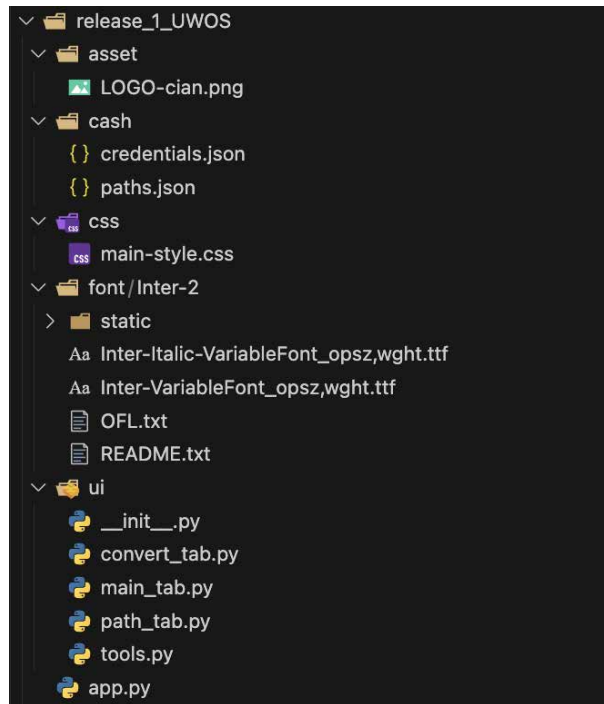


Figure 10.4: Software project structure

The architecture is centered around a main file, `app.py`, which manages the different interface modules and relies on several dedicated directories:

- `ui/`: contains the layout files as well as the core interface functions;
- `css/`: groups the style sheets used for visual customization;
- `cache/`: stores SSH credentials and recently accessed paths.

This modular separation improves code readability, facilitates future developments, and allows for the reuse of individual components in other projects.

To-Do

At the time of writing, the final application is not yet fully functional. For more details on the current status and the source code, the reader is invited to consult the GitHub repository associated with the project.

Note

The development of this graphical interface could, on its own, be the subject of a complete technical report. However, since this work was not the core objective of the internship, this section is limited to a general overview of its architecture and main functionalities.

PART IV

Field Mission in Chile – Experimental Deployment

11 Fieldwork Context and Scientific Collaboration

This chapter presents the objectives and collaborative framework of the field mission in Chile. Beyond the purely technical validation of the UWOS prototype, the deployment was embedded in a broader interdisciplinary and human context. It involved close interactions with local universities, NGOs, and field actors, as well as the discovery of the ecological and social specificities of the study sites. These exchanges provided essential insights for adapting the project to local conditions and for ensuring that the instrument design addressed real scientific and conservation needs. The following sections describe the scientific objectives pursued during the mission, the associated logistical and environmental considerations, and the interdisciplinary collaborations that shaped this experience.

11.1 General Context

This mission in Chile is part of an international CNRS research project (IRP), led by Hervé Glotin, Julie Patris, and Ivan Hinojosa. The main objective, as detailed in Section 4, is the in-depth study of Narrow-Band High-Frequency (NBHF) dolphin species.

However, the instrument developed during this internship has a broader application potential and can be employed in other scientific contexts. Before deploying it in long or complex campaigns dedicated to the study of NBHF dolphins, it was deemed relevant to test it in a more controlled environment, particularly within the framework of studies conducted by Dr. Ivan Hinojosa.

The coastal region of Concepción provides an ideal testing ground: it is rich in endemic marine species and hosts a university active in marine sciences, with associated laboratories and a seaside biological station. This environment supports both the finalization of the instrument and the implementation of experiments aimed at assessing its performance and limitations.



Figure 11.1: Mission stopover points in Chile

Concepción – Lenga Biological Station

The Lenga Biological Station is located on the Pacific coast, in San Vicente Bay, near Concepción. It is managed by the Universidad Católica de la Santísima Concepción (UCSC), which conducts various research projects there as well as science outreach activities, particularly for school audiences.



Figure 11.2: Lenga Biological Station

This site plays a central role in the mission, thanks to its facilities that allow both controlled tank immersions and real marine environment tests.

Caleta Chañaral de Aceituno

Located in northern Chile, the Caleta Chañaral de Aceituno region holds major interest for the study of marine ecosystems due to its immediate proximity to the Chañaral Islands National Reserve, a protected area belonging to the Humboldt Archipelago. This site hosts remarkable biodiversity, including colonies of sea lions, fur seals, and various pelagic cetaceans such as migratory whales, regularly attracted by the

local productivity. In addition, the preservation measures implemented over the past two decades have contributed to maintaining the integrity of the kelp forest (“bosque de algas”). This ecosystem plays a crucial role in supporting a wide diversity of fish and invertebrate species, especially rocky reef fishes, which are emblematic of the area. The coexistence of protected habitats, high marine productivity, and active local management makes Caleta Chañaral de Aceituno a unique site for both acoustic and visual observations under natural conditions.



Figure 11.3: Caleta Chañaral

11.2 Logistics & Ecological Footprint

Reflections on the Ecological Footprint of Scientific Transport

In order to ensure consistency between research objectives and contemporary environmental challenges, the issue of transportation was critically addressed from the earliest stages of mission preparation. An alternative to air travel was considered, particularly through the use of decarbonized maritime transport. However, scheduling constraints imposed by the different partner institutions (CNRS, UCSC, etc.) did not allow such a solution to be implemented for this specific mission.

Despite this, initial contacts were established with sailing cargo companies such as Zéphyr et Borée (based in Lorient) and Grain de Sail (Saint-Malo). The prospect of partnerships between these innovative private structures and research institutions opens up interesting perspectives for the low-carbon transport of scientific equipment.

At a time when globalization largely depends on maritime flows, a return to sail transport for certain goods represents an opportunity to rethink logistics chains. It thus seems relevant to consider a synergy between these sustainable transport actors and the scientific community, in order to address the environmental challenges of the 21st century.

This reflection is rooted in a personal conviction: doing science in no way exempts us from ecological awareness in our professional practices, including logistics.



(a) Grain de Sail 2



(b) Canopée – Zéphyr et Borée

Figure 11.4: Examples of sailing cargo vessels**Note**

The estimated carbon footprint of this internship is calculated in Section V, Chapter 17.

Transport of Equipment from France

All instrumentation, including the UWOS and the QHB board, was transported in my personal luggage. Despite a total weight of nearly 10 kg, this choice simplified customs procedures, reduced logistical costs, and ensured that the equipment arrived on time without jeopardizing the field mission.

11.3 Interdisciplinary Approach and Local Engagement

During this mission in Chile, I had the opportunity to move beyond the strictly technical scope of the project by meeting local actors committed to protecting life. These exchanges, at the intersection of science, civil society, and environmental issues, enriched my understanding of the broader context in which marine scientific research takes place in Chile.

11.3.1 Meeting with Civil Society Actors: Camila – Concepción**Background and Research Themes**

Camila Calderon-Quirgas is an author, environmental activist, and manager of the independent publishing house *Ermitaño Book*. She is also the founder of the feminist association *Soplo a la Vista*, based in the Concepción region, which advocates for cetacean protection through an intersectional approach combining ecology, feminism, social justice, and cultural heritage. Her work lies at the crossroads of science communication, youth awareness, and environmental advocacy.

Interdisciplinary Exchange

Our meeting provided an opportunity to discuss the many dimensions of marine conservation in Chile and the crucial role played by civil society in protecting biodiversity. Thanks to her dual editorial and activist perspective, Camila develops accessible narratives around marine biology, environmental history (such as the legacy of whaling), and local ecological resistance. Her approach helped me better understand the challenges of knowledge transmission beyond academic circles, particularly through publishing and artistic practices.

Caleta Chome and Cetacean Observation

As part of her association's activities, Camila regularly organizes marine observation outings. Together, we visited Caleta Chome, a small cove located on the Hualpén Peninsula, formerly the site of a now-

defunct whaling factory. This location, with its strong historical and ecological significance, is a well-known observation point for marine mammals.

The area is influenced by a submarine canyon close to the coast, which generates nutrient upwelling that fosters a high concentration of plankton and krill, attracting numerous marine species. During our outing, we were fortunate to observe two fin whales (*Balaenoptera physalus*) using binoculars and a camera.



(a) Fin whale – 16/07/2025



(b) Camila, at Chome Point

Figure 11.5: View of Caleta Chome with Camila

11.3.2 Scientific Collaboration and Field Observation with Mauricio — Valdivia

Background and Research Topics

Mauricio Soto Gamboa is a researcher at the Universidad Austral de Chile, specializing in coastal marine mammals. He has collaborated several times with Julie Patris and Franck Malige. His work focuses mainly on populations of Chilean dolphins (*Cephalorhynchus eutropia*) and Peale's dolphins (*Cephalorhynchus australis*), with particular attention to their spatial distribution, behavior, and the anthropogenic pressures threatening their habitats. These research topics directly align with the objectives of the project.



Figure 11.6: Universidad Austral de Chile — Isla Teja Campus, Valdivia

Meeting and Interdisciplinary Exchanges

During two days spent in Valdivia, I had the opportunity to engage in in-depth discussions with Mauricio about his research protocols. He presented his dolphin monitoring devices, acoustic and visual detection methods, as well as the specific constraints of the Chilean coastal environment: currents, background noise, and human activities (shipping, artisanal fishing). We discussed the challenge of detecting NBHF signals in shallow and dynamic waters, which fully justifies the use of a flexible and lightweight instrumental approach.

I was also able to attend a laboratory meeting, a valuable moment in which a team member presented their work to the group. These interdisciplinary exchanges allowed me to situate my project within an active scientific community where engineering, acoustics, and ecology are closely interconnected.

At-Sea Excursion and Cetacean Observation

On the second day, we embarked on a 4-meter zodiac for a sea excursion off Valdivia, in the gulf of the same name. This area is of major ecological interest for cetacean research: the semi-enclosed configuration of the gulf, the proximity of estuaries and upwelling zones, and the nutrient-rich environment create highly favorable conditions for observing coastal dolphins and other marine species. Several studies have documented resident populations of Chilean dolphins (e.g., [Malige et al. \[26\]](#)), as well as seasonal occurrences of orcas, humpback whales, and pilot whales.

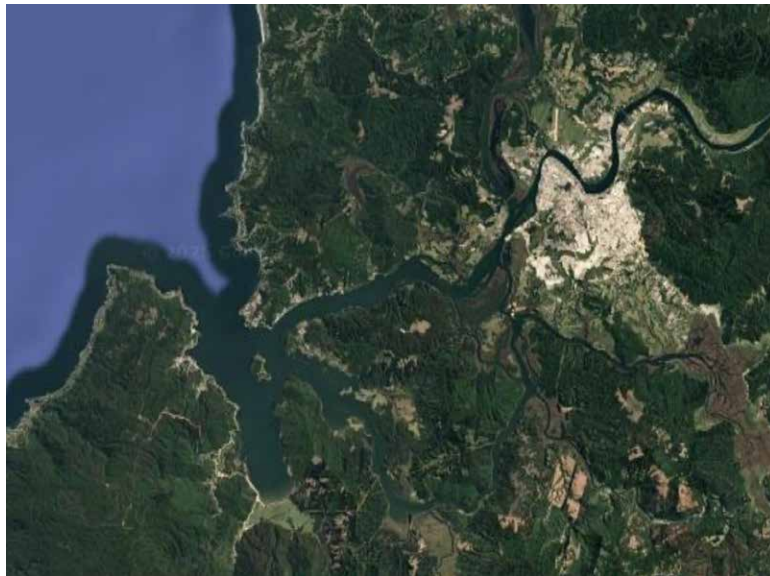


Figure 11.7: Satellite view of the Gulf of Valdivia

Despite rough sea conditions (with waves close to two meters), we patrolled the bay for about four hours, scanning the surface with binoculars and a camera. Although no dolphin groups were identified that day, we observed several groups of Magellanic penguins (*Spheniscus magellanicus*), a colony of South American sea lions (*Otaria flavescens*), and two large dorsal fins spotted in the distance. Mauricio suspected that they belonged to a pair of orcas, but we were unable to approach before they dived.

Beyond the observations themselves, this outing allowed me to gain a very concrete understanding of the field conditions under which monitoring instruments are deployed. It also reinforced the value of having a compact, robust, and rapidly deployable device such as the one we developed.



(a) Launching the zodiac from the fishing port



(b) Observation break and convivial moment on board

Figure 11.8: Pictures of the sea outing with Mauricio

12 Preliminary Tests and Technical Challenges

The transition from laboratory design to real-world deployment inevitably reveals unexpected constraints and difficulties. This chapter presents the preliminary tests carried out in Chile and the associated technical challenges. Although not all of the planned experiments could be completed, the opportunity to test the prototype under realistic conditions proved essential for validating key aspects of its design, such as waterproofing, ballast stability, acoustic performance, and optical imaging.

12.1 Field Tests in Lenga

Context and Objectives

Field validation is a crucial step to assess whether the instrument can withstand realistic deployment conditions, and to confirm that both the acoustic and optical modules operate reliably in a marine environment. The main objectives of the trials in Lenga and Ramucho were to verify waterproofness and buoyancy of the housing, to test the proper functioning of the audio and video acquisition systems in seawater, and to evaluate the practicality of handling the instrument during deployment and recovery.

Methodology and Implementation

The first attempts took place at the Lenga Marine Biology Station. Because of strong winds and heavy swell, full offshore deployment could not be achieved. Instead, preliminary experiments were conducted in the controlled basins of the station, which provided a safe environment to verify the basic functionalities of the system.



(a) Controlled basins of the station - with Christopher



(b) Underwater Image in lenga

Figure 12.1: Filed tests in lenga

Results and Limitations

The tests, although limited, provided useful insights. The flotation and ballast system ensured satisfactory stability even under moderate agitation. The acoustic recordings confirmed that the acquisition chain was functional, though some background noise. The optical module produced clear underwater images, although their quality strongly depended on turbidity and ambient lighting.

Nevertheless, the short duration of the tests did not allow for systematic evaluation of long-term stability or calibration. Further deployments under more favorable weather conditions will be necessary to confirm endurance and robustness in operational scenarios.

Note

For the long-term tests carried out at Caleta Chañaral, the image orientation was corrected to ensure consistent video rendering and TDOA^a calculation.

^aTime Difference of Arrival

To-Do

It was observed that the image focus is not optimal when operating underwater, although this issue does not appear in air. This limitation may be due to the dome not being perfectly hemispherical, which could cause a slight offset of the sensor from the optical center.

12.2 Ballast and Handling

General Principles

The mounting configuration and the adjustment of ballast are critical to ensure both stability and functionality of the instrument during deployment. The housing must remain properly oriented so that hydrophones and cameras can operate efficiently, while also resisting disturbances caused by waves, currents or handling.

Ballast and Buoyancy Control

The ballast configuration was determined empirically. Initial trials were conducted using 18 D-cell batteries combined with 2 kg of ballast weight, which provided sufficient stability in calm water conditions. A second configuration was then tested with 24 batteries and 4 kg of ballast. This setup proved to be highly stable, even in shallow water exposed to wave action. (see Figure.14.4)

Based on these results, the latter configuration was selected as the optimal compromise, offering both robustness against environmental disturbances and practicality in terms of transport and deployment.

12.3 Technical Issues Encountered

12.3.1 QHB Board Failure

Observed Symptoms

One morning, when connecting the QHB board to an external power supply, abnormal behavior was observed. The previous day, an oscilloscope had confirmed the presence and waveform of the UART2 signal. However, on this occasion, the board repeatedly rebooted into standby mode without initiating the expected capture process.

A faint smell of burning also emanated from the board, suggesting possible overheating or a short circuit. The anomaly appeared to be localized near one edge of the printed circuit board.

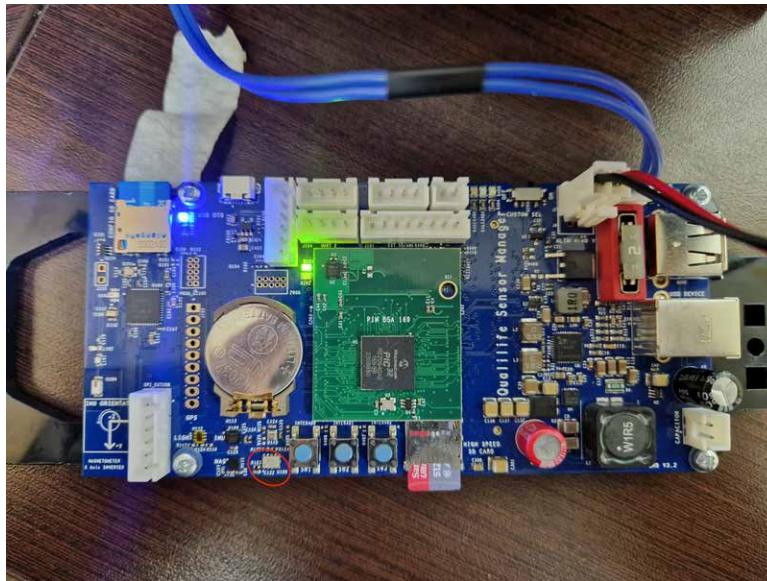


Figure 12.2: Defective QHB board

Local Diagnosis

As the host laboratory was mainly focused on marine biology, it lacked both the necessary equipment (soldering station, electronic microscope, etc.) and the electronics expertise required to perform a detailed diagnosis. Nevertheless, I was able to retrieve the Systrace files, which indicated that the board had remained locked in standby mode. The configuration SD card also appeared to be damaged, making it impossible to read the JConfig.cfg file.

Solution Implemented: Exchanges with SMIoT

After several days of communication with the LIS and SMIoT teams in Toulon, it was decided to return the defective QHB board to France. A new, fully tested and functional board was then shipped to Chile, allowing the continuation of the mission.

This setback, however, delayed the schedule by several weeks, particularly impacting the experimental tasks related to synchronization.

12.3.2 Other Minor Issues Encountered

UART Communication

The day before deployment, we updated the QHB's RTC via a computer interface. However, this procedure caused the UART connection between UWOS and the QHB to fail, possibly due to a grounding issue... To restore proper UART data transmission, UWOS had to be disassembled and a temporary loop was created between the RX and TX lines before reconnecting it to the QHB.

Low Battery Voltage

Another noteworthy issue was observed when using slightly depleted batteries (1.3 V instead of the nominal 1.5 V). In such conditions, the QHB occasionally exhibited startup problems, which could also compromise UWOS power supply. This highlights the importance of ensuring fully charged or fresh batteries to maintain reliable system operation.

12.3.3 Scientific Discussion

Those types of incidents highlight the importance of incorporating flexibility margins into any field-embedded project. Despite rigorous preparation, unforeseen technical issues can arise and temporarily compromise the smooth progress of missions. These experiences illustrate the critical role of responsive technical support, clear documentation, and effective fallback procedures.

12.4 Additional Experimental Designs for Future Work

Several experiments in marine environments were planned to further assess the instrument's performance in marine environments; however, time and resource constraints prevented their execution. Detailed experimental designs, including factor tables and setup schematics, are provided in Appendix D.

Interference Tests

The objective of this experiment was to verify that connecting the Raspberry Pi directly to the QHB power supply would not introduce electrical interference, potentially causing noise in the audio signal. To investigate this, a Taguchi $L8$ plan was proposed, considering factors such as power voltage, power source, wireless usage, recording state, and the presence of an audio signal (see Appendix D.2).

Resistance to Wave Action

To ensure that the addition of the camera module would not destabilize the base system, a Taguchi $L9$ plan was designed to evaluate the influence of camera presence, QHB orientation, wave amplitude, wavelength, and current strength at three levels (see Appendix D.3).

Optical Distortion Characterization

Optical distortion caused by the acrylic dome and the air/water interface was planned to be quantified using a calibration grid and a Taguchi $L4$ plan. The goal was to determine the distortion coefficient k in both air and water, and to assess the relevance of image correction models depending on the medium. A schematic of the planned camera setup is shown in Figure D.3, and the experimental factors are summarized in Table D.3 (see Appendix D.4).

Although none of these experiments were performed, the detailed planning demonstrates the intended methodology and ensures that the procedures are well-documented for future work.

13

Scuba Diving Training

13.1 Context and Objective

Given the nature of the project, which involves submerged sensors and operations in the marine environment, I was offered the opportunity to complete scuba diving training. The objective was to facilitate future deployment, maintenance, or recovery operations of submerged acoustic instruments. This training had a dual purpose:

- to acquire the necessary skills to work on subaquatic equipment within a scientific framework;
- to better understand the physical and logistical constraints of working underwater, in order to improve the design of future embedded devices.

13.2 Training Content

The training was conducted by SSI-certified instructors¹ from the ProSub LTDA diving center². It was structured around both theoretical and practical modules:

- Theory: diving principles, gas physics, risk management, dive planning.
- Pool practice: buoyancy control, mask clearing, emergency management.
- Open water dives: underwater navigation, observation of local species, adaptation to current, reduced visibility, and cold conditions.

The training took place over several weeks in Concepción, in parallel with my research activities. Supervised in particular by Mauricio González Valderrama, it allowed me to acquire the theoretical and practical foundations necessary to operate safely in an underwater environment.

¹Scuba Schools International is a for-profit organization that teaches the skills required for scuba diving and freediving.

²SSI-certified center based in Talcahuano



(a) Mask removal exercise



(b) Buoyancy adjustment

Figure 13.1: Photos of various exercises in the ProSub swimming pool

At the end of this training, I obtained an Open Water Diver certification, enabling me to dive autonomously to a depth of 18 meters. This skill is an asset for future scientific campaigns requiring direct intervention on equipment at sea.

In addition, this experience strengthened my understanding of the constraints associated with the underwater environment: reduced visibility, currents, equipment handling underwater, non-verbal communication, etc. All of these factors must be taken into account when designing and deploying scientific instruments at sea.

14

Experiment at Isla Chañaral

The field experiment in Isla Chañaral, Chile, aimed to test COPEPOP¹ (the UWOS optical module under real deployment conditions in combination with the QHB acoustic recorder), which marked both the first real-world test of the instrument and a genuine scientific field mission. The objective was not only to validate the system's technical performance under operational conditions, but also to collect preliminary ecological observations that could provide valuable insights for ongoing research in the Humboldt Archipelago. The following sections describe the context of the mission, the materials and methods employed, and the first results obtained.

14.1 Introduction

Chañaral de Aceituno, located within the Humboldt Current system, is a hotspot for marine biodiversity, hosting a wide variety of fish, invertebrates, and several cetacean species.

For this first real-world deployment, the objective was to test whether UWOS could contribute to associating specific sounds with fish species, with a particular focus on the pejeperro (*Semicossyphus darwini*). This large wrasse, known to maintain a regular territory, had been repeatedly observed at the entrance of a small underwater cave near the Reserva Nacional Pingüino de Humboldt, at a depth of approximately 9 m.

Beyond this target species, the richness of the local ecosystem offered the opportunity to record additional biological sounds and contextual observations, illustrating the potential of multimodal monitoring in such environments.

This experiment was made possible thanks to the support of César Villarroel — diving instructor, director of the ExploraSub center, and underwater photographer — whose in-depth knowledge of the Caleta area and the reserve was invaluable. The deployment also contributes to a documentary film he is currently producing about the pejeperro.

¹Coordinando Opto-acustica para el Estudio de Peces en el Oceano Pacífico



(a) Male Pejeperro



(b) Female Pejeperro

Figure 14.1: Photo by Eduardo Sorensen

The objective of this mission was to:

- Deploy the combined QHB + UWOS system on a fixed mooring to collect synchronized acoustic and optical data.
- Assess video quality in a natural rocky reef environment with moderate ambient light.
- Collect reference footage of the pejeperro for later evaluation of detection algorithms.

Why Fishes?

Although the instrument was initially designed for observing high-frequency narrow-band (NBHF) cetaceans, its potential applications are far broader. While marine bioacoustics has traditionally focused on marine mammals, it is now well established that many fish species also produce sounds, typically in the low- to mid-frequency range. These acoustic emissions, associated with behaviours such as reproduction, territorial defence, foraging, and social interactions, form a key component of the marine soundscape.

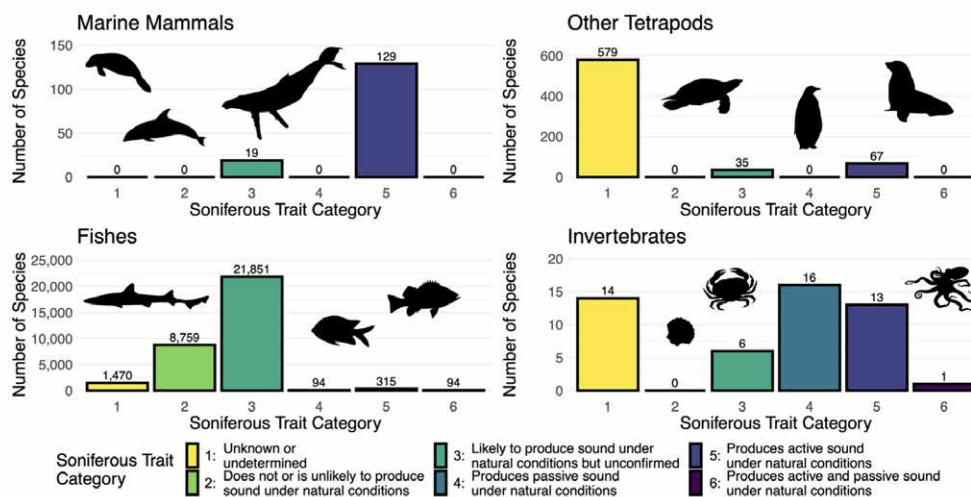


Figure 14.2: Distribution of aquatic and semi-aquatic species and subspecies identified as soniferous, categorized by acoustic trait. Panels separate each taxonomic group, with representative illustrations. (Adapted from Looby et al. [4])

As illustrated in Figure 14.2, adapted from Looby et al. [4], more than **21,000 fish species** are considered either confirmed or potential sound producers. This highlights the vast, yet still largely unexplored, scope of research on fish acoustics. Documenting these emissions is of dual importance: firstly, to improve

our understanding of the behaviour and ecology of these species; and secondly, to assess their vulnerability to anthropogenic pressures such as noise pollution, habitat alteration, and climate change.

This need is further supported by the work of Ivan Hinojosa and César Villarroel, who emphasize the importance of a detailed characterization of the coastal acoustic landscape in Chile — including fish species — to better inform conservation strategies and the sustainable management of fisheries resources.

14.2 Materials and Methods

The complete COPEPOP instrument — consisting of the QHB acoustic recorder and the UWOS optical module — was deployed for this experiment. The deployment team comprised César Villarroel, Ivan Hinojosa, Julie Patris, Franck Malige, and the author.



Figure 14.3: Ivan Hinojosa, Julie Patris and Matthieu Jung before departure.

14.2.1 Choice of Experimental Parameters

The choice of experimental parameters is a determining factor for the success of the deployment. Since the system could not be optimized in situ, estimations were made beforehand to select realistic and robust values. Three key aspects were considered: acoustic localization performance (TDOA), energy autonomy, and data storage capacity.

14.2.1.1 TDOA

The Time Difference of Arrival (TDOA) is a classical technique used to estimate the direction of an acoustic source by comparing the arrival times of the same signal at two or more spatially separated sensors. When the distance between sensors is known, the TDOA provides a direct constraint on the possible source locations, which geometrically correspond to hyperbolas in two dimensions (or hyperboloids in three dimensions). The achievable angular resolution of the system strongly depends on both the baseline length between sensors and the sampling frequency of the acquisition system.

The baseline between the two hydrophones is 36 cm. From this distance, the theoretical Time Difference of Arrival (TDOA) can be derived. For a maximal angle of incidence of 180° , the TDOA is:

$$\Delta T = 24 \cdot 10^{-5} \text{ s.} \quad (14.1)$$

In practice, the system geometry restricts the effective angular aperture to 120° due to the camera FOV². The corresponding maximal TDOA is therefore:

$$TDOA_{COPEPOP}(120^\circ) = \cos(30^\circ) \cdot 24 \cdot 10^{-5} \approx 0.2 \text{ ms.} \quad (14.2)$$

From this value, the resolution in the reconstructed acoustic image can be estimated, i.e. the number of distinguishable source positions (hyperbolas):

$$n = 2 \cdot f_s \cdot TDOA_{COPEPOP}(120^\circ), \quad (14.3)$$

where:

f_s : audio sampling frequency.

Note

The factor of 2 accounts for both positive and negative TDOA values.

Sampling frequency	Number of hyperbolas n
128 kHz	51
256 kHz	102

Table 14.1: Number of possible source positions (hyperbolas) as a function of sampling frequency

For subsequent experiments, a sampling frequency of 256 kHz was selected in order to maximize the number of distinguishable source positions (n) and thus improve the angular resolution of the system.

14.2.1.2 Energy Autonomy

The system was powered by 24 D-cell batteries. Assuming an average capacity of 25.5 Wh per unit, this corresponds to a total available capacity of 612 Wh. According to the QHB user manual, its power consumption is approximately 2.30 W for 2 channels and 256 kHz. For UWOS, laboratory measurements (see Chap. 8) indicated a maximum consumption of 1.78 W (see Table D.1). The combined power draw is therefore estimated at 4 W, leading to a total autonomy of:

$$T = \frac{612}{4 \cdot 0.9} \approx 170 \text{ h,}$$

where the factor 0.9 accounts for the 90% duty cycle of video acquisition (9 minutes of recording per 10 minutes).

By applying a conservative safety margin of 20%, the effective operational autonomy can be reasonably estimated at 136 hours.

14.2.1.3 Storage Autonomy

Both QHB and UWOS have 512GB cards for data recording:

QHB. The QHB generates data files of fixed size, regardless of content, at approximately 3 GB/h. This corresponds to 170 hours of continuous recording, i.e. sufficient for the expected deployment duration.

²Field of View

UWOS. The video module produces files of variable size depending on light conditions, turbidity, and motion. From the final laboratory tests before deployment, the average size was estimated at 1.2 MB/s, i.e. 4.32 GB/h. Considering that the 512 GB SD card also hosts the operating system of the Raspberry Pi, the effective available storage is reduced to about 430 GB. The corresponding storage autonomy is therefore:

$$\frac{430}{4.32} \approx 99 \text{ h.}$$

UWOS's storage capacity is thus the **limiting factor** of the mission. On this basis, the experimental parameters can be defined as follows:

14.2.2 QHB & UWOS configuration

The deployment system consisted of:

- **QHB acoustic recorder** — 2-channel, 256 kHz sampling rate, ~2.3 W consumption, provided by the SMIoT laboratory [Barchasz et al. \[8\]](#).
- **UWOS module** — Raspberry Pi Zero 2W, ArduCam IMX708, Witty Pi 4 Mini, enclosed in a BlueRobotics watertight housing.
- **Hydrophones** — two units provided by the SMIoT laboratory.
- **Mechanical integration** — modules mounted on the dual-support structure described in Chapter 9, with fixed orientation.
- **Power supply** — 24 D-cell batteries located in the central QHB tube, providing an autonomy of approximately 136 hours of recording.

The UWOS module was configured to record 9 minutes of video every 10 minutes, in synchrony with the QHB recording cycles, as specified in the configuration file below:

```
[Settings_UWOS]
output_file = islachanaral.h264
width = 1920
height = 1080
framerate = 30
codec = h264
bitrate = 10000000
nopreview = True
disable_display = True
mission_name = islachanaral_090825

[Settings_QHB]
Shutdown_Duration = 60
Preparing_Duration = 20
Recording_Duration = 540
Stopping_Duration = 10
```

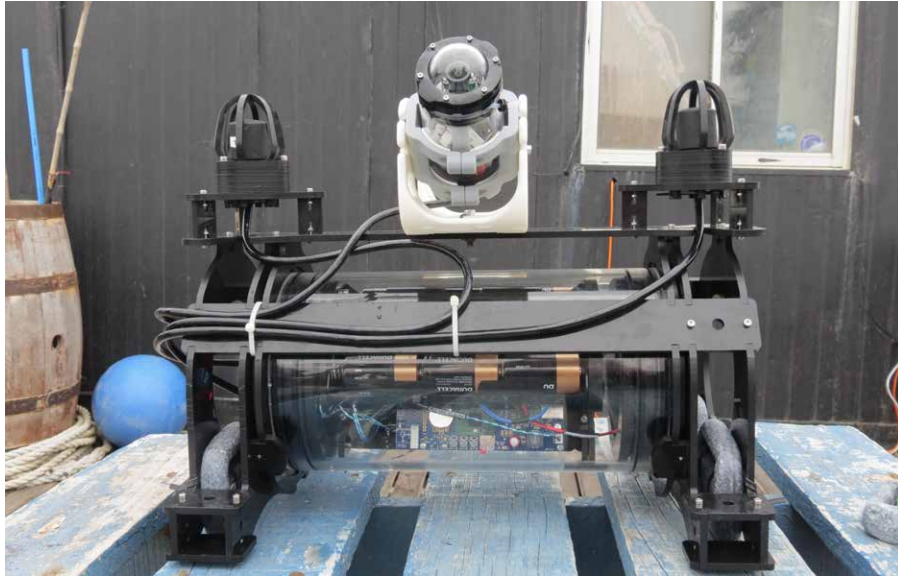
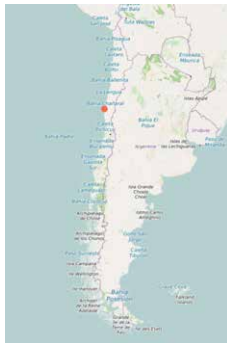


Figure 14.4: The COPEPOP instrument during the 2025 deployment.

Prior to deployment, the instrument was fully loaded with batteries and equipped with an additional 4 kg of ballast to ensure stability in the water column (Fig. 14.4). The ballast mass was determined empirically during pre-deployment trials. The system was secured to a nearby rock with a rope, positioned near the entrance of the target cave.

14.2.3 Deployment site

The islet of Chañaral is located off the coast of Chañaral de Aceituno, in the Atacama Region of Chile (Fig. 14.5).



(a) Reserva Humboldt – Caleta Chañaral



(b) Instrument position near the islet

Figure 14.5: Location of the experimental site.

14.2.4 Deployment procedure

Deployment was carried out from the *Explorasub* vessel. In addition to COPEPOP, short-term hydrophones and a 360° camera were deployed by Ivan Hinojosa. Upon arrival at the site, the vessel was anchored and the dive team (I. Hinojosa, C. Villarroel, and the author) equipped themselves with scuba gear.

The first descent, to approximately 10 m depth, was used to position the short-term hydrophone and 360° camera. The team then surfaced briefly before conducting a second descent to install COPEPOP at the entrance of the cave. As the two descents were separated by less than 10 minutes, they were considered a single dive.



Figure 14.6: Ivan Hinojosa (left) and César Villarroel (right) during COPEPOP installation.

Not being a professional diver but nevertheless being autonomous, my role in this dive was that of a simple observer, and if possible to take some images of the installation using a GoPro.



Figure 14.7: Final position of COPEPOP at the entrance of the cave.

The instrument remained in place from Saturday, August 9, 2025, to Wednesday, August 13, 2025.

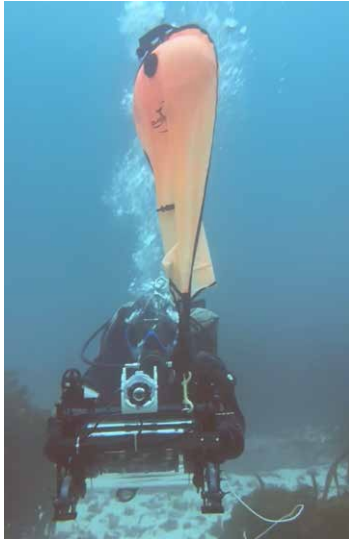


Figure 14.8: César Villarroel – End of test

At the end of the recording period, the instrument needs to be retrieved. However, with its nearly 14 kg weight, it is not easy for the divers to lift it. As shown in Figure 14.8, César Villarroel uses a lift bag to bring the instrument to near-neutral buoyancy.

14.3 Preliminary results

The field deployment at Isla Chañaral provided the first proof of concept for the COPEPOP system. Although only preliminary analyses could be conducted within the timeframe of the internship, the data collected already highlight the potential of combining acoustic and optical modalities.

Acoustic Recordings

SoundTrap

On the same day and location (within a few meters), a SoundTrap hydrophone deployed by Iván Hinojosa recorded several biological sounds. One example is shown in Fig. 14.9.

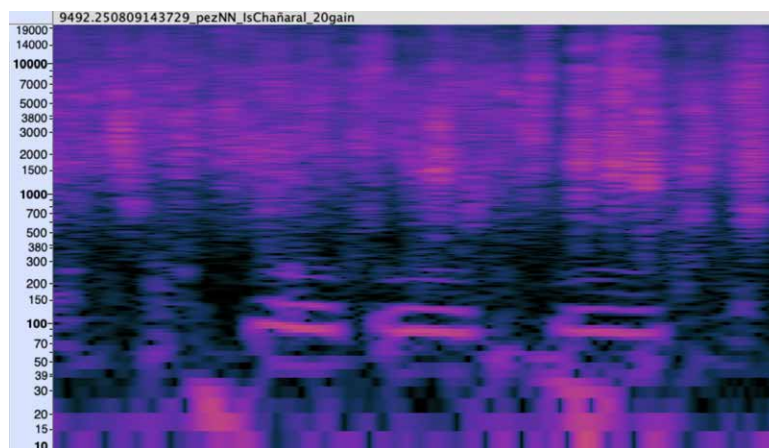
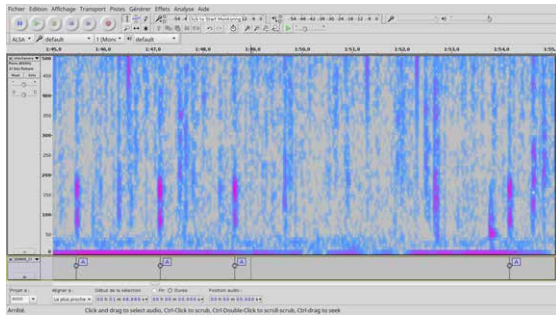


Figure 14.9: Example of a biological signal around 100 Hz recorded with SoundTrap (source: Iván Hinojosa).

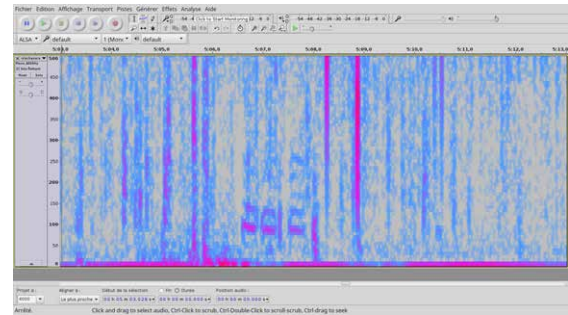
Three emissions around 100 Hz are clearly visible.

COPEPOP

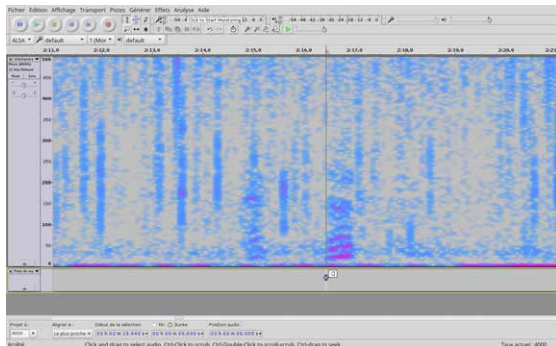
A rapid examination of the files recorded with COPEPOP, conducted by Julie Patris and Franck Malige, also revealed the presence of sounds that can be associated with fish activity. These signals could be tentatively classified into different categories of vocalizations, as illustrated in Fig. 14.10.



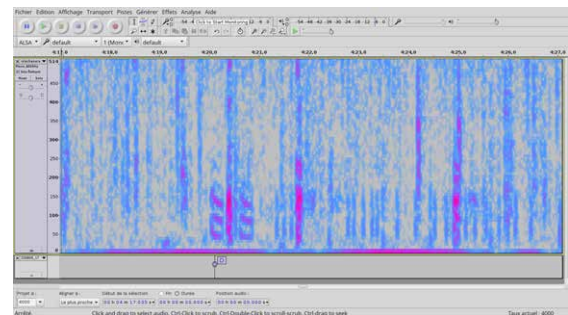
(a) Fish vocalizations A



(b) Fish vocalizations B



(c) Fish vocalizations C



(d) Fish vocalizations D

Figure 14.10: Spectrogram of a fish sound recorded with COPEPOP

Vocalization B (Fig. 14.10b) closely resembles the sound previously recorded with the SoundTrap (see Fig. 14.9). However, the video analysis has not yet been completed, which makes it impossible at this stage to relate these acoustic signals to a specific species or to a particular behavior. This limitation clearly highlights the scientific potential of the project: by combining synchronized acoustic and optical recordings, UWOS aims to enable unambiguous source attribution, a key step towards understanding fish communication and behavior in situ.

Video Recordings

The UWOS modules successfully captured underwater video during the deployment. Although a detailed review of this dataset has not yet been completed, the recordings represent a valuable resource for future analyses. They will enable correlation of visual cues with the acoustic signals described above, thus making it possible to identify species and investigate behavioral patterns in natural conditions.



(a) Exemple of video with external light (August 9, 2025)



(b) Exemple of video with natural light (August 9, 2025)

Figure 14.11: Example of underwater frames captured by UWOS during deployment.

However, the videos appeared somewhat dark under natural light conditions inside the cave. This suggests that adopting a more light-sensitive sensor, such as the IMX462, would be advisable, allowing future deployments to operate without external lighting and thus maintaining a fully non-intrusive observation system.

Outlook

These preliminary results confirm the operational functionality of COPEPOP under real field conditions and underline its scientific potential. The complementary use of acoustics and imagery opens the way to reliable, multimodal documentation of fish and cetacean activity, a crucial step for advancing our understanding of underwater communication and behavior.

PART V

Results and Discussions

15

Results and Technical Feedback

The opportunity to deploy the prototype in real field conditions provided a unique occasion to evaluate its actual performance beyond laboratory tests. This chapter presents a synthesis of the results obtained during these deployments, highlighting both the strengths and the limitations of this first version. These technical feedback elements form the basis for discussing the reliability, robustness, and perspectives for future improvements of the system.

15.1 System Performance Summary

The COPEPOP system was successfully deployed and retrieved during the field mission at Chañaral Island. The QHB acoustic module operated for a total of **99 hours**, producing **300 GB** of high-resolution audio data at 256 kHz across two channels. The UWOS optical module followed the programmed acquisition for a total of **596 video segments**, corresponding to **99 hours** of footage at 1080p/30fps, for **135 GB**.

No critical failures occurred during the mission, and no errors were recorded in the .log file. No thermal shutdowns or sensor malfunctions were reported.

15.2 Robustness and Reliability Assessment

From a mechanical perspective, the watertight housings maintained full integrity throughout the deployment, with no evidence of water ingress. The ballast configuration (4 kg) ensured satisfactory stability in moderate current conditions in the cave, as confirmed by video footage showing no displacement.

From an electrical and electronic standpoint, power consumption remained within the predicted budget, enabling the system to meet the expected autonomy of ~150 hours. The mechanical attachment system of the UWOS's support could be reinforced to further reduce micro-vibrations observed in certain optical sequences.

15.3 User Feedback

Feedback was collected from the deployment team using a structured questionnaire (see Annex E). Overall, users reported that:

- **Transport and setup:** rated as *Neutral* (average score 2.5/5), with positive feedback highlighting the compactness and portability of the system, but the weight of the instrument is a hindrance to its installation.
- **Deployment procedure:** rated as *Neutral* (average score 2.5/5).
- **Data retrieval:** rated as *easy* (average score 3.5/5), with no major issues reported when accessing files; however, opening the camera casing was noted as slightly difficult.

- **Data quality:** acoustic recordings were reported as very good, although several comments mentioned the limited visibility and low-light conditions inside the cave as a drawback for video analysis (see Fig. 14.11). Some issues with focus were also noted (see Fig. 12.1).

Suggested improvements included:

1. Integration of a clear visual indicator to confirm system readiness before deployment, rather than relying solely on the small green LED of the UWOS.
2. Addition of adaptive exposure control for optical recordings in low-light conditions, or alternatively, replacement of the current sensor.

15.4 Global Evaluation against Initial Objectives

Table 15.1 summarises the degree to which the initial project objectives (as defined in Chapter 4) were met.

Objective	Achievement level	Remarks
Design and build a compact, integrated optical-acoustic platform	Fully achieved	Mechanical and electronic integration validated; no water ingress observed.
Achieve >96 h operational autonomy	Achieved	100 h autonomy recorded; 150 h estimated.
Collect synchronised optical and acoustic data in the field	Partially achieved	All the data is there and seems to match, but none of it has been synced yet.
Ensure stable deployment in moderate currents	Achieved	Stable positioning observed; minimal displacement in footage.
Provide usable data for target species detection	Achieved	Acoustic data high quality; optical data limited under very low light.

Table 15.1: Summary of project objectives and evaluation of achievement.

In conclusion, the COPEPOP instrument met the majority of its initial objectives, demonstrating robust performance, high-quality data acquisition, and operational reliability. The main areas for improvement concern the low-light optical optimisation, the focus issue, and vibration damping for certain deployment conditions.

16

Limitations, Potential Improvements and Perspectives

16.1 Hardware and Structural Improvements

The current mechanical design of COPEPOP proved robust during deployment, with no evidence of water ingress or structural damage. Nevertheless, several limitations were identified, suggesting avenues for improvement:

- **Vibration damping:** Minor micro-vibrations were visible in some optical sequences, likely caused by hydrodynamic forces acting on the housing. Reinforcing the mounting plate of the UWOS support and implementing a cross-shaped attachment system could significantly reduce this effect.
- **Attachment system:** While the rope-and-rock anchoring system was functional, a modular quick-release mount would enable faster and safer installation and retrieval, particularly in strong current conditions.
- **Lighting:** In low-light environments, optical recordings suffered from reduced visibility. The integration of low-power LED arrays with adaptive brightness control could enhance video quality without significantly impacting power consumption.
- **Dedicated SD slot for UWOS:** Currently, the video data are written directly to the Raspberry Pi's main SD card, which also hosts the operating system. Adding a dedicated SD slot for data storage would simplify file recovery and reduce the risk of OS card corruption.
- **Focus adjustment:** The UWOS module was slightly out of focus during this mission. Switching to a slightly longer tube (150 mm) would provide a wider adjustment margin and allow more precise focusing.
- **Low-light sensor upgrade:** Since this first COPEPOP deployment took place in a cave with reduced illumination, upgrading the optical sensor to the IMX462 would be highly beneficial for future missions. This improvement may also require the transition to a longer (150 mm) tube.

16.2 Software and Interface Improvements

From a user perspective, the current configuration process relies on manual editing of text-based configuration files. While functional, this approach has limitations in terms of accessibility, usability, and error prevention. The following improvements are suggested:

- **User interface:** A lightweight, cross-platform graphical interface would enable intuitive parameter selection (recording cycles, resolution, sensor activation) without requiring direct code modifications.

- **On-site feedback:** Implementing real-time system status indicators (e.g., LED sequences) would allow divers and operators to confirm correct initialization before deployment.
- **Error handling and diagnostics:** Automated logging with structured error classification (e.g., “battery low”, “storage full”, “sensor not detected”) would accelerate troubleshooting and simplify post-mission analysis.

16.3 Synchronization Accuracy and Data Processing

UART Communication

As discussed in Chapter 9, the current implementation of the QHB only transmits UART frames once per second. This resolution is insufficient for ensuring precise synchronization between the QHB (acoustic module) and the UWOS (optical module). Two possible improvements can be considered:

- **Higher transmission rate:** Increasing the frequency of status frames (e.g., every millisecond) would provide a finer temporal resolution, thereby reducing uncertainty in the alignment of acoustic and optical recordings.
- **Event-based signaling:** Instead of relying solely on periodic messages, the QHB could transmit dedicated frames exactly at the start and stop of each recording. This would guarantee that both systems register the same reference points, minimizing drift and latency effects.

Software-based Synchronization

Given that QHB log files record the precise timing of each recording, it is possible to develop a post-processing pipeline to automatically align acoustic and optical data streams. Such software could merge logs with the video metadata (timestamps) and audio recordings, producing a single, synchronized dataset.

This approach would not only simplify data handling for end-users but also reduce the risk of human error during manual alignment.

16.4 Potential for AI Integration

The large, multi-modal datasets generated by COPEPOP present opportunities for artificial intelligence applications:

- **Acoustic analysis:** Machine learning models could automate the detection and classification of biological sounds (fish calls, marine mammal clicks) and anthropogenic noise.
- **Optical analysis:** Computer vision techniques could identify species, track movements, and quantify biomass from video sequences.

16.5 Low-Power Operation

Although the current system meets its autonomy target (~ 150 hours), several strategies could further extend operational life. One key improvement concerns **software optimisation**. As discussed in Section 17, and according to a recent study by [Pereira et al. \[27\]](#), Python consumes approximately 74 times more energy than C. Rewriting the UWOS software in C would therefore significantly reduce computational overhead and power consumption.

Reducing energy requirements would not only increase deployment duration but also allow the use of smaller battery packs, thereby lowering the overall weight and environmental footprint of the system.

17 Carbon Footprint Assessment of the Internship

In the context of ecological transition and growing awareness of climate issues, it has become essential to integrate environmental considerations into all aspects of engineering activities, including scientific projects.

The internship I completed between France and Chile involved several activities potentially emitting greenhouse gases: long-distance travel, energy consumption, manufacturing a prototype incorporating electronic components, and so on.

The aim of this chapter is to provide a simple yet rigorous estimate of the carbon footprint associated with the internship by identifying the main sources of emissions. This approach aligns with eco-design principles and scientific responsibility, reflecting the values of sustainable engineering.

17.1 Assessment Methodology

The chosen approach is based on a macroscopic estimation of the internship's carbon footprint, using order-of-magnitude data from public databases and widely available tools. This is not a full life cycle assessment (LCA¹), but rather a simplified quantification intended to contextualize the project's environmental impacts.

17.1.1 Sources and Tools Used

The emission factors used for this assessment mainly come from the ADEME Carbon Database², which provides official data on emissions associated with various consumption categories. For structuring the calculations, two complementary tools were used:

- **Nos Gestes Climat**: a public simulator developed by ADEME, useful for estimating personal footprint components (housing, food, local transport).
- **MicMac**: a simplified project carbon footprint assessment tool, allowing the modeling of customized scenarios.

17.1.2 Scope Definition

The carbon footprint evaluated in this study is structured around two main axes. The first concerns the personal scope, which covers impacts related to daily life during the three months in France followed by three months in Chile, such as housing, food, and local transportation. The second corresponds to the professional scope, encompassing emissions related to the research activities themselves, notably those linked to instrument fabrication, electrical consumption of the equipment, as well as transportation of scientific gear.

¹Life Cycle Assessment

²<https://base-empreinte.ademe.fr/>

17.1.3 Assumptions Made

To simplify the calculation, several assumptions were made:

- The round-trip flight between France and Chile is taken in economy class, with no layovers.
- Housing (in both France and Chile) is of a standard type, without air conditioning, with moderate electricity use, and similar emission profiles.
- The diet is vegetarian, with a significant share of local products.
- The electronic components used for the prototype are mostly new, with a lump-sum estimate of their impact based on their weight and type.
- Transportation within Chile is done by bus.

These assumptions allow for a robust estimation at the project scale while maintaining sufficient clarity of the results.

17.2 Emission Sources Estimation

The carbon footprint assessment of the internship is based on identifying the main emission sources, to which standard emission factors (as mentioned above) from the ADEME Carbon Database are applied. Table 17.3 presents a global estimate of greenhouse gas emissions for the entire internship, broken down by source.

17.2.1 International Transport

The largest emission source is long-haul air travel between France and Chile. A round-trip in economy class emits on average about 4 tonnes of CO₂ equivalent (t CO₂e), according to the ADEME Carbon Database. This alone accounts for more than half of the internship's total emissions.

17.2.2 Internal Transport

All internal travel was done by bus, a relatively low-emission mode of transport. This impact is detailed in the table below (Table 17.1), based on 29.4 g CO₂e per km for diesel buses and 259 g CO₂e per km for short/medium-haul flights.

Route	Mode	Estimate (Kg CO ₂ e))
Santiago -> Concepción	Bus	14,6
Concepción -> Valdivia (A/R)	Bus	27,2
Concepción -> Chañaral	Bus	43
Chañaral -> Santiago	Bus	28,3
Total estimé	–	113,1

Table 17.1: Estimated carbon footprint of internal travel

Ultimately, the planned mission to Punta Arenas could not take place, which, incidentally, meant avoiding domestic air travel. As a result, the various bus journeys accounted for an estimated 113 kg CO₂e.

17.2.3 Daily life

- **Accommodation:** The carbon footprint of accommodation was estimated at 0.6 t CO₂e over six months, assuming modest housing without excessive heating or air conditioning, and with reasonable electricity consumption.
- **Food:** A vegetarian diet, based on locally sourced products, significantly reduces the carbon footprint. The estimate used here is 0.6 t CO₂e over six months, compared to around 1 t for an omnivorous diet.
- **Local transport:** All internal travel was carried out by bus, on foot, or by bicycle, representing an estimated 0.1 t CO₂e over six months.

17.2.4 Professional activity

The method used consisted in estimating the carbon footprint in kilograms of CO₂ equivalent (kgCO₂e) for each component or phase in the instrument's life cycle.

The following relationship applies:

$$\text{Instrument footprint (kgCO}_2\text{e)} = \sum_{i=1}^n (M_i \times F_i) \quad (17.1)$$

where:

M_i : Mass or quantity of component i (e.g. 1 Raspberry Pi, 0.1 kg PLA)

F_i : Emission factor of component i (in kgCO₂e/unit)

Emission source	M_i	F_i	Estimate (kgCO ₂ e)
2 Hydrophones (manufactured in Toulon)	2 units	10.0	20.0
Housing 6" (manufactured in Toulon)	1 unit	3.0	3.0
Housing 2" (Blue Robotics)	1 unit	4.0	4.0
QHB audio electronic board	1 unit	6.0	6.0
12 alkaline D-cells	28 units	0.3	8,4
PLA prints	500 g	1.1/kg	0.55
Raspberry Pi 4	1 unit	8.0	8.0
2× Arducam cameras	2 units	4.0	8.0
Witty Pi 4	1 unit	4.0	4.0
Pololu DC-DC converter	1 unit	1.1	1.1
Estimated total			63,05

Table 17.2: Simplified estimate of the instrument's carbon footprint

17.3 Overall results and visualisation

Data	Estimate (t CO ₂ e)
Round-trip air travel France–Chile	4.0
Accommodation (6 months)	0.6
Vegetarian diet (6 months)	0.6
Internal travel (bus)	0.1
Energy consumption (IT equipment, tools, etc.)	0.2
Instrument manufacturing	0.1
Estimated total	5.6

Table 17.3: Estimated carbon footprint of the internship

The estimated carbon footprint of the internship amounts to approximately **5.6 tonnes of CO₂ equivalent (t CO₂e)**. This figure provides a representative order of magnitude for the environmental impact of an international scientific internship involving long-distance travel, technical activities, and prototype manufacturing.

The breakdown of emissions by category reveals a marked imbalance: air travel alone accounts for approximately **70%** of the total footprint. This highlights the decisive influence of project location on overall environmental impact.

For comparison:

- The average French citizen emits approximately **9 t CO₂e/year**.
- The individual carbon neutrality target is **below 2 t CO₂e/year**.
- This internship therefore represents around **62%** of the average annual French footprint, in just six months.

Figure 17.1 shows the estimated distribution of emissions by major category.

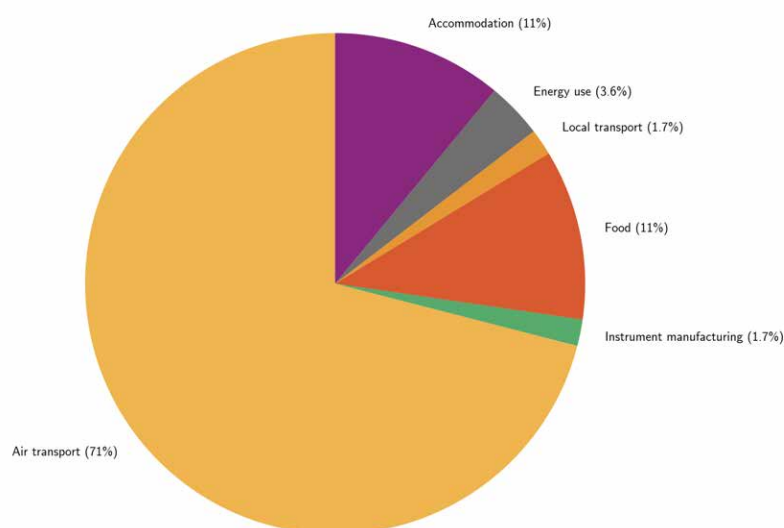


Figure 17.1: Estimated breakdown of the internship's carbon footprint

17.4 Critical analysis and eco-design perspectives

Limitations of the assessment

The evaluation presented here is based on order-of-magnitude estimates derived from public databases and calculators. While it provides a realistic approximation, it has several methodological limitations:

- **Simplified approach:** This is not a full Life Cycle Assessment (LCA), which would require precise data on material origins, manufacturing processes, component lifespans, and end-of-life scenarios.
- **Generic data:** Several items were estimated using generic values (e.g. instrument manufacturing) due to the absence of detailed information on components and their supply chains.
- **Limited scope:** Some indirect impacts were not included, such as future use of the instrument, server hosting of collected data, or long-term maintenance of the equipment.

Despite these limitations, this evaluation serves to raise awareness of the approximate magnitude of emissions generated by a scientific project, and to identify the most significant leverage points.

Key insights and reduction strategies

The results clearly show that **air travel** remains the dominant factor, raising questions about the geographical location of internships and scientific missions in a carbon-constrained world. Alternatives such as remote collaboration, mission clustering, or extended stays to amortize the impact of flights should be considered.

Regarding the equipment developed, several eco-design strategies can be applied to reduce environmental impact:

- **Component reduction:** optimise electronics to limit the use of raw materials.
- **Reusability and repairability:** design the instrument to be easily disassembled, repaired, and upgraded.
- **Material choices:** favour more durable or lower-impact components (eco-PCBs, recycled plastic housings, etc.).
- **Local or shared manufacturing:** reduce long-distance deliveries and dispersed purchases by centralising production.
- **Energy-efficient software:** optimise embedded code to reduce energy consumption.

To-Do

In terms of software optimisation, one improvement we did not have time to implement was rewriting all embedded code in C. According to a recent study by [Pereira et al. \[27\]](#), Python consumes approximately 74 times more energy than C.

Incorporating these considerations at the earliest stages of development, through a responsible design approach, helps train engineers who can combine technical performance with sustainability.

This reflection on the internship's carbon footprint shows that even a simplified environmental approach can provide concrete insights for both understanding and action. Analysing emission sources fosters a culture of sustainable engineering and encourages rethinking certain technical, logistical, or organisational choices from a low-carbon perspective.

18

Skills Acquired and Personal Development

18.1 Technical and Interdisciplinary Skills

Throughout the COPEPOP project, a wide range of technical skills were developed, spanning hardware design, software configuration, and field deployment. On the hardware side, this included the integration of acoustic and optical modules into a compact, watertight housing; ballast configuration; and troubleshooting of mechanical stability in challenging aquatic environments. On the software side, skills were acquired in embedded system configuration (Raspberry Pi Zero 2W, sensor interfacing), camera control, and optimisation of power scheduling for extended deployments.

This project also required significant interdisciplinary work, combining principles of electronics, marine biology, mechanical engineering, and environmental science. Collaboration with marine ecologists provided insights into target species behaviour and habitat constraints, directly influencing deployment strategy and data acquisition settings.

18.2 Project Management

From initial design to final deployment, the project followed a full engineering lifecycle: requirements analysis, specification, prototyping, validation, and operational testing. Time and resource management were critical, particularly when coordinating between multiple partners, including the SMIoT laboratory, local field teams and the technical issue encountered!

The field campaign demanded adaptability to changing environmental and logistical conditions, such as weather constraints, equipment availability, and last-minute changes.

18.3 Fieldwork and Diving Operations

An essential aspect of this project was the ability to perform underwater operations during deployment and retrieval of the COPEPOP system. Although not a professional diver, I am an autonomous recreational diver, which allowed me to actively participate in the installation process alongside experienced marine biologists and technicians.

This required adapting technical tasks to the constraints of the underwater environment:

- Performing precise positioning of equipment in low-light, current-affected conditions.
- Maintaining situational awareness with respect to both the instrument and the surrounding ecosystem.
- Communicating effectively with dive partners through hand signals and pre-defined procedures.
- Managing personal buoyancy and air consumption while handling sensitive, sometimes bulky, equipment.

Working underwater also reinforced the importance of preparation: ensuring that all tools, mounting hardware, and safety measures were ready before descent minimised dive time and reduced risks for both divers and equipment. The experience provided valuable insight into the practical challenges faced by researchers who operate in aquatic environments, highlighting the need for robust, ergonomic, and diver-friendly instrument designs.

18.4 Scientific Communication and Collaboration

The project provided multiple opportunities to communicate scientific and technical concepts to a variety of audiences:

- **Within the research team:** regular progress updates, technical documentation, and troubleshooting reports.
- **For external partners:** clear explanations of the instrument's capabilities, constraints, and deployment procedures.
- **In academic contexts:** preparation of figures, methodological descriptions, and preliminary results for integration into reports and potential publications.

Working alongside engineers, divers, and marine biologists reinforced the importance of a shared vocabulary and mutual understanding of objectives. Feedback from users and collaborators during and after the deployment informed improvements to both hardware and workflow, fostering a culture of iterative development.

18.5 Adaptation to an International Research Context

Beyond the technical and scientific aspects, this internship also represented a personal challenge of adaptation to a new cultural and linguistic environment. Conducting part of the project in Chile required daily communication mostly in Spanish, strengthening both my language skills and my ability to convey technical concepts across linguistic barriers.

Working within an international and interdisciplinary team meant adapting to different work practices, academic cultures, and logistical realities. These conditions required flexibility, problem-solving, and the capacity to quickly establish effective collaboration with new partners.

This immersion abroad also provided valuable insights into the role of engineers as mediators between cultures and disciplines. It highlighted the importance of interpersonal skills, open-mindedness, and resilience in ensuring the success of complex scientific projects carried out in diverse contexts.

18.6 Reflection on the Role of Engineer in Environmental Research

This project highlighted the multifaceted role of engineers in environmental research:

- **Problem-solvers:** designing and adapting tools to operate reliably in unpredictable, often harsh marine conditions.
- **Facilitators:** enabling researchers from other disciplines to collect high-quality data efficiently, thereby accelerating ecological understanding.
- **Innovators:** integrating emerging technologies—such as low-power electronics, AI-assisted analysis, and sustainable materials—into field-ready solutions.

Yet, being an engineer in this context extends beyond technical performance. It also requires a sense of responsibility regarding the long-term impacts of our creations. In a time of global climate change and

biodiversity loss, developing new technologies cannot be dissociated from their environmental footprint. This means asking critical questions: Do we really need this system? Can it be made with less energy or fewer resources? How can it be deployed without amplifying the problems we aim to study?

This internship underscored that engineering for science does not mean we can—or should—do everything that is technically possible. Instead, it calls for a balanced approach where innovation is aligned with ecological responsibility. By embedding sustainability and life-cycle thinking into design choices, engineers can act not only as enablers of scientific progress but also as guardians of the environment.

In this sense, the role of the engineer is to ensure that technological development contributes meaningfully to ecological knowledge and conservation, without creating disproportionate costs for the very ecosystems we aim to protect.

18.7 Personal Development

Beyond the skills acquired, this internship was a deeply rewarding personal experience. One of the aspects I appreciated the most was being able to contribute at all levels of the project, from early design sketches and hardware assembly to software integration and real-world deployment.

I particularly enjoyed the technical dimension of the project: building, testing, troubleshooting, and improving the prototype step by step. At the same time, the fieldwork in Chile was an equally valuable part of the experience. Deploying the system in real conditions, diving to install it, and collaborating with scientists on site provided both strong technical learning and unforgettable human encounters.

This balance between technical engineering and field application confirmed my motivation to pursue work that bridges design and practice, and to contribute to projects that have both scientific impact and environmental meaning.

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A

Video Conversion Software

Note

This software was developed during the very first camera tests. In the new interfaces currently under development, the video conversion feature is now integrated as a dedicated tab only.

Within the development of the solution, an ancillary need was identified. Indeed, the videos recorded by the Raspberry Pi are stored in the H264 format. While this format is common, it is poorly supported by most standard video players. Therefore, it appeared useful to implement a small Python script relying on the open-source FFmpeg library in order to automatically convert all videos from a H264 folder into MP4, making them easier to handle and process.

To make the tool more user-friendly and accessible for people unfamiliar with command-line scripts, a simple graphical interface was developed using Tkinter. The script was then packaged into a Windows .exe executable, allowing it to be launched directly from a folder without requiring additional setup. No executables were generated for other operating systems, under the assumption that users on Linux in particular are generally comfortable with terminal-based tools.

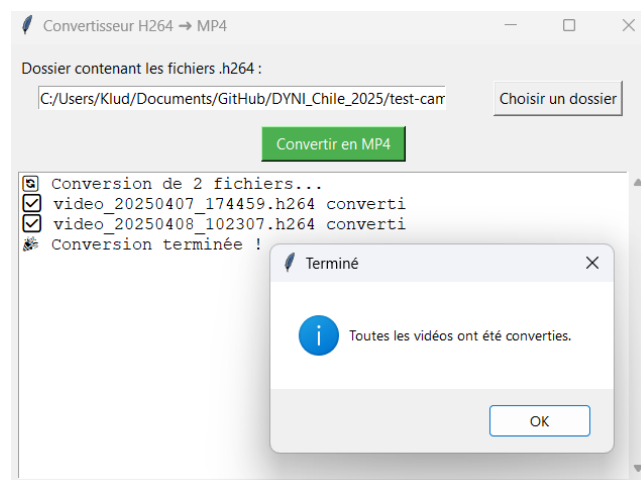


Figure A.1: Software Converter H264 to MP4

The software is available on GitHub: <https://github.com/JungMatthieu/UWOS>
The pseudocode and implementation are presented below:

Algorithm 4 H264 to MP4 Converter

Input: A folder containing .h264 files

Output: Converted .mp4 files stored in a converted subfolder

```

1 Wait for the user to select a folder

2 if the folder is invalid then
3   | Display an error and stop
4 end
5 Create the converted folder if it does not exist
   List all files ending with .h264

6 if no file is found then
7   | Display a message and stop
8 end
9 for each .h264 file do
10  | Generate the output name in .mp4
    | Build the ffmpeg command
    | Execute the command
    | if success then
11  | | Display "conversion successful"
12  | else
13  | | Display "error"
14  | end
15 end
16 Display "conversion completed"

```

B

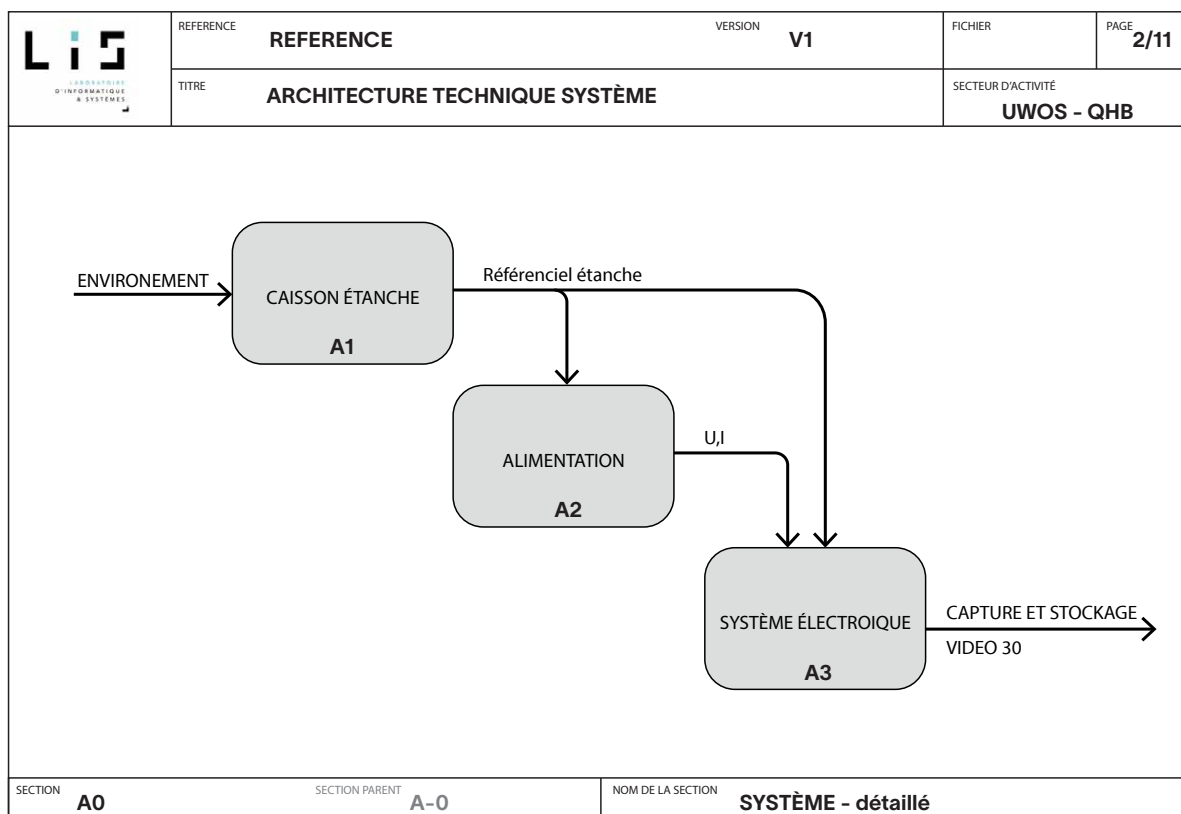
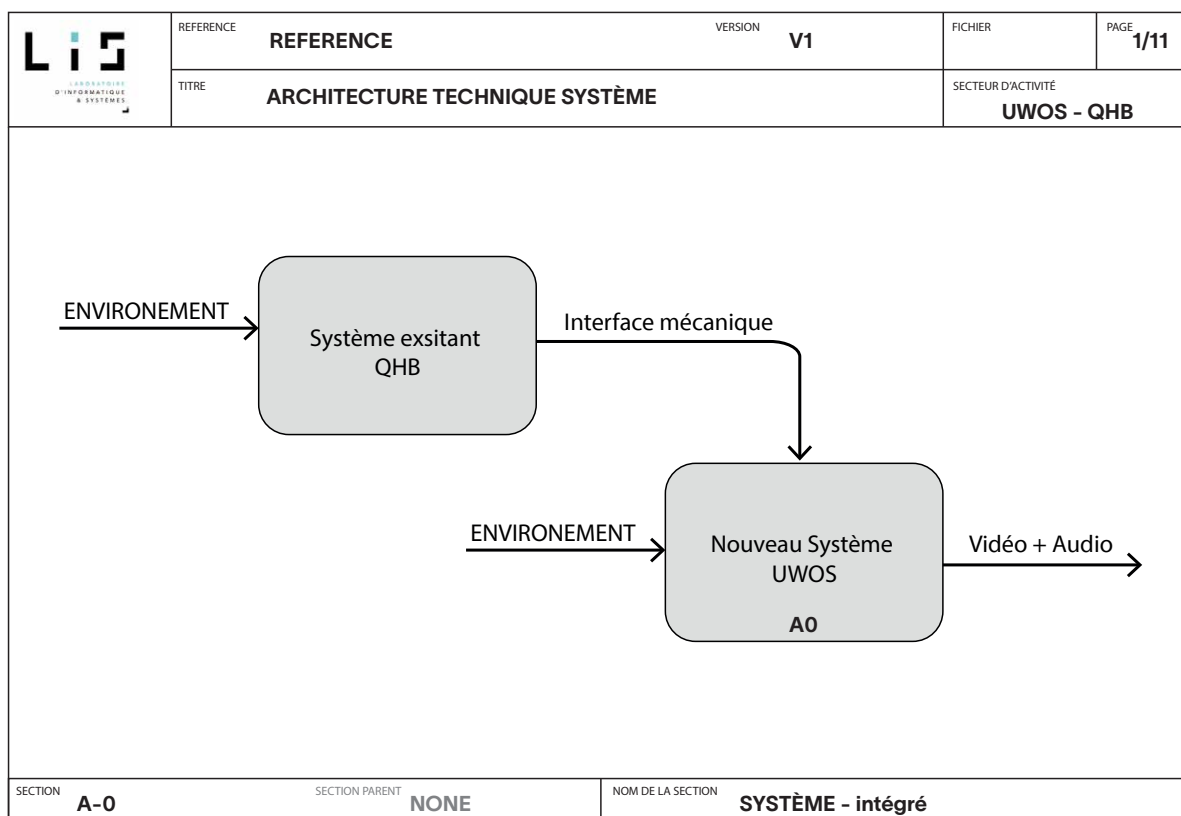
Specifications

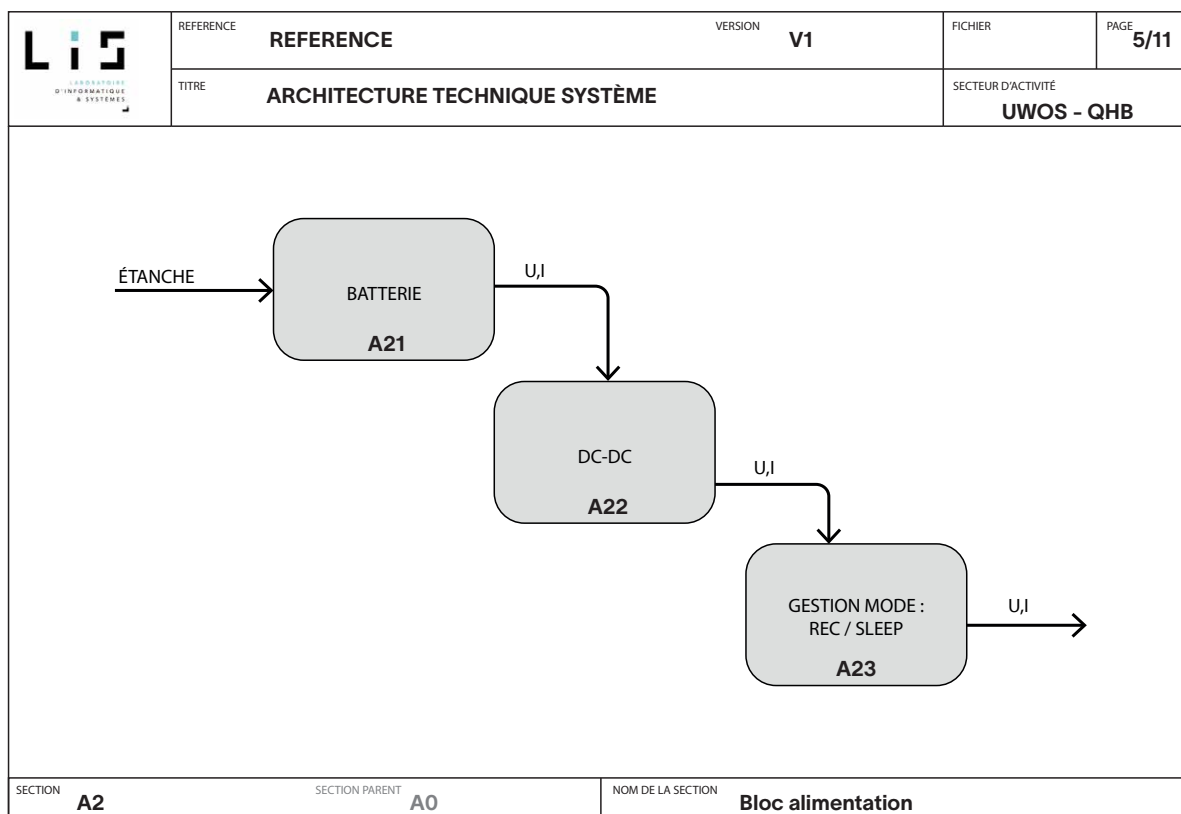
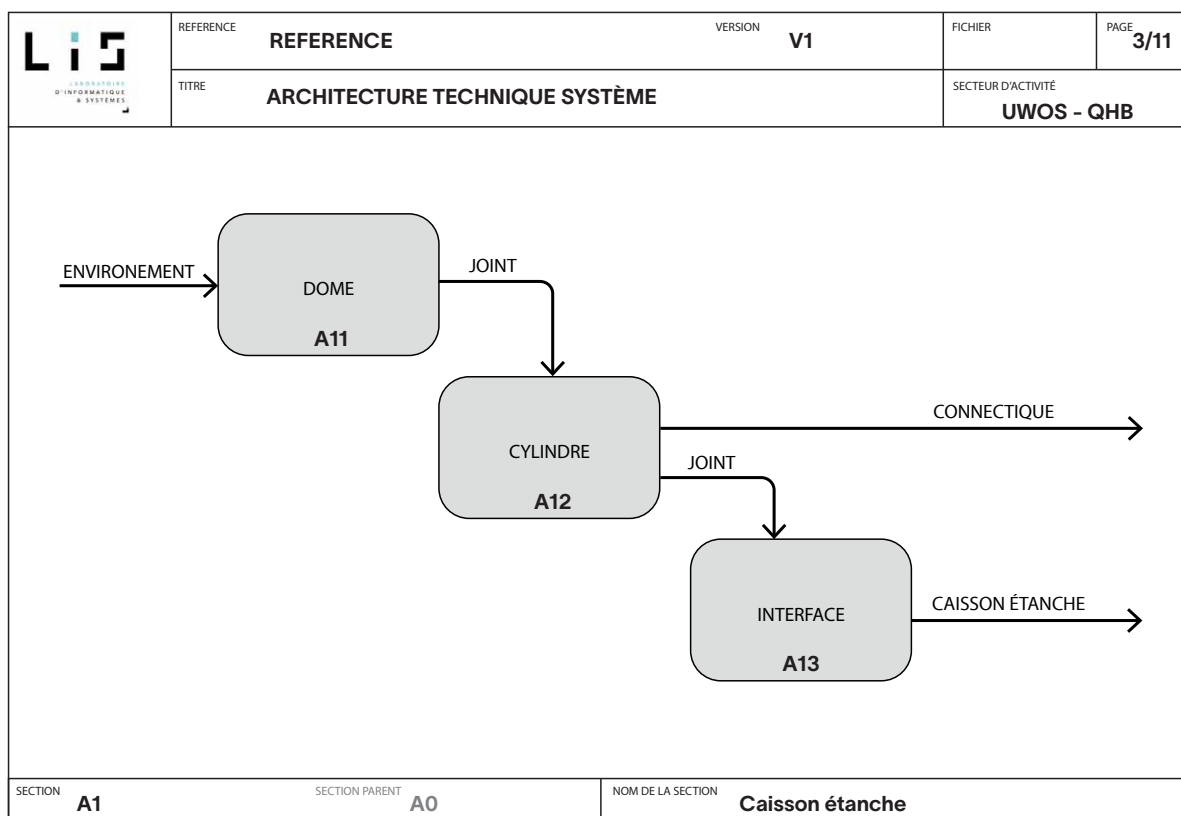
Note

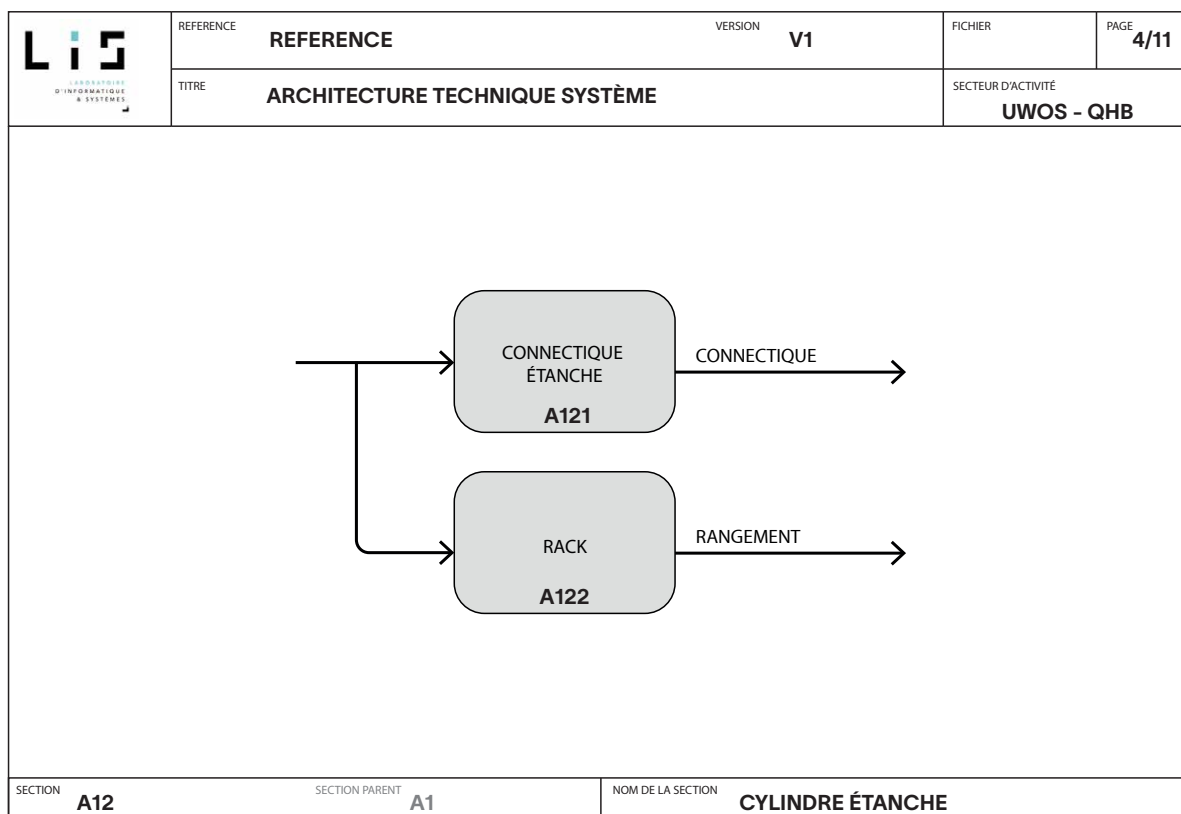
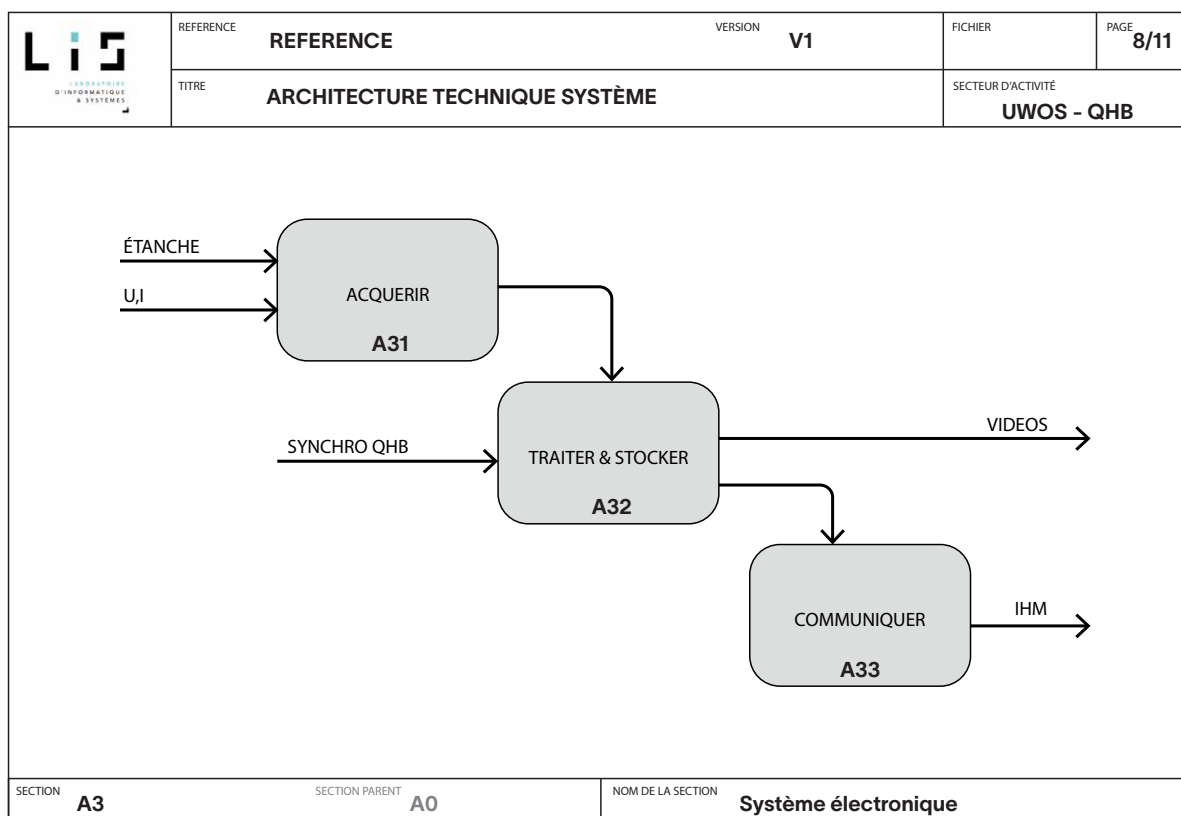
To avoid overloading this book, the complete specifications are available on GitHub <https://github.com/JungMatthieu/UWOS>. Only the tables and the technical system architecture are listed here. ↓↓↓

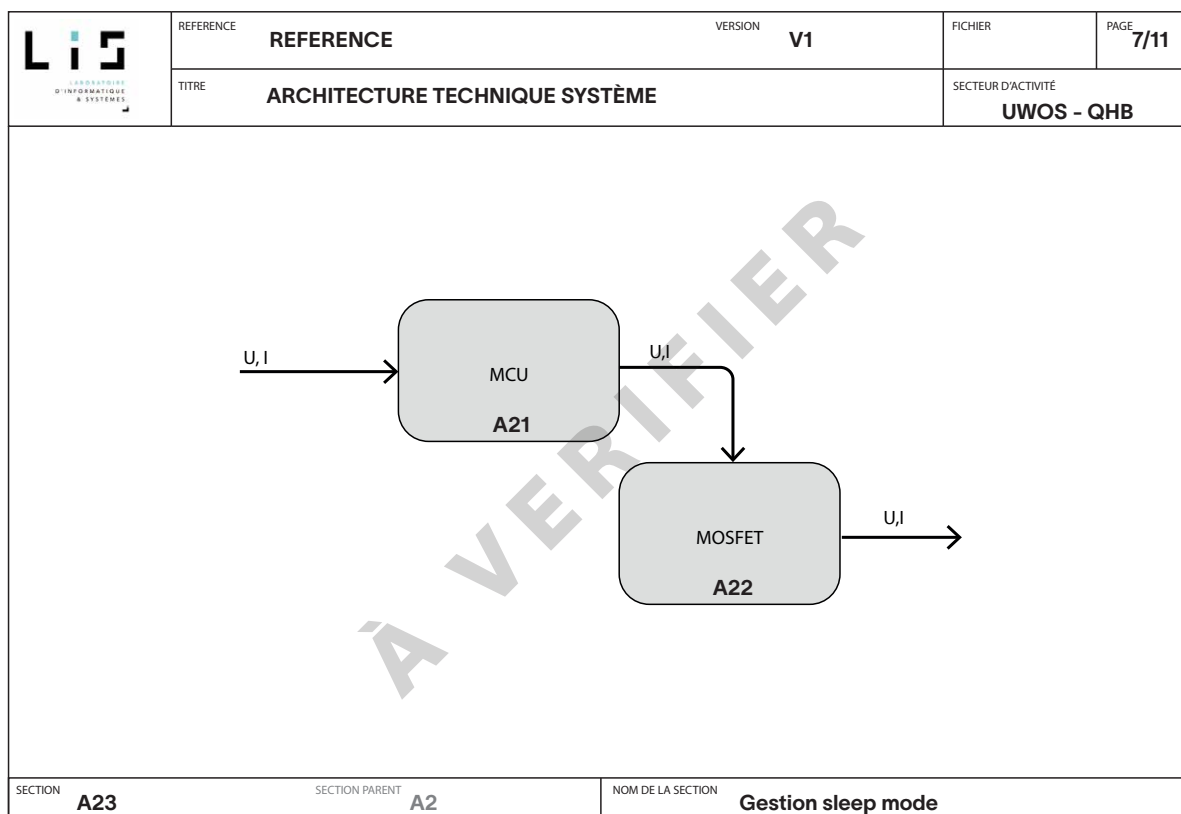
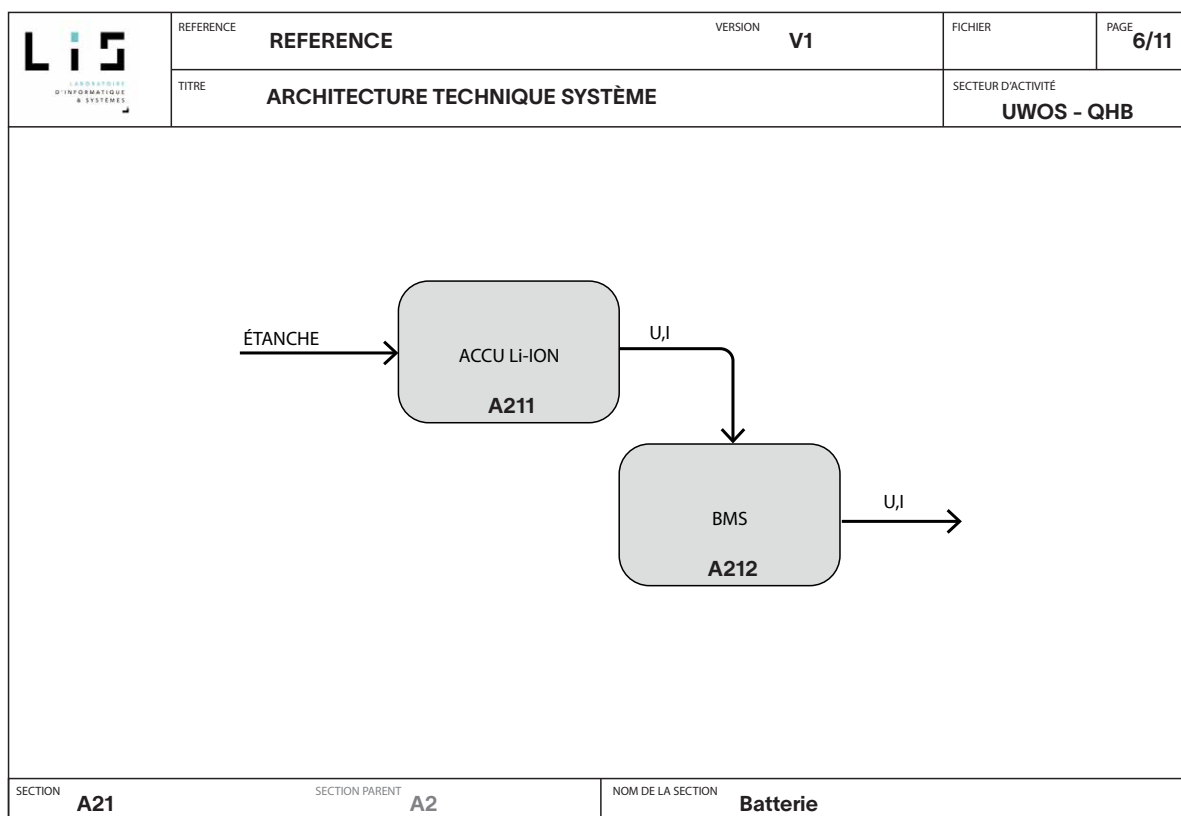
Fonctions		Critères d'acceptation		Niveaux d'acceptation		FLEXIBILITE	
		Fonctions principales				Classe	Limites
FP1	Capturer des vidéos sous-marines	• Résolution d'image		Full HD	F0	HD	
		• Frame Rate		24 fps	F0	15 fps	
		• Stocker les vidéos		H.264	F0	lisible en post-processing	
		• Filmer en grand angle		objectif 180°	F0	120°	
		• Sensibilité à la lumière		0.01 Lux	F0	0.5 Lux	
		• Filmer en continu sur des périodes de plusieurs heures		12h	F1	6h	
		• Synchro		1ms	F0	100ms	
FP2	Déclenchement des vidéos	• Adapter la sensibilité capteur		Avec luxmètre	F0		
		• Gestion des modes REC / SLEEP		--	F0	--	
FP3	Informier l'utilisateur de l'état de l'enregistrement	• Etat		Rec / pas Rec	F0		
		• Moyen d'information		Visuel : LED	F2	--	
Fonctions secondaires							
FC1	Résistance mécanique	• Etanche		40m - IP68 ou plus	F0	30m	
		• Chocs		Chute 1m	F0		
		• Température d'utilisation		-10° to 40°	F0	0° to 30°	
		• Resistance a la corrosion		Acrylique ou Inox	F0		
		• Masse		<1kg	F1	<1.5kg	
FC2	S'interfacer avec l'existant	• Taille		depend of sensor	F1		
		• Type et Niveau d'énergie		5V sur D alkaline	F0		
		• Orientation module		5 choix d'orientation	F1	3 choix	
		• connectique		1 sortie étanche min	F0		
		• Consommation		<10W	F1	±20 %	
FC3	Autonomie	• Gestion des mode REC et Veille			F0		
FC4	Batterie	• Alimenter par batterie		Li-Ion ou D Alkaline	F1		
FC5	Stockage	• Tout enregistrer		512Go to 1To / V30	F1	512Go / V30	
		• Ne pas chauffer			F2		
FC6	Coût	• Ne doit pas dépasser le budget		1 000€	F2	1 000€	
FC7	Maintenance	• Facilement ouvrable		CAO	F2		
FC8	Exploitation	• Récupération des données		Donnée formaté	F0		
		• Transportable et manipulable par des plongeurs			F1		
		• Système fiable		Eviter la corruption des données	F0		
FC9	Être furtif	• Ne dois pas déranger l'environnement		Ne pas emettre de lumière, son	F2		

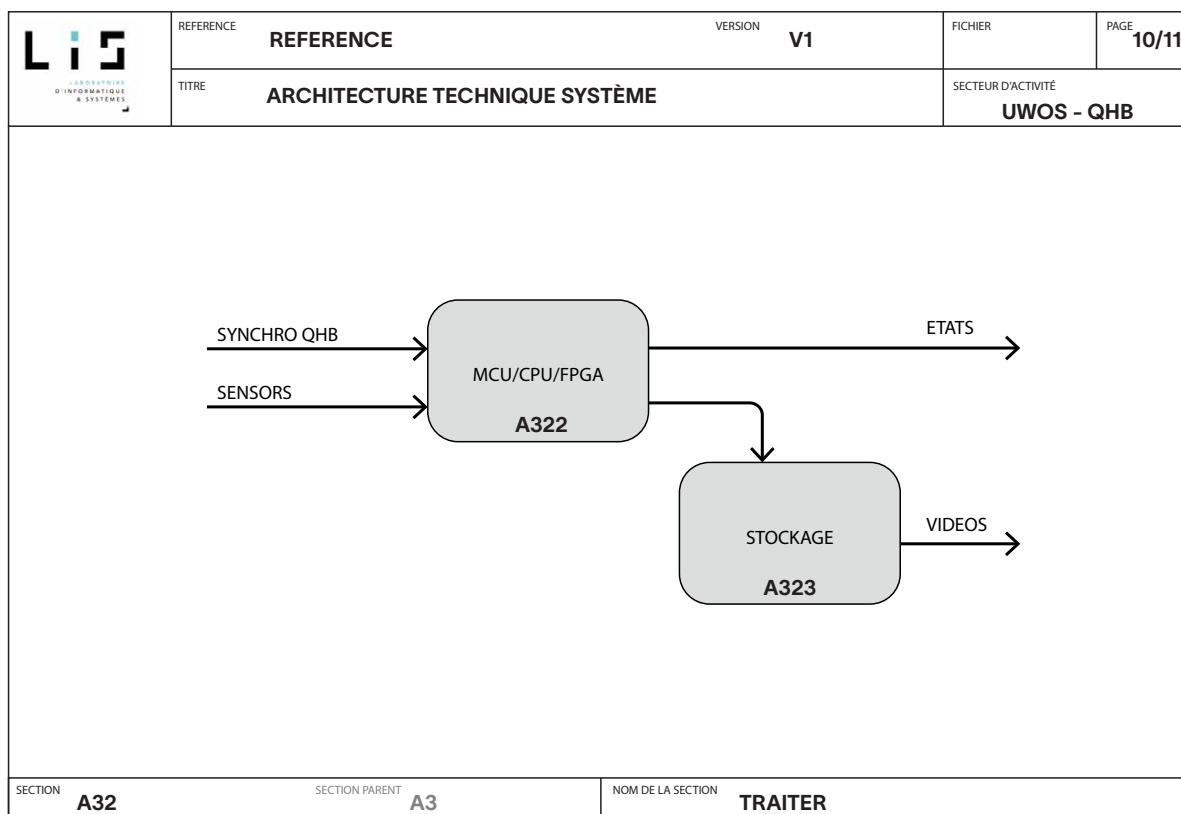
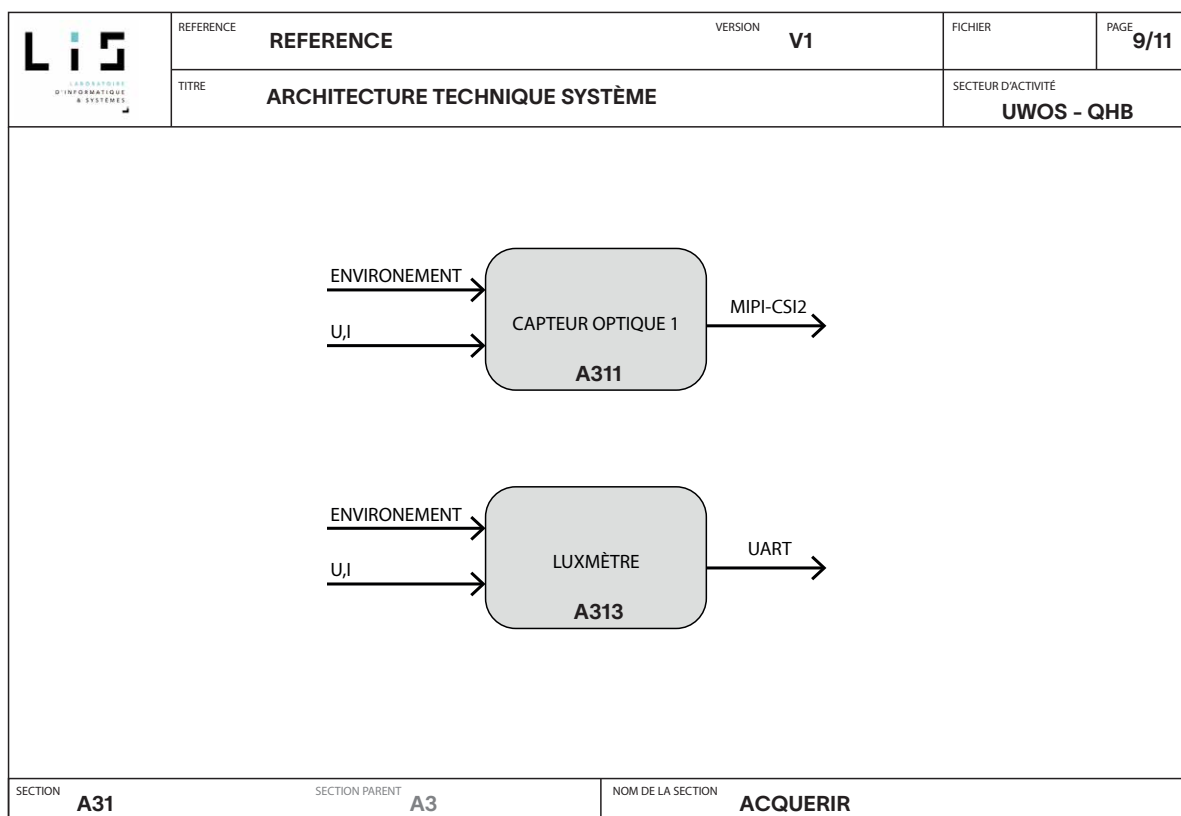
Fonction	Mode de défaillance	Phase de vie	Événement redouté (fonction)	Événement redouté (utilisateur)	Gravité	Mesure à prévoir
Acquisition optique	Absence de fonction	Exploitation	Aucune image capturée	Pas d'identification des espèces	1,00	Vérification des connexions, redémarrage automatique
	Fonction dégradée	Exploitation	Images floues ou bruitées	Mauvaise interprétation des données	0,60	Calibration automatique, algorithme de correction
Synchronisation Audio/Vidéo	Perte de fonction	Exploitation	Décalage temporel entre son et image	Analyse incohérente	0,80	Algorithme de synchronisation et buffer de correction
Stockage des données	Fonction intempestive	Exploitation	Écriture en boucle, perte de données	Données perdues	1,00	Vérification mémoire, mise en tampon redondante
	Perte de fonction	Exploitation	Aucun enregistrement sauvegardé	Analyse impossible	1,00	Redondance du stockage, tests périodiques
Alimentation électrique	Perte de fonction	Exploitation	Arrêt du capteur	Capteur inutilisable	0,60	MAJ du système
	Fonction dégradée	Exploitation	Faible autonomie	Temps de capture réduit	0,60	Optimisation de la gestion énergétique
Étanchéité	Fonction dégradée	Exploitation	Infiltration d'eau, court-circuit	Panne définitive	1	Test de pression, joints renforcés
Interface utilisateur	Absence de fonction	Exploitation	Aucune indication d'état	Mauvaise compréhension du statut	0,1	none
Système de fixation	Fonction intempestive (vibrations)	Transport/Installation	Mauvaise stabilité, angles incorrects	Mauvaise captation des images	0,33	Redondance fixation
	Perte de fonction	Exploitation	Mauvaise stabilité, angles incorrects	Analyse difficile	0,5	Redondance fixation
Résistance mécanique	Fonction dégradée	Exploitation	Déformation de la structure	Détérioration prématurée	0,33	Utilisation de matériaux renforcés
Résistance aux chocs	Fonction intempestive	Exploitation	Chute entraînant des dégâts internes	Dysfonctionnement prématuré	0,5	Tests de choc, mousse de protection
Gestion thermique	Fonction dégradée	Exploitation	Surchauffe	Risque de panne et de condensation	0,66	Dissipation thermique, optimisation énergétique
Gestion des modes REC / SLEEP	Fonction intempestive	Exploitation	Passage en mode veille inopiné	Perte d'enregistrement	0,66	Algorithme de gestion optimisé

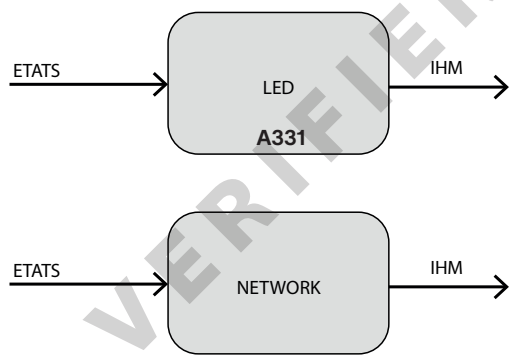










	REFERENCE	REFERENCE	VERSION	V1	FICHIER	PAGE	11/11
	TITRE	ARCHITECTURE TECHNIQUE SYSTÈME			SECTEUR D'ACTIVITÉ UWOS - QHB		
							
SECTION	A33	SECTION PARENT	A3	NOM DE LA SECTION COMMUNIQUER			



Sensors Benchmark

Sensor	IMX378	IMX219	IMX462	IMX519	IMX708 - M12	IMX477R	VD55G1
Optical size	1/2.3"	1/4" (IMX219)	1/2.8"(6.46mm sensor diagonal	1/2.53"	1/2.43"	1/2.3"	1/2.8"
Chroma		RGB	RGB	RGB	RGB	RGB	Monochrome
Pixel size	1.55 µm	1.12µm(H)×1.12µm(V)	2.9µm × 2.9µm	1.22µm×1.22µm	1.4µm × 1.4µm	1,55 µm × 1,55	2.16µm
Resolution	12.3 MP (4056×	8MP - 3264*2(H)×2464(V)	1920(H) × 1080(V)@60fps	16MP	11.9 Megapixels //	12,3 MP - 4056(	0.56MP - 804x7
Frame Rates		1600×600@21fps // 3840×1080@17fps		1080p30 // 720p60	2304 × 1296p56 //	2028 × 1520p4(	
Ouput format		RAW10	RAW10	RAW 10/RAW 8, COM	RAW10	RAW12/10/8, C	RW8 RAW10
ISP output format		JPG, YUV420, RAW, DNG; MJPEG, H.264					
IR snsitivity	650nm IR-cut fil	Integral IR Filter, visible light only	Integral IR-cut Filter, visible light	Integral IR-cut Filter, v	Integral IR-cut Filter	Integral IR-cut F	Clear
Lens Type	M12 Lens	M12 Lens	M12 Lens	M12 Lens	M12 Lens	M12 lens	M12 lens
HDR	Oui	Non	oui	oui	oui	oui	non
light sensitivity	Bonne (Grands	Moyenne	Excellente (starvis)	Bonne	Bonne	Bonne	Bonne
Camera board size		25mm x 24mm	25 × 24mm	25 × 24mm	25 × 24mm	25 × 24mm	30mm x 35mm
Price	≈ 40€	≈ 20€	≈ 40€	≈ 37€	≈ 37€	≈ 50€	
	https://www.dfr	https://www.uctronics.com/camera-mod	https://www.uctronics.com/car	https://www.uctronic	https://www.uctron	https://www.uc	

Figure C.1: Sensors Benchmark

D

Design of Experiments

D.1 Power Consumption

Run	Capteur	Display	Résolution	FPS	Light	
1	IMX462	OFF	720p	15	Faible	
2	IMX462	OFF	1080p	30	Moyenne	
3	IMX462	OFF	2k	15	Fort	
4	IMX462	ON	720p	30	Moyenne	
5	IMX462	ON	1080p	15	Fort	
6	IMX462	ON	2k	30	Faible	
13	IMX462	OFF	720p	15	Fort	
14	IMX462	ON	1080p	30	Faible	
17	IMX462	OFF	1080p	15	Moyenne	
7	IMX708	OFF	720p	15	Moyenne	
8	IMX708	OFF	1080p	30	Fort	
9	IMX708	OFF	2k	15	Faible	
10	IMX708	ON	720p	30	Fort	
11	IMX708	ON	1080p	15	Faible	
12	IMX708	ON	2k	30	Moyenne	
15	IMX708	OFF	2k	15	Moyenne	
16	IMX708	ON	720p	30	Moyenne	
18	IMX708	ON	2k	30	Fort	

#	Run	Capteur	Display	Résolution	#	FPS	Lumens	Consom(W)	#	Nom du fichier	Yolo score	Détection ok ?
1	IMX708	OFF	1080x720	15	Moyenne (20%)	0.175 - 0.220A	122146	OK				
2	IMX708	OFF	1920x1080	30	Fort (80%)	0.340A	122340	OK				
3	IMX708	OFF	1920x1080	15	Faible (0%)	0.250A	135628	NON				
4	IMX708	ON	1080x720	30	Fort (80%)	0.250A	124046	OK				
5	IMX708	ON	1920x1080	15	Faible (1%)	0.270A	124215	OK				
5.5	IMX708	ON	1920x1080	30	Faible (0%)	0.300A	135752	NON				
6	IMX708	ON	1920x1080	30	Moyenne (20%)	0.350A	124452	OK				
7	IMX708	OFF	1920x1080	15	Moyenne (20%)	0.270A	124805	OK				
8	IMX708	ON	1080x720	30	Moyenne (20%)	0.250A	125007	OK				
9	IMX708	ON	1920x1080	30	Fort (80%)	0.347A	125236	OK				
10	IMX462	OFF	1080x720	15	Faible(0%)	0.245A	132458					
11	IMX462	OFF	1920x1080	30	Moyenne (20%)	0.342A	132707					
12	IMX462	OFF	1920x1080	15	Fort (80%)	0.255A	133004					
13	IMX462	ON	1080x720	30	Moyenne (20%)	0.298A	133141					
14	IMX462	ON	1920x1080	15	Fort (80%)	0.271A	133420					
15	IMX462	ON	1920x1080	30	Faible(0%)	0.353A	133630					
16	IMX462	OFF	1080x720	15	Fort (80%)	0.238A	133838					
17	IMX462	OFF	1920x1080	30	Faible(0%)	0.363A	134103					
18	IMX462	OFF	1920x1080	15	Moyenne (20%)	0.270A	134237					

Courant en veille :		5	0.001	0.005
V	A	W		

Figure D.1: Design of Experiments - Power Consumption

D.2 Interference

Note

Reminder, due to time constraints and various delays this experiment was not carried out !

This experiment aims to verify whether UWOS introduces any interference in the QHB audio recordings.

Setup, Method, and Results

In this case, we will use a much smaller experimental plan, as the number of factors is lower. We use a Taguchi $L_8 (2^4, 4^1)$ table.

$$L_8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 \\ 2 & 1 & 1 & 2 & 2 \\ 2 & 2 & 2 & 1 & 1 \\ 3 & 1 & 2 & 1 & 2 \\ 3 & 2 & 1 & 1 & 1 \\ 4 & 1 & 2 & 2 & 1 \\ 4 & 2 & 1 & 1 & 2 \end{pmatrix} \quad (D.1)$$

with the following factor table:

Factor	Levels	Type	Taguchi Slot
Power voltage	2.8/5/12.4/22 VDC	4 levels	Col1, A_i
Power Source	Separate / Same	2 levels	Col2, B_j
Wireless	ON / OFF	2 levels	Col3, C_k
Recording	ON / OFF	2 levels	Col4, D_l
Audio signal	With / Without	2 levels	Col5, E_m

Table D.1: Interference: L8 plan mapping

Method

As in Chap.8, the *TENMA 72-10495 Digital-control DC Power Supply 0-30V 5AX2* was used. To evaluate interference, we analyze the audio recordings from the QHB, examining both the signal-to-noise ratio and the noise level. A sound emitter, in this case a *JBL Go* speaker, is placed in front of a microphone connected to the QHB. The QHB and UWOS are either connected to the same power supply (TENMA) or to a separate battery pack of D-cell alkaline batteries.

D.3 Resistance to Wave Action

Note

Reminder, due to time constraints and various delays this experiment was not carried out !

This experimental plan is proposed to ensure that the presence of the camera module does not destabilize the base system. All factors are tested at three levels:

Factor	Levels	Type	Taguchi Slot
Camera presence	NO / YES / YES	3 levels	Col1, A_i
QHB orientation	0 / $\pi/4$ / $\pi/2$	3 levels	Col2, B_j
Wave amplitude	Low / Medium / High	3 levels	Col3, C_k
Wavelength	15 fps / 20 fps / 30 fps	3 levels	Col4, D_l
Current	Weak / Medium / Strong	3 levels	Col5, E_m

Table D.2: L9 plan mapping

D.4 Optical Distortion Characterization

Note

Reminder, due to time constraints and various delays this experiment was not carried out !

Context and Objectives

In our case, optical distortion arises mainly from the acrylic dome and the air/water interface. To quantify distortion, we use a uniform distortion function applied to an orthogonal grid, with coefficient k , following Pelletier et al. [17]. This function moves the grid nodes closer or further from the image center based on k , as described by the following equations:

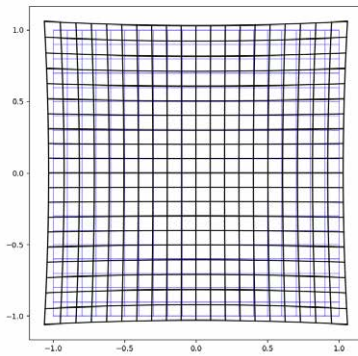
$$x_{distorted} = x_{real}(1 + k \cdot r^2) \quad (D.2)$$

$$y_{distorted} = y_{real}(1 + k \cdot r^2) \quad (D.3)$$

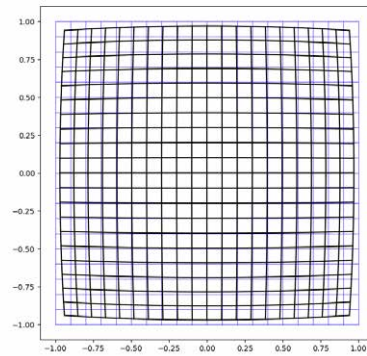
where:

$$r^2 : x_{real}^2 + y_{real}^2$$

k : distortion coefficient (positive for a "fish-eye" effect, negative for pincushion distortion)



(a) Example with $k = 0.03 > 0$



(b) Example with $k = -0.03 < 0$

Figure D.2: Distorted (black) and regular (blue) grids for two distortion coefficients.

The goal of this experiment is to quantify the optical distortion coefficient k introduced by the presence of a port in air and water, and to compare the calibration matrices obtained in both media to assess the relevance of image correction models depending on the usage context.

Setup, Method, and Results

Component	File Type	License
UWOS (Raspberry Pi 4) + Arducam	Electronic	–
Optical dome 2" (BlueRobotics)	Physical	–
3D-printed camera mount	SolidWorks file	–
Calibration target (25 mm checkerboard)	PDF file	CC BY-SA
Water tank (3x2x1 m)	Physical	–

Table D.3: Components used for optical characterization

Method

The camera system is securely mounted inside the housing to maintain a fixed position relative to the port (cf. Chap. 5). The lens glass is aligned with the dome's optical center. The camera is placed on top of the instrument. An external tripod holds the calibration target at various distances (20 cm to 1 m) and angles.

Lighting is natural but controlled to minimize reflections on the dome. Images are captured in both media (air then water) without modifying the camera's position in its housing.

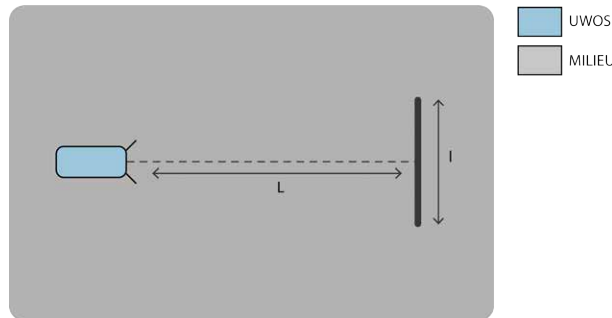


Figure D.3: Schematic of the optical characterization experiment

With:

L : Distance between the camera and the wall (here 240 cm)

I : Width of the calibration target

For this experiment, the number of factors being limited, the simplest Taguchi table L_4 (2^3) is used:

$$L_4 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \quad (D.4)$$

Factor	Levels	Type	Taguchi Slot
Medium	Air / Water	2 levels	Col1, A_i
Target distance	0.7 m / 3 m	2 levels	Col2, B_j
Target tilt	0° / 45°	2 levels	Col3, C_k

Table D.4: Optical distortion: L4 plan mapping



User feedback Form

Date: _____

Name / Team: _____

Deployment location: _____

Role during deployment: Diver Technician Data Analyst Other: _____

Section 1 – Deployment & Setup

1. How easy was it to transport the instrument to the deployment site?

Very easy Easy Neutral Difficult Very difficult

Comments: _____

2. How easy was the assembly and pre-deployment setup (mounting, ballast, cable connections)?

Very easy Easy Neutral Difficult Very difficult

Comments: _____

3. How would you describe the handling of the instrument underwater?

Very easy Easy Neutral Difficult Very difficult

Comments: _____

4. Is the weight of the instrument a hindrance to its installation?

Yes Neutral No Other: _____

Section 2 – Operation

5. Was the start-up and configuration process straightforward?

Yes Mostly No

If no, specify the main difficulties: _____

6. Were the indicator signals (LEDs, logs, acoustic cues) clear enough to confirm correct operation?

Yes No

Comments: _____

7. How well did the instrument remain stable in its deployed position?

Very stable Stable Slight movement Significant drift

Section 3 – Data Retrieval & Quality

8. How easy was it to retrieve the instrument after deployment?

Very easy Easy Neutral Difficult Very difficult

9. Were the data files accessible and in the expected format?

Yes Partially No

10. Rate the quality of the acoustic recordings:

Excellent Good Acceptable Poor Unusable

Comments: _____

11. Rate the quality of the optical recordings:

Excellent Good Acceptable Poor Unusable

Comments: _____

Section 4 – Suggestions

12. Which features worked particularly well?

13. Which improvements would you suggest for future deployments?

14. Are there additional features or capabilities you would like to see integrated?

Enhancing Marine Acoustic Research with Vision

This internship report presents the design and implementation of an autonomous, open-source, low-cost, and energy-efficient optical sensor, developed to complement passive acoustic monitoring of NBHF (Narrow-Band High-Frequency) marine species. Integrated into the Qualilife HighBlue (QHB) platform from the University of Toulon, the system enables continuous underwater image acquisition for up to several days, providing a visual counterpart to acoustic recordings.

The prototype was first tested in laboratory and coastal conditions in Toulon, before being deployed during scientific missions off the coast of Concepción, Chile. By capturing synchronized video and acoustic data, the sensor extends the scope of marine biodiversity monitoring, with potential applications in ecological research, education, and citizen science.

Beyond the technical development, this work also reflects on the broader responsibilities of engineers in environmental research. A full carbon footprint assessment of the internship was conducted, highlighting the importance of minimizing the ecological impact of scientific fieldwork and international collaborations. This dual perspective—technological innovation and ecological responsibility—illustrates how engineering can contribute not only to advancing science, but also to protecting the very ecosystems it seeks to understand.

Autor: Matthieu Jung

Supervisors: Julie Patris, Hervé Glotin, Iván Hinojosa, Lyudmyla Yushchenko.

In collaboration with : Valentin Gies, Valentin Barchasz, Franck Malige.

With the participation of: Mauricio Soto Gamboa, Camila Calderon-Quirgas, Mauricio Gonzalez Valderrama, César Villarroel...

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Host Institution: LIS CIAN Laboratory & SMIoT Laboratory of the University of Toulon, Universidad Catolica de la Santissima Concepcion, Universidad Austral de Chile

Engineering School: SeaTech